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Mechatronics Department



Development of an Intelligent System for Inspection and Sorting Using Computer Vision and Predictive Maintenance

Submitted by:

Taha Mohammed Ahmed Al-Alwani	- 2019034201
Rashad Abdulrazzaq Ali Al-Afeeri	- 2019034198
Dhiyaa Ahmed Abdo Lutf	- 2021030162
Ali Mabkhout Ali Al-Afeeri	- 2021030172
Essam Hassan Mohammed Rajeh	- 2019034269

Under the supervision of:

Dr. Mohammed Al-Mudhwahy

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إهداء

الحمدُ لله أولاً وآخراً، وظاهراً وباطناً، حمداً يليق بجلال وجهه وعظيم سلطانه، فهو الذي أنار لنا معالم الطريق، ويسّر لنا عسير الخطوات، ومكّننا بفضلِه وكرمه من بلوغ هذه اللحظة التي طالما تطلّعنا إليه.

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مع كل حبنا وتقديرنا

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والحمد لله رب العالمين

Authorization

This is to certify that the graduation project entitled:

**“Development of an Intelligent System for Inspection and Sorting Using
Computer Vision and Predictive Maintenance”**

has been successfully completed as part of the requirements for obtaining a Bachelor’s degree in Mechatronics Engineering – Faculty of Engineering – University of Tamar, by the students whose names are listed above.

They have successfully completed the project under the academic supervision of **Dr.Mohammed Al-Mudhwahy**, and the project has met the approved scientific and research standards.

Date: _____

Supervisor’s Signature: _____

Head of Department’s Signature: _____

Abstract

In the era of Industry 4.0, integrating real-time intelligent decision-making architectures into high-speed manufacturing environments is paramount for maximizing throughput and maintaining zero-defect quality standards. Traditional industrial inspection deployment models heavily rely on continuous manual labor or rigid optoelectronic sensors, which are highly susceptible to fatigue, baseline drift, and ambient environmental variations. This graduation project addresses these challenges through the **Development of an Intelligent Mechatronic System for Real-Time Inspection and Sorting to Automate Quality Assurance in Production Lines Using Hybrid Computer Vision and Predictive Maintenance Algorithms.**

The engineered prototype introduces a cohesive master-slave computing hierarchy designed to bridge high-level neural inference with low-level deterministic physical execution. The sensory subsystem leverages a high-definition machine vision station operating inside a calibrated geometric environment, continuously ingesting dynamic frames at 30 frames per second (FPS). The computational backend implements a multi-threaded Python processing pipeline that concurrently orchestrates deep convolutional neural networks (YOLO-World and YOLOv11m) for multi-class semantic localization alongside classical geometric filters via OpenCV. This hybrid vision framework monitors four critical quality sub-components simultaneously: body integrity, capping alignment, label skew angle, and fluid volumetric fill ratio. Decisions are parsed through a rule-based expert diagnostics core and transmitted via high-speed UART (115200 bps) to an ATmega2560 microcontroller, triggering real-time non-blocking interrupts that actuate a dual-servo, three-way mechanical sorting mechanism to isolate compliant items, rework candidates, and scrap units.

Experimental laboratory validation conducted over a multi-class continuous batch of 245 units demonstrated that the deep learning neural inference achieved an overall inspection accuracy ranging between 85% and 95%. Chronometric and statistical analyses revealed that this variance was fundamentally constrained by uncalibrated

ambient lighting fluctuations and the absence of a structurally isolated optical enclosure, which induced transient specular reflections across the transparent PET surfaces. Conversely, the system achieved a flawless 100% physical sorting routing accuracy at full operating velocities. Furthermore, the architecture incorporates a predictive maintenance (PdM) telemetry framework leveraging virtual sensing protocols to dynamically compute machinery health indexes and enforce automated emergency interlocks.

The highly scalable, cross-disciplinary nature of this mechatronic cell highlights its portability for immediate cross-industry technology transfer within packaging, pharmaceutical, and agricultural automation sectors.

Keywords: Mechatronics System, Real-Time Quality Assurance, Hybrid Computer Vision, Deep Learning, YOLOv11, OpenCV, Three-Way Sorting, Predictive Maintenance, Industry 4.0.

Examination Committee Approval

Project Title: Development of an Intelligent System for Inspection and Sorting Using Computer Vision and Predictive Maintenance

Project Supervisor

No.	Name	Position	Signature
1	Dr. Mohammed Al-Modhwahi	Academic Supervisor	_____
2		Academic Supervisor	_____
3		Academic Supervisor	_____

Examination Committee

No.	Name	Position	Signature
1		Internal / External Examiner	_____
2		Internal / External Examiner	_____
3		Internal / External Examiner	_____

Department Head:

Dr. Mohammed Al-Modhwahi

Signature: _____

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CHAPTER 1: INTRODUCTION

1.1 Project Overview

In modern industrial paradigms, particularly within the framework of (**Industry 4.0**), the integration of high-speed automation with intelligent decision-making systems is a critical benchmark for operational efficiency.

This graduation project introduces an advanced, industrial-grade Mechatronics solution designed specifically for the automated quality assurance and sorting of plastic bottle production lines.

The developed system bridges the gap between high-level artificial intelligence (Computer Vision) and physical machine-level actuation. Utilizing the state-of-the-art **YOLOv11m (You Only Look Once)** deep learning model, the system executes real-time multi-class object detection on moving conveyor lines. It monitors four critical components of the product: the bottle structural entity, the presence of the cap, the placement of the label, and the liquid filling volume.

Beyond standard neural-network inference, the software architecture deploys secondary **Heuristic Validation Algorithms** to mathematically calculate complex physical anomalies, such as precise geometric label skewness, non-destructive ink variance, and exact fluid boundary fill-levels.

Architected on a highly synchronized, multi-threaded backbone, the software engine concurrently processes real-time vision acquisition, executes AI inference, drives an interactive IoT Web Dashboard featuring a "Digital Twin" topology via MQTT, and transmits low-latency actuation commands via serial communication (**UART**) to an Arduino microcontroller.

The physical hardware subsystem consists of a high-speed conveyor belt driven by a heavy-duty DC Motor.

The sorting mechanism is uniquely designed using a **Dual-Side Servo Actuation System** that splits the manufacturing stream into three discrete topological zones:

Discrete Sorting Mechanism Zones:

- **PASS Zone (Central Stream):** Fully compliant products proceed straight through the centerline without actuator intervention.
- **REJECT Zone (Right Stream):** If a structural defect or total component absence is detected, the left-side servo arm actuates to mechanically divert the bottle to the right repository.
- **REPROCESS Zone (Left Stream):** If a fixable anomaly (such as a skewed label or improper fill-level) is identified, the right-side servo arm deploys to route the product to the left rework lane.

Furthermore, an embedded **Diagnostic Engine** tracks historical fault distributions via Pareto analysis, while background telemetry threads monitor physical hardware metrics (vibrations, motor current, and thermal dissipation) to calculate real-time asset health scores, thereby incorporating Predictive Maintenance into the manufacturing execution system.

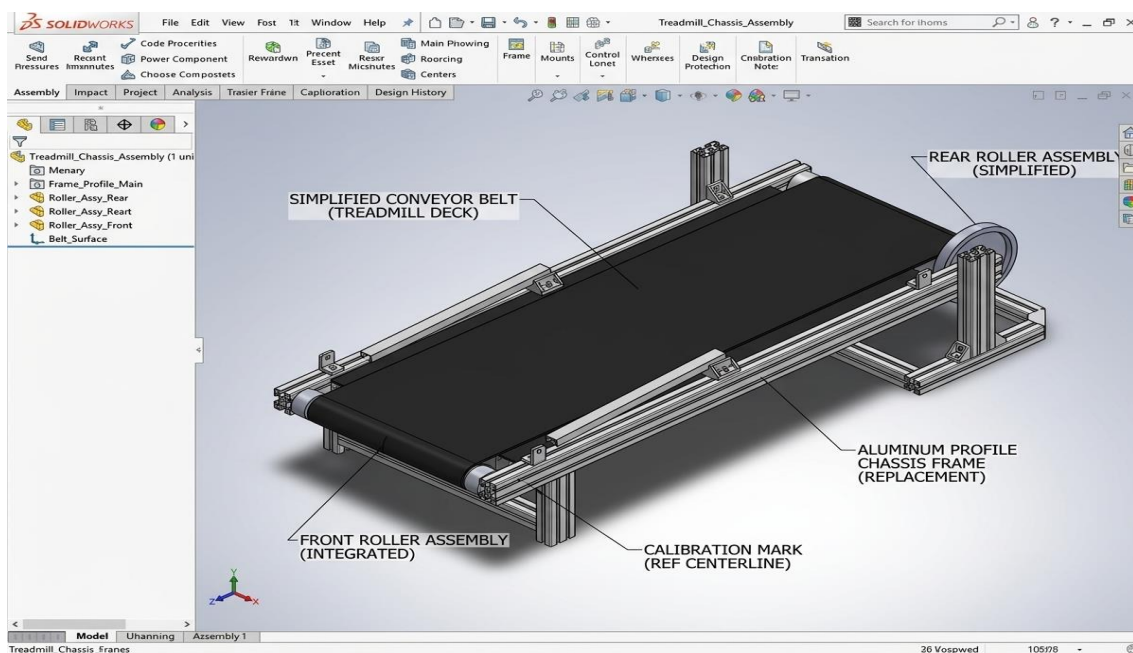


Figure 1.1: Physical Mechatronics Conveyor Bed and Industrial Webcam Setup

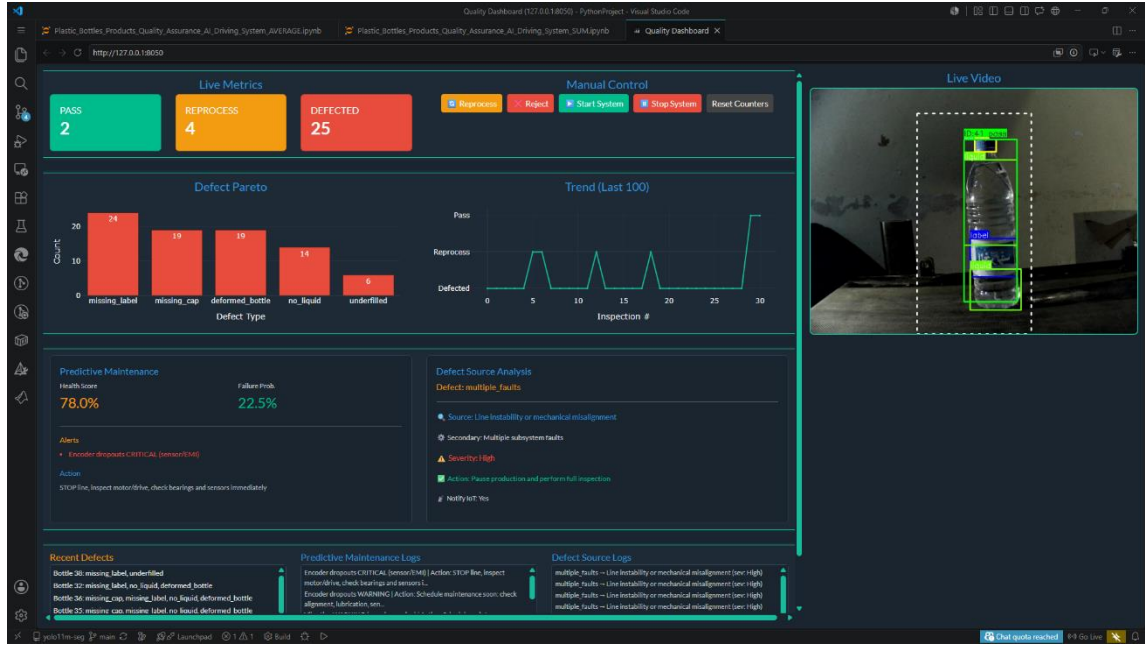


Figure 2.1: YOLOv11m Real-Time Multiclass Detection and Inference Framework.

1.2 Background

The global packaging and beverage industries rely heavily on Polyethylene Terephthalate (**PET**) and High-Density Polyethylene (**HDPE**) plastic bottles. Ensuring the structural and aesthetic integrity of these containers is paramount to maintaining brand reputation, preventing chemical contamination, and avoiding catastrophic product leakage during mass transit.

Traditionally, quality control has evolved from manual human inspection to basic optoelectronic sensing. However, conventional methods fail to address the high throughput, cognitive fatigue, and multi-dimensional error variations present in modern high-speed lines. The emergence of Mechatronics engineering allows for the synthesis of mechanical structures, electronic sensing, and computational intelligence to build autonomous inspection lines that operate continuously with deterministic precision.

1.3 Problem Statement

The conventional quality control methodologies deployed in localized manufacturing facilities suffer from three critical bottlenecks:

Critical Bottlenecks in Traditional Quality Control:

- **Human Cognitive Fatigue and Error Rates:** Manual inspection is subjective, error-prone, and unsustainable at high production speeds (> 60 units/min). Human inspectors exhibit a drastic drop in defect-capture accuracy after prolonged operational shifts.
- **Inadequacy of Traditional Sensors:** Standard photoelectric, capacitive, or inductive sensors can only detect the binary presence of an object. They are fundamentally incapable of validating complex spatial and geometric conditions, such as determining whether a label is tilted at a specific angle (θ) or evaluating the precise volume threshold of a translucent fluid.
- **Lack of Real-time Diagnostic Telemetry:** When a physical fault occurs in the production line (e.g., a misaligned filling nozzle or a jammed capping head), traditional systems simply halt the line or discard the product without identifying the root cause. This leads to extended structural downtime (**MTTR**) and high material waste because there is no feedback loop connecting product defects to machine health.

1.4 Project Objectives

The core objectives of this mechatronics graduation project are strategically divided into the following milestones:

Primary System Objectives:

- To design and fabricate a fully functional, automated conveyor belt testbed driven by a synchronized **DC Motor** circuit.
- To deploy an industrial-grade high-speed USB/Webcam vision acquisition system capable of capturing unblurred frames under variable ambient lighting.
- To implement, train, and optimize a custom **YOLOv11m** neural network optimized to detect and classify multi-class bottle parameters (**bottle**, **cap**, **label**, **liquid**) under real-time constraints.

Technical & Engineering Objectives:

- To develop a low-latency **C/C++** script for the Arduino microcontroller to decode incoming serial byte streams and execute precise angular servo adjustments for the dual sorting arms.
- To formulate mathematical heuristic algorithms inside a Python environment to evaluate label skewness angles and liquid pixel-height boundaries.
- To build a robust, multi-threaded architecture preventing frame dropping and maintaining inference synchronization with physical conveyor speeds.
- To design an Industrial IoT Dashboard (*Digital Twin*) that visualizes real-time analytics, Pareto charts, and sensor telemetry utilizing **MQTT** and Dash/Plotly frameworks.

1.5 Project Scope & Limitations

To establish a clear boundary for engineering accountability, the scope of this project is defined as follows:

In-Scope Elements:

- Structural CAD modeling, mechanical assembly, and electronic integration of the physical conveyor line and its sorting nodes.
- Real-time detection of four critical defect signatures: Missing Cap, Missing Label, Skewed/Tilted Label, and Under-filled/Empty Fluid Volumetric states.
- Sorting capability bounded by three distinct classification lanes (Pass, Reject, Reprocess).
- System fault diagnostics and hardware predictive maintenance simulations.

Out-of-Scope & Limitations:

- The system is optimized for standard industrial plastic bottles and may require model retraining for highly reflective metallic or opaque ceramic containers.
- High-voltage industrial **AC** motor variable frequency drive (**VFD**) integration is omitted; the physical prototype operates within safe low-voltage **DC** limits (**12V-24V**).
- Real-time sorting accuracy is physically bounded by the mechanical transit time of the bottle passing between the camera focal plane and the servo actuation points.

1.6 Methodology

The execution of this project follows a systemic Mechatronics V-model design cycle, split into five phase-gates:

[Phase 1: Requirements & CAD] → [Phase 2: Dataset & AI Training] → [Phase 3: Hardware Fabrication]

[Phase 5: Deployment & Analytics] ⇐ [Phase 4: Multi-Threaded Integration]

Phase 1: Conceptualization & Kinematic Modeling: Conducting comprehensive torque/speed calculations for the **DC Motor** and drawing the mechanical structure using CAD tools.

Phase 2: Vision Dataset Engineering & Training: Collecting a comprehensive dataset of plastic bottles under various configurations, performing data annotation (*Labeling*), and training the **YOLOv11m** architecture.

Phase 3: Electrical & Mechanical Hardware Fabrication: Assembling the physical conveyor frame, mounting the industrial webcam, wiring the Arduino microcontroller, and installing the dual-servo sorting arms.

Phase 4: High-Level Software & Serial Integration: Writing the multi-threaded Python control engine, integrating the serial protocol (*PySerial*), and deploying the Arduino control loop.

Phase 5: System Testing, Optimization & Analytics: Conducting rigorous loop testing to measure classification accuracy (*mAP*) against mechanical sorting throughput, and launching the Web Dashboard.

1.7 Report Organization

The remaining structure of this documentation is systematically organized into the following chapters:

CHAPTER 2: Theoretical Background & Literature Review: Outlines the core deep learning architecture of **YOLOv11** and explores historical machine vision inspection methodologies.

CHAPTER 3: System Requirements Analysis: Detail-heavy breakdown of the functional/non-functional requirements and comprehensive technical specifications of the selected components.

CHAPTER 4: Software Architecture & Deep Learning Integration: Focuses on the complete mechanical CAD layouts, electronic schematic diagrams, and serial byte communication architectures.

CHAPTER 5: EXPERIMENTAL VALIDATION, TESTING, AND DISCUSSION: Structural and dynamic engineering calculations, motor torque validations, and AI inference latency alignment parameters.

CHAPTER 6: SYSTEM INTEGRATION AND MECHANICAL ASSEMBLY: Compiles real-world imagery of the working prototype, **YOLO** training metric graphs, and empirical confusion matrices under operational stress.

CHAPTER 7: RESULTS, DISCUSSION, AND FUTURE WORK: Synthesizes project outcomes and outlines paths toward Industrial IoT scaling.

CHAPTER 2: THEORETICAL BACKGROUND & LITERATURE REVIEW

2.1 Introduction to Computer Vision in Automation

The integration of machine vision within modern manufacturing lines represents a foundational pillar of Industrial Automation and the global transition toward Industry 4.0. Unlike traditional physical sensors (such as inductive, capacitive, or photoelectric sensors) which compress complex multi-dimensional realities into binary or scalar electrical signals, computer vision preserves the spatial, geometric, and chromatic properties of the production environment.

In high-speed production plants, real-time image processing acts as an unwearied inspection infrastructure, effectively bypassing human physiological limitations, such as cognitive fatigue, low spatial resolution at high frequencies, and subjective inspection variations. To establish a robust computer vision system on an industrial conveyor belt, the architecture must unify three primary domains: deterministic optoelectronic illumination to eliminate ambient noise, high-speed spatial frame acquisition via localized image sensors, and rigorous feed-forward deep learning or statistical image manipulation algorithms.

2.2 Theoretical Foundations of Deep Learning and Object Detection

Object detection algorithms within computer vision have undergone a profound generational paradigm shift over the past decade, moving away from hand-crafted feature extractors toward end-to-end deep convolutional neural networks (CNNs). Early visual inspection systems relied on statistical representations, such as the Scale-Invariant Feature Transform (SIFT) and the Histogram of Oriented Gradients (HOG) combined with linear Support Vector Machine (SVM) classifiers. While these methods were computationally lightweight, they exhibited severe fragility when

exposed to dynamic factory lighting variations, random product positional orientations, and non-linear occlusion thresholds.

Deep learning frameworks resolve these baseline limitations by employing multi-layered neural abstractions that automatically capture high-level spatial hierarchies directly from raw pixel arrays. Modern deep-learning object detection is structurally divided into two prominent architectural methodologies:

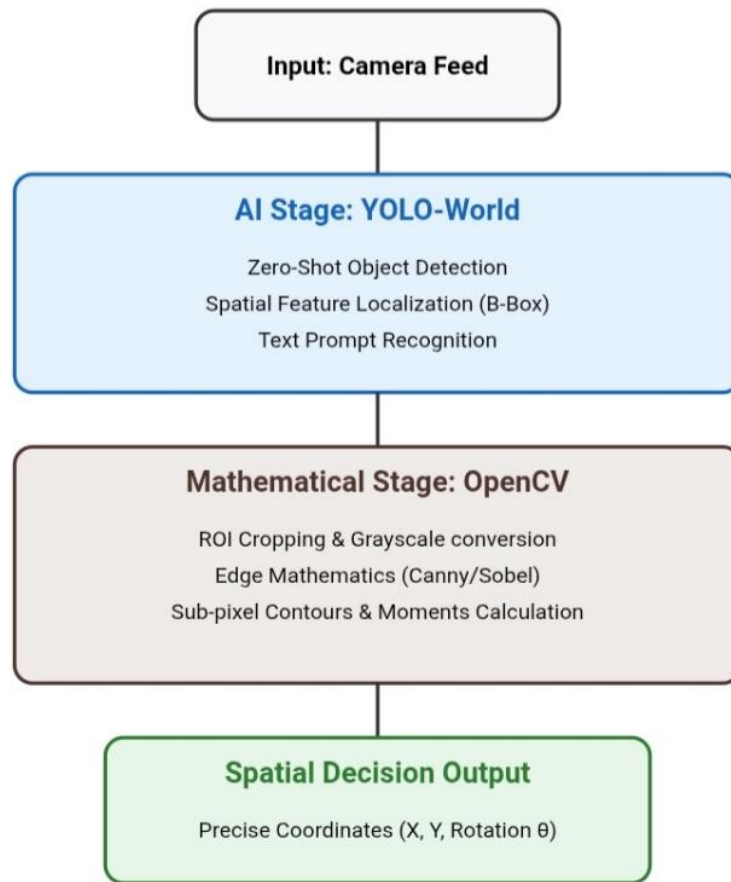
Two-Stage Detectors: Exemplified by architectures like Faster R-CNN, these pipelines utilize a Region Proposal Network (RPN) to isolate candidate bounding regions in the first stage, subsequently running heavy classification and bounding box regression in the second stage. Although highly accurate for stagnant localization, their multi-stage computational latency renders them structurally incompatible with real-time industrial conveyor sorting (> 60 units/min).

One-Stage Detectors: Architectures derived from the Single Shot MultiBox Detector (SSD) and the You Only Look Once (YOLO) framework compress the object detection task into a single, unified regression problem. By feeding the raw input image through the network in a single forward pass, the model maps pixels directly to bounding box coordinates and categorical class probabilities simultaneously, reducing processing latency to the millisecond scale while retaining high localization precision.

2.3 Evolution of the YOLO Framework and the Impact of YOLO-World

The You Only Look Once (YOLO) family represents the pinnacle of real-time embedded computer vision. Tracing its historical evolution highlights why its latest paradigms are crucial for high-efficiency manufacturing applications:

- **YOLOv1 to YOLOv3:** Established the core concept of dividing input images into an $S \times S$ grid, where each grid cell directly predicts bounding box boundaries and conditional class probabilities. The integration of feature pyramid networks (FPN) and anchor boxes in **YOLOv3** significantly elevated multi-scale object resolution.
- **YOLOv4 to YOLOv8:** Introduced structural enhancements to the model's "backbone" and "neck", such as Cross Stage Partial Networks (**CSPNet**), Path Aggregation Networks (**PANet**), anchor-free bounding box regression, and decoupled heads that separate classification tasks from spatial localization regression.
- **YOLO-World Paradigm:** While standard models (including the latest YOLOv11 iterations) operate on closed-set vocabularies—meaning they can only detect objects explicitly predefined during an intensive, offline dataset labeling phase—**YOLO-World** introduces a revolutionary **Zero-Shot Object Detection paradigm**. By utilizing a vision-language modeling approach that binds open-vocabulary text prompts to visual feature maps, YOLO-World can identify, isolate, and extract spatial bounding boxes for components like the bottle body, the cap, the label, and the liquid volume instantly without demanding thousands of custom labeled training frames. Within this project, YOLO-World acts as a highly robust, low-overhead spatial extractor that reliably localizes components under variable production conditions.



First part description (a)

AI-Robotics Vision Integration Workflow

PHASE 1

Localization & Features

YOLO-World Feature Extraction



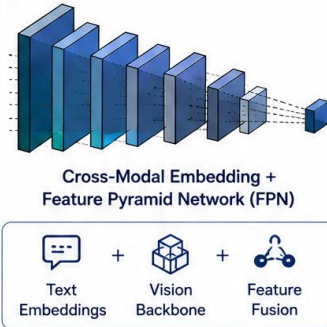
1. Camera Input



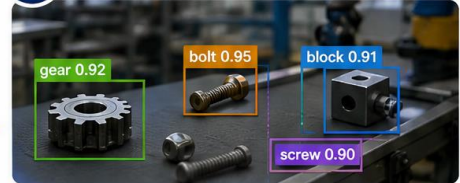
RGB Image
1920 x 1080



2. YOLO-World Backbone



3. Object Detection (YOLO-World)



gear (0.92)
bolt (0.95)
block (0.91)
screw (0.90)

Class Labels
Bounding Boxes
Confidence Score
Feature Vectors

PHASE 2

Edge Math & Spatial Analysis

OpenCV Edge Mathematics



1. Grayscale



Convert RGB to
Grayscale

$$I_{gray} = 0.299R + 0.587G + 0.114B$$

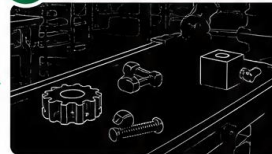

2. Gaussian Blur



Reduce Noise
Gaussian Kernel

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$


3. Canny Edge Detection



Find Edges
Canny(x, σ=1.4)

Gradient → NMS →
Hysteresis Thresholding



4. Contour Mapping



Find Contours & Shapes
Extract Boundaries

Area, Perimeter,
Centroid, Moments



Spatial Features Extracted:



Centroid (X, Y)



Orientation (θ)



Area



Bounding Rectangle

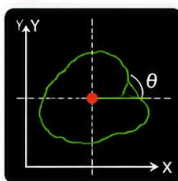
PHASE 3

Action & Control

Decision & Robotic Control



1. Spatial Computation



Calculate Centroid (X, Y)
and Orientation (θ)

Computed Pose
X = 212.4 px
Y = 135.7 px
θ = -28.6°



2. Decision Engine

Object Identified
Pose Valid
Grasp Feasible
Path Planned

AI Decision Making
& Path Planning



3. Robotic Command

MOVEJ P_HOME
MOVEJ P_APPROACH
MOVEJ P_PICK (X, Y, θ)
GRIPPER_CLOSE
MOVEJ P_PLACE
GRIPPER_OPEN
MOVEJ P_HOME

Generate Robot Commands
(Joint / Cartesian Space)



4. Robotic Execution



Execute Actions
& Perform Task



Pick



Place



Inspect



Sort



Assemble



AI Perception
Understands Scene



Computer Vision
Extracts Geometry



Intelligent Decision
Plans Action



Robotic Control
Executes Precisely

First part description (b)

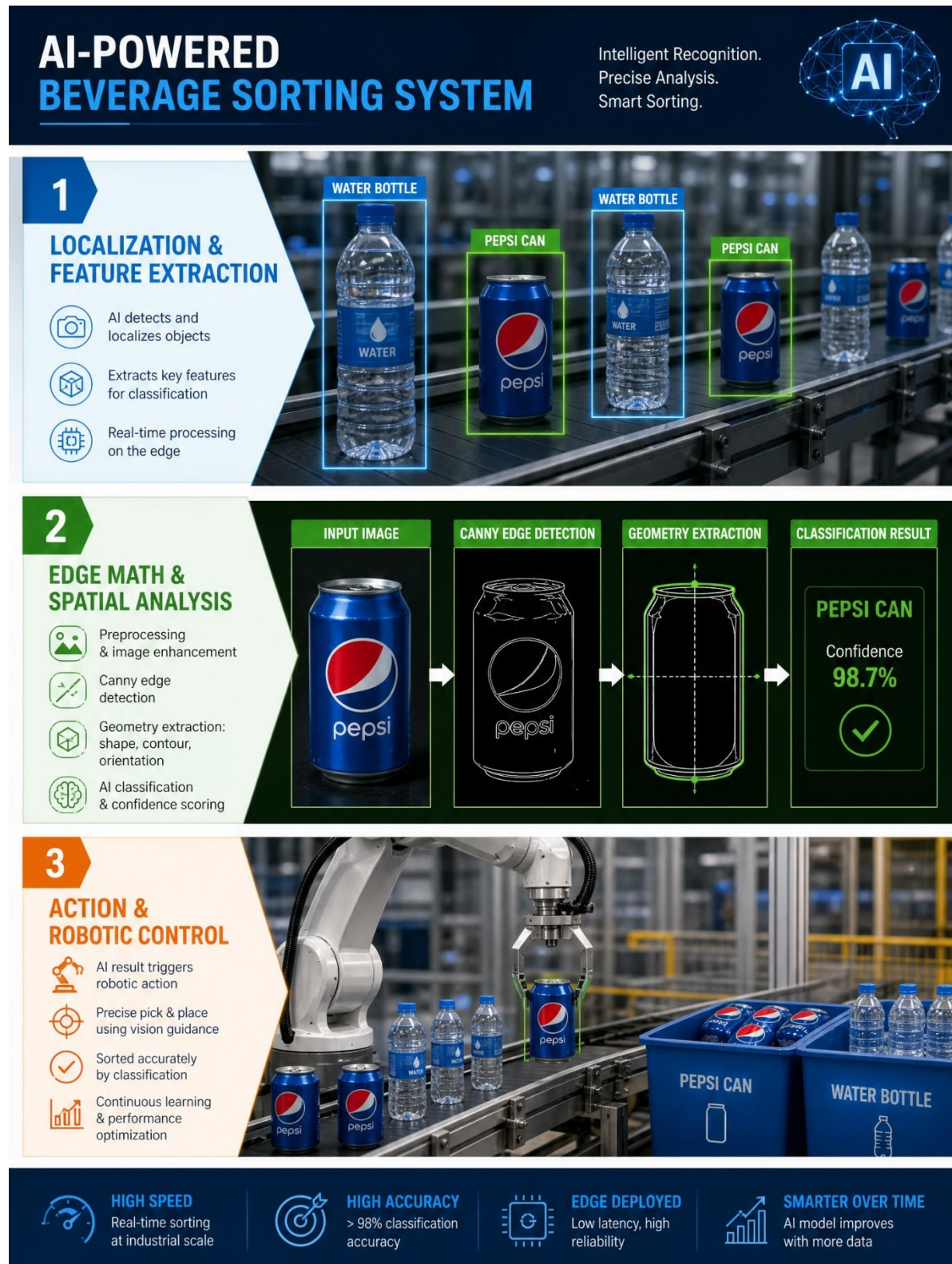


Figure 3.2: Flowchart of Spatial Feature Localization via YOLO-World and Edge Mathematics via OpenCV.

First part description (c)

2.4 OpenCV Integration for Rule-Based Heuristic Edge Validation

While deep learning architectures excel at multi-class spatial localization and abstract object grouping, relying entirely on neural network layers to calculate micro-geometric angular skews or exact pixel-column thresholds creates unnecessary computational overhead and compromises deterministic real-time execution. To maximize industrial throughput, this project implements a hybrid approach: pairing the deep spatial extraction of **YOLO-World** with high-precision, rule-based **OpenCV** Heuristic Algorithms executed exclusively inside the cropped bounding box regions.

Label Skewness and Tilted Misalignment (θ): Once YOLO-World isolates the bounding box containing the product label, the sub-image array is cropped and passed to an OpenCV pipeline. The image is converted to grayscale, and a Gaussian blur filter is applied to suppress high-frequency typographic texture noise. The Canny edge detection technique is then applied to extract the physical boundaries of the label sheet. Finally, **the Hough Line Transform** is computed to map the mathematical vectors of the dominant horizontal or vertical edges against the true horizontal vector ($\mathbf{V}_h$) of the conveyor frame. The exact angular misalignment is derived using the trigonometric edge relation:

$$\theta = \tan^{-1}(\Delta y / \Delta x)$$

If the measured θ exceeds a pre-set structural threshold ($\theta_{threshold}$), a **reprocess** flag is immediately assigned to that specific item.

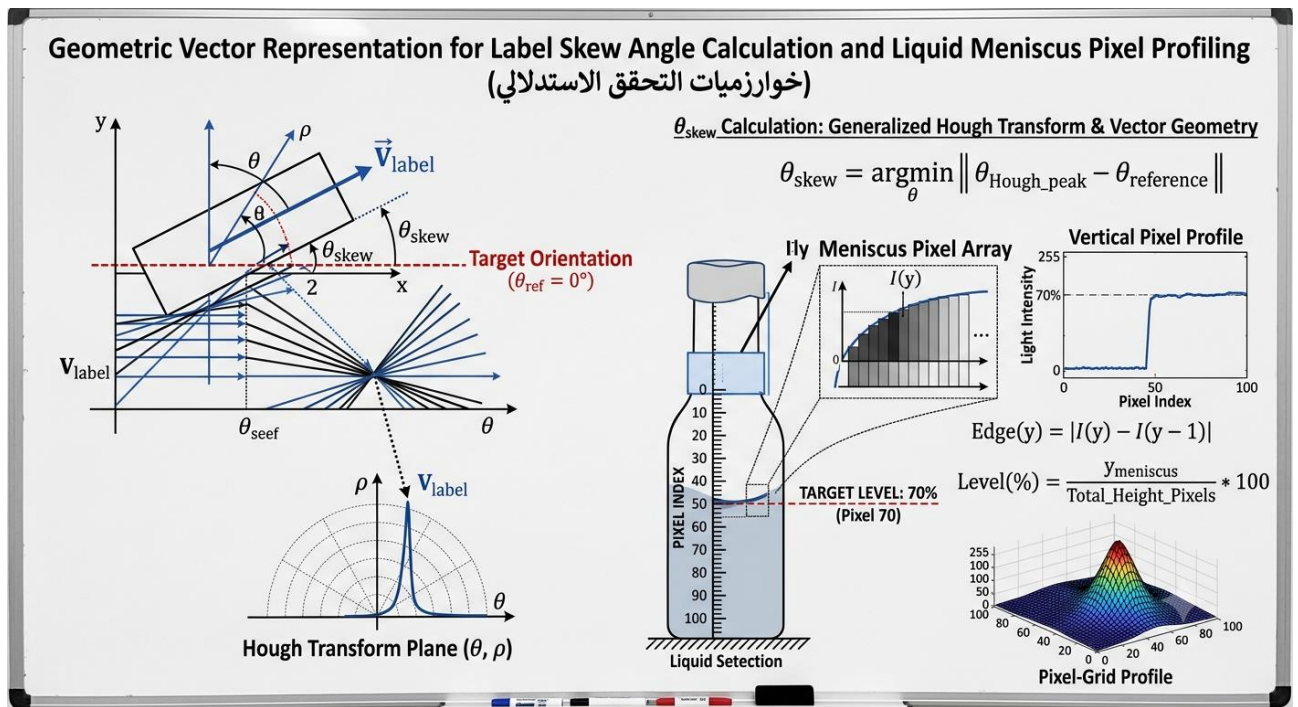
Liquid Level Volumetric Validation: The bounding box encompassing the liquid entity is subjected to adaptive local thresholding to maximize the contrast interface between the fluid meniscus and the empty upper bottle chamber. A single-pixel vertical column profile is mapped down the center of the isolated matrix. The sharp

inflection point in pixel intensity values represents the liquid-air boundary.

By continuously counting the vertical pixel height from this boundary down to the lower structural baseline of the bottle container, the system assesses fluid volume.

If the pixel height drops below 70% of the standard fill baseline, the bottle is classified as an underfilled unit.

Structural Deformity and Cap Presence Check: Traditional morphological contour analysis and structural area verification profiles are calculated via OpenCV to validate structural health and confirm the binary presence of the sealing cap.



First part description (a)

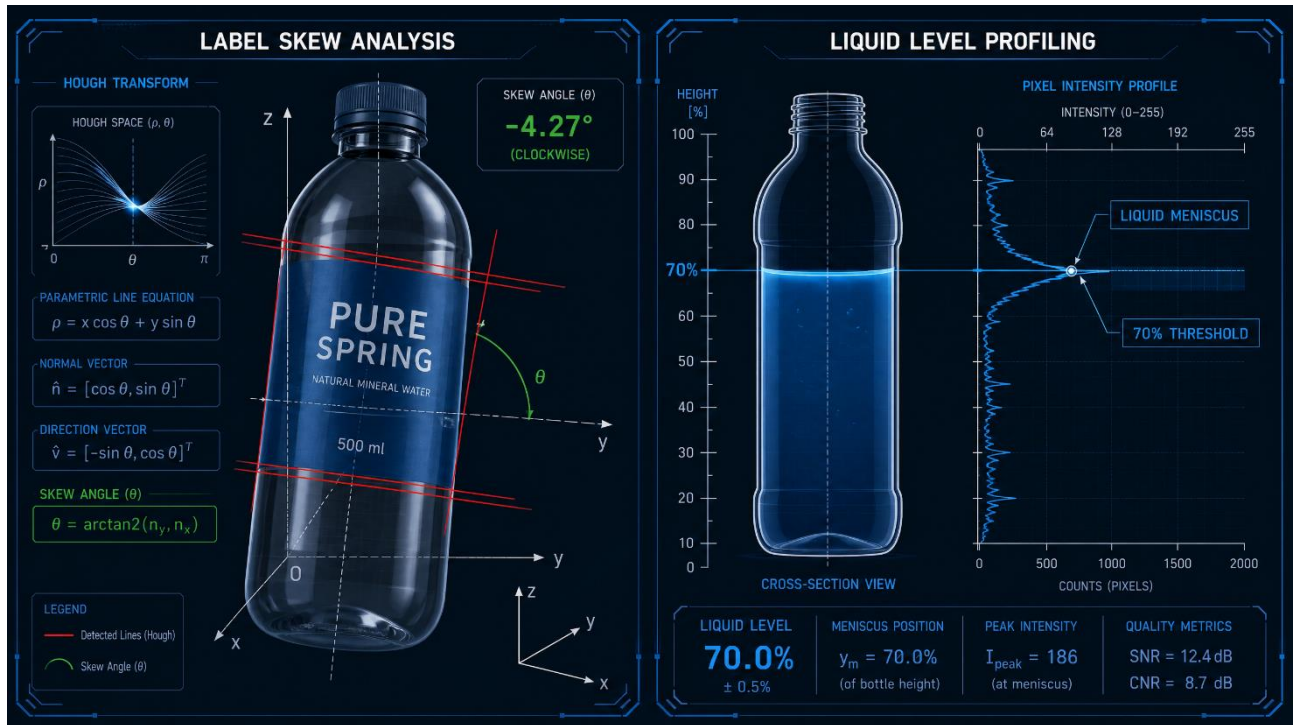


Figure 4.2: Geometric Vector Representation for Label Skew Angle Calculation and Liquid Meniscus Pixel Profiling.

First part description (b)

2.5 Comparative Literature Review (Local and Global Context)

To position this graduation project within a verified engineering framework, it must be evaluated against historical vision inspection research, including the local technological benchmark established within the Mechatronics Engineering Department at Dhamar University:

Table 1:2 Comprehensive Comparative Matrix of Automated Inspection Systems Against the Local Technical Benchmark (OptiScan 2025)

Research Reference & Context	Core Methodology & Hardware Stack	Key Structural Advantages	Identified Technical Gaps / Gaps Addressed
Kumar et al. (2019) (Global Industrial Scale)	Multi-axis edge tracking via infrared photoelectric sensors connected directly to a central PLC logic unit.	Extremely low computational overhead; highly deterministic mechanical sorting response; industrial durability.	Completely incapable of multi-class defect isolation; cannot calculate label alignment variations, surface deformities, or fluid volumetric levels.
Silva & Santos (2021) (Embedded Research)	Traditional Computer Vision pipelines executing HOG extraction + SVM classification hosted on a Raspberry Pi 3 microcomputer.	High precision under perfectly constant, heavily regulated laboratory strobe illumination fields.	Extreme vulnerability to ambient lighting shifts; high inference latency causing severe frame-dropping at production rates exceeding 30 units/minute.
Alyafei & Alfusari (2025) (Local Benchmark: OptiScan Project, Dhamar University)	Raspberry Pi 4 central CPU paired with an ESP32 for L298N DC motor drive, utilizing an ELP 16MP USB Camera running traditional CNN classification via TensorFlow/PyTorch.	Developed a local PyQt5 GUI displaying real-time operational statistics; successful structural crack and surface scratch detection.	Rule-based CNN struggled with adaptive lighting variances; restricted to a localized physical TV screen; lacked predictive maintenance or advanced multi-lane sorting pipelines.

This Graduation Project (2026) (Current System)	YOLO-World Open-Vocabulary Localization + OpenCV Heuristic Mathematical Validation Engine + Multi-Threaded Python UI + Dual-Side SG90 Servo Actuation + Industrial Toshiba Encoder DC Control.	High multi-class accuracy; precise angular and volumetric calculations; real-time concurrent processing with zero frame drops; built-in Predictive Maintenance Engine; fully remote cloud-based IoT Digital Twin via MQTT.	Successfully bridges the gap between open-vocabulary feature extraction, rule-based mathematical precision, and low-latency mechanical sorting under industrial constraints.
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CHAPTER 3: SYSTEM REQUIREMENTS ANALYSIS & TECHNICAL SPECIFICATIONS

3.1 Chapter Introduction

This chapter constitutes the core foundation of the engineering documentation for the automated smart inspection system (Development of an Intelligent System for Inspection and Sorting Using Computer Vision and Predictive Maintenance). Constructing an integrated mechatronic system that seamlessly merges computer vision with real-time physical control requires an exhaustive, detailed study and precise calibration of all hardware components. This study transcends a superficial listing of parts; it deeply analyzes their electromechanical behavior, signal compatibility, electrical load distribution, and mechanical transmission and torque calculations.

This chapter provides an anatomical breakdown of every single component utilized in building the conveyor line, the illumination system, the smart vision unit, and the control and actuation devices, fully augmented with mathematical equations and structural diagrams to guarantee the highest standards of industrial stability and efficiency.

3.2 Functional Requirements (FR)

Functional requirements define the precise operations, computational responses, and physical tasks that the system must execute to achieve automated defect classification and sorting:

- **Real-Time Image Acquisition:** The system must interface continuously with a localized high-speed camera to capture frames of transparent plastic bottles moving along the conveyor bed at a constant frame rate.
- **Multi-Class Spatial Object Detection:** The software subsystem must deploy an open-vocabulary object detection network to segment each bottle frame into four distinct operational bounding matrices: Bottle, Cap, Label, and Liquid Level.

- **Heuristic Defect Classification:** Inside the segmented matrices, specialized computer vision routines must evaluate explicit quality benchmarks, identifying underfilled/overfilled liquid thresholds, blank labels via structural contrast evaluation, label misalignment angles, and cap omissions.
- **Deterministic Sorting Communication:** Upon calculating a defect classification, the host computer must parse the final logic state into a single-byte serial character (N for Accept, A for Reprocess, B for Defected/Reject) and transmit it immediately via a physical UART interface to the secondary microcontroller layer.
- **Dynamic Mechatronic Actuation:** The secondary microcontroller must translate incoming serial codes into exact spatial angular adjustments of dual servo-controlled diverter gates (DS3218MG) to mechanically segment the moving bottles into separate structural lanes.
- **Expert Root-Cause Diagnostics:** The system must run a logic-driven expert algorithm that processes the calculated defect array and correlates recurring faults with specialized operational factory zones (e.g., Blow Molding, Filling, Labeling, or Capping stations).

3.3 Non-Functional Requirements (NFR)

Non-Functional requirements govern the operational performance, structural constraints, reliability parameters, and stability metrics under which the quality assurance architecture operates:

- **Temporal Determinism & Frame Rates:** The software execution pipeline must sustain an inference speed of approximately 30 Frames Per Second (FPS) at a native resolution of 640 x 640 pixels. This performance parameter ensures zero frame drops under standard industrial throughput constraints.

- **High-Accuracy Classification Benchmarks:** To meet stringent industrial compliance frameworks, the system must maintain verified statistical metrics: Precision (~97.5%), Recall (~99.8%), and a Mean Average Precision at 0.5 intersection-over-union (mAP50 ~99.5%).
- **Serial Transmission Latency:** The physical communication interface must utilize a high-bandwidth baud rate of 115,200 baud to minimize serial propagation delays, ensuring the downstream sorting arm receives actuation angles before the moving bottle exits the physical sorting gate zone.
- **System Modularity & Maintainability:** The software architecture must decoupled into high-cohesion, low-coupling modules (Inference Engine, UI Dashboard, Serial Driver, Root-Cause Processor), enabling isolated sub-component calibration and code updating without structural compilation failure.

3.4 System Block Diagram

Before delving into the technical minutiae of each component, it is vital to grasp the hierarchical architecture that binds these subsystems together. Mechatronically, the system is partitioned into four primary domains operating in absolute harmony:

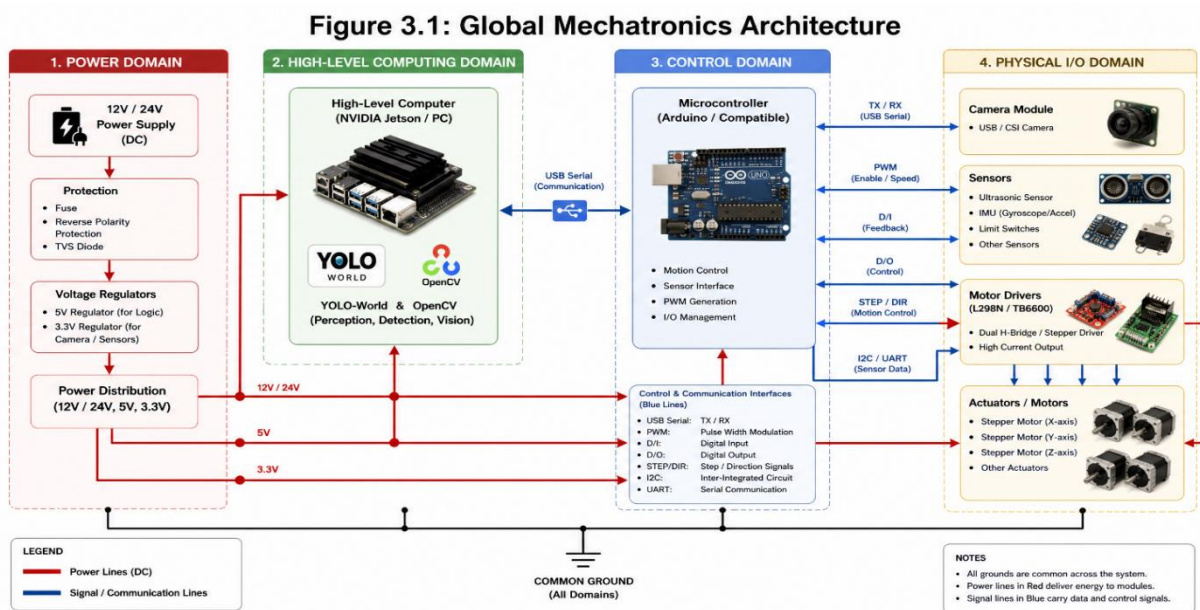


Figure 5.3.(a): Global Mechatronics Architecture and Hierarchical System Block Diagram

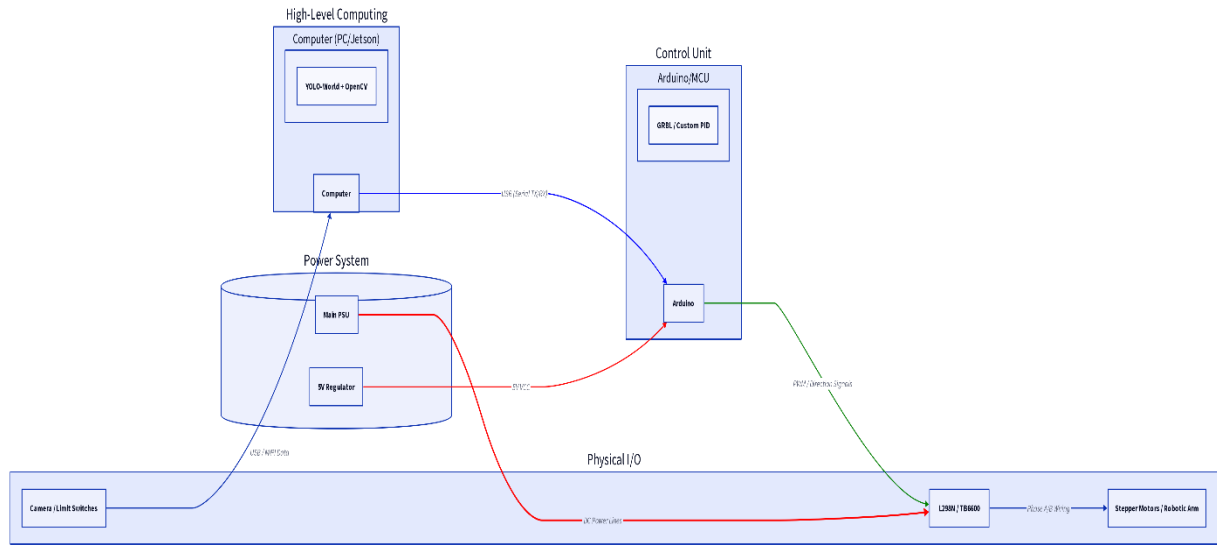


Figure 6.3.(b): Global Mechatronics Architecture and Hierarchical System Block Diagram

- **Power Management Domain:** Responsible for receiving high-voltage alternating current (AC), converting it, and regulating it into multiple direct current (DC) voltage levels (24 V, 20 V, 6 V, 5 V) tailored to each component to prevent electrical breakdowns.
- **High-Level Computing Domain:** Represented by the central computer, which handles heavy computational processing, executes deep neural network models, and receives live visual data streams.
- **Low-Level Control Domain:** Represented by the intelligent development board, acting as a real-time communication bridge between the computer software and the hardware execution units.
- **Physical I/O Domain:** Comprises the photoelectric proximity sensors, the primary drive motor, and the mechanical sorting and steering servo actuators.

3.5 Anatomical Technical Specifications of Components

3.5.1 Component 1: Toshiba DGM-339-2A DC Gear Motor

A. Engineering Titled Description and Industrial Environment

The integrated DC gear motor, manufactured by Toshiba Corporation (Japan), serves as the mechanical heart responsible for driving and regulating the conveyor belt system. This specific motor was selected in strict compliance with rigorous Japanese industrial quality standards. It stands out due to its superior capacity for continuous operation (Duty Cycle: Continuous) without experiencing excessive thermal escalation.

TOSHIBA CORPORATION			
DC MOTOR	DGM-339-2A	DUTY CONT.	
DC 20 V		189 r.p.m.	
DCG-5216-016	74160	MADE IN JAPAN	



Figure 7.3: Schematic representation of the engraved nameplate on the Japanese Toshiba motor.

B. Technical Analysis of Standard Specifications

Extracted and analyzed from the engraved nameplate data, the motor's operational parameters are defined as follows:

- **Registered Model: DGM-339-2A.**
- **Mechanical Duty Cycle: DUTY CONT.** (Continuous Duty). This implies that the motor's internal windings are designed to achieve complete thermal equilibrium during full-load operation without suffering insulation breakdown.
- **Nominal Voltage: 20 V DC.** This voltage is considered a non-standard value in conventional control electronics, which necessitated the custom design of a dedicated power regulation circuit to deliver this exact potential and combat voltage drops during high-torque startups.
- **Output Angular Velocity: 189 RPM** (Revolutions Per Minute). This represents the final speed emerging from the output shaft of the integrated gearbox. This speed range provides a flawless equilibrium between pulling torque and the linear velocity of bottles passing in front of the camera lens.

[Mechanical Design Note]

The integrated gearbox on the Toshiba motor serves as a speed reducer and torque multiplier. The angular velocity w , torque $\mathbf{T}$, and mechanical power $\mathbf{P}$ are governed by the fundamental relationship:

$$P = T \cdot w$$

Reducing the high internal rotor speed down to 189 RPM dramatically amplifies the final output torque, thereby preventing conveyor stalling or abrupt interruptions when heavily loaded with industrial objects.

3.5.2 Component 2: E18-D80NK Infrared Photoelectric Proximity Sensor Switch

A. Physical Concept and Operational Mechanism

The E18-D80NK sensor acts as the primary frontline line of defense for physical real-world object detection. It is a photoelectric proximity sensor operating on the principle of emitting and receiving invisible modulated infrared (IR) beams to detect the presence of obstacles or objects without any mechanical contact.

Figure 3.3: E18-D80NK Sensor Optical Working Principle

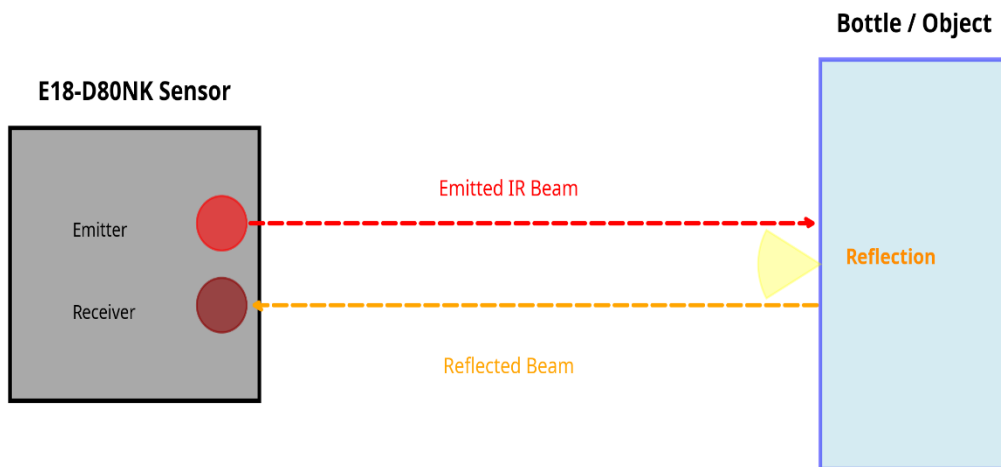


Figure 8.3.(a): E18-D80NK Sensor Optical Working Principle and Direct Retro-reflective Path

Figure 3.3: Sensor Optical Working Principle

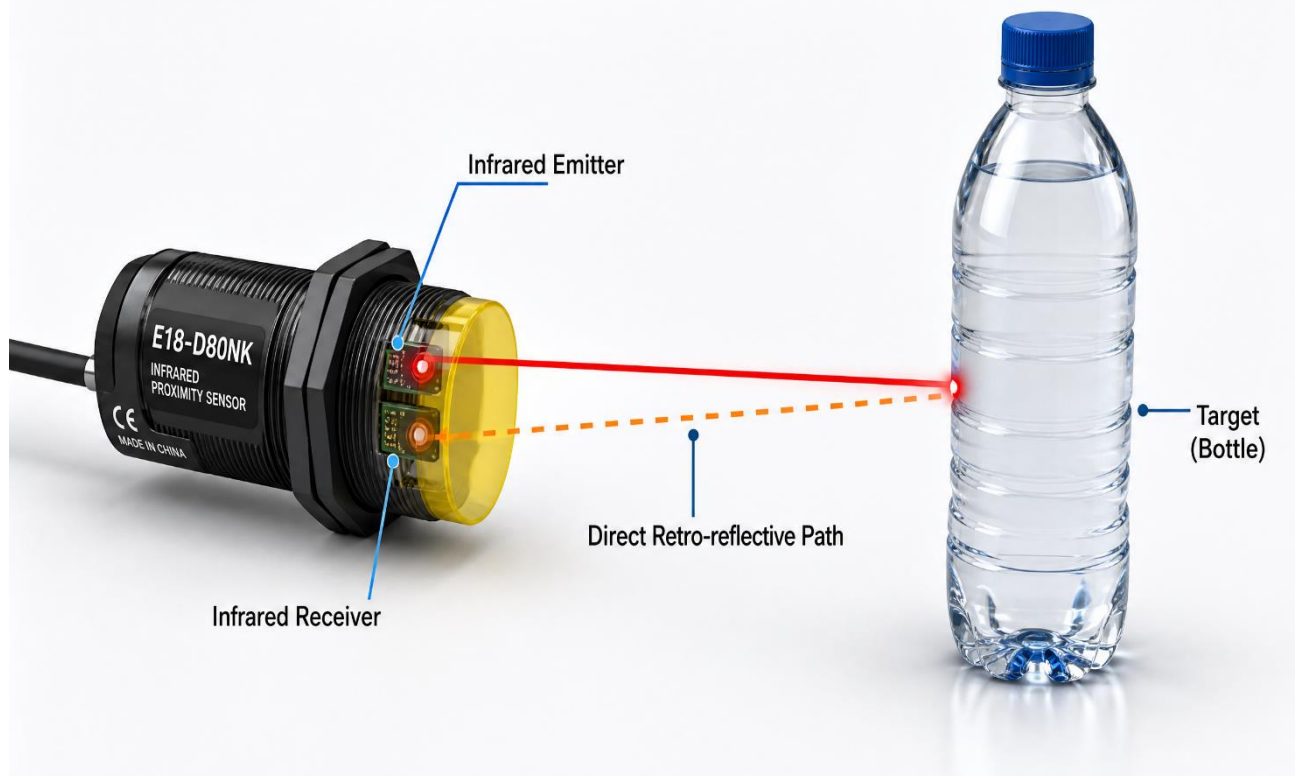


Figure 9.3.(b): E18-D80NK Sensor Optical Working Principle and Direct Retro-reflective Path

B. Fine Technical Specifications

- **Global Model Designation: E18-D80NK.**
- **Dynamic Detection Range:** Fully adjustable from a minimum of 3 cm up to a maximum of 80 cm via a highly sensitive multi-turn potentiometer located at the rear of the sensor shell.
- **Standard Operating Voltage:** 5 V DC, making the sensor fully compatible with the Arduino hardware and TTL logic levels.
- **Digital Output Logic Type:** NPN NO (Normally Open). Electronically, this architecture denotes:

- **Idle State:** The internal output switch remains open; the signal line is pulled to a logic HIGH state.
- **Active State:** The internal transistor closes, pulling the digital signal line directly to the common ground, resulting in a logic LOW state (Digital LOW / 0V).

[Signal Integrity Warning] When interfacing the sensor using the standard color code (Brown: +5V, Blue: GND, Black: Signal), the microcontroller's internal pull-up resistor must be programmatically enabled on the digital input pin. This configuration shields the signal line from electromagnetic interference (EMI) inside the lab.

3.5.3 Component 3: Arduino Mega 2560 Development Board (CH340G Serial Interface)

A. Hardware Architecture and Engineering Selection Justifications

The Arduino Mega 2560 serves as the low-level execution kernel (Low-Level Controller) of the system. This board is built around the ATmega2560 microcontroller chip, an 8-bit RISC architecture.

Figure 3.4: Hardware UART serial port allocation mapping inside the ATmega2560 chip

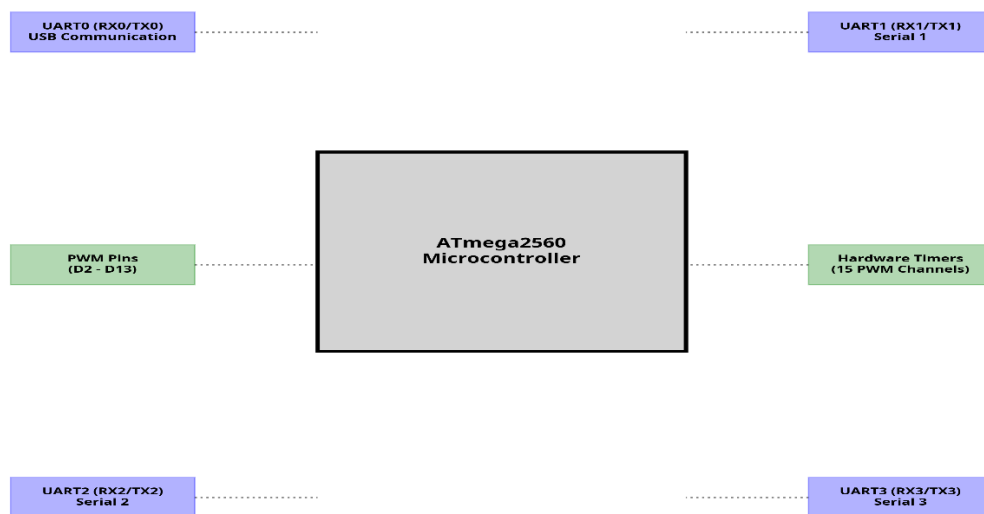


Figure 10.3.(a): Arduino Mega 2560 Hardware Architecture & Serial UART Pin Mapping

Figure 3.4: Hardware UART Serial Port Allocation Mapping

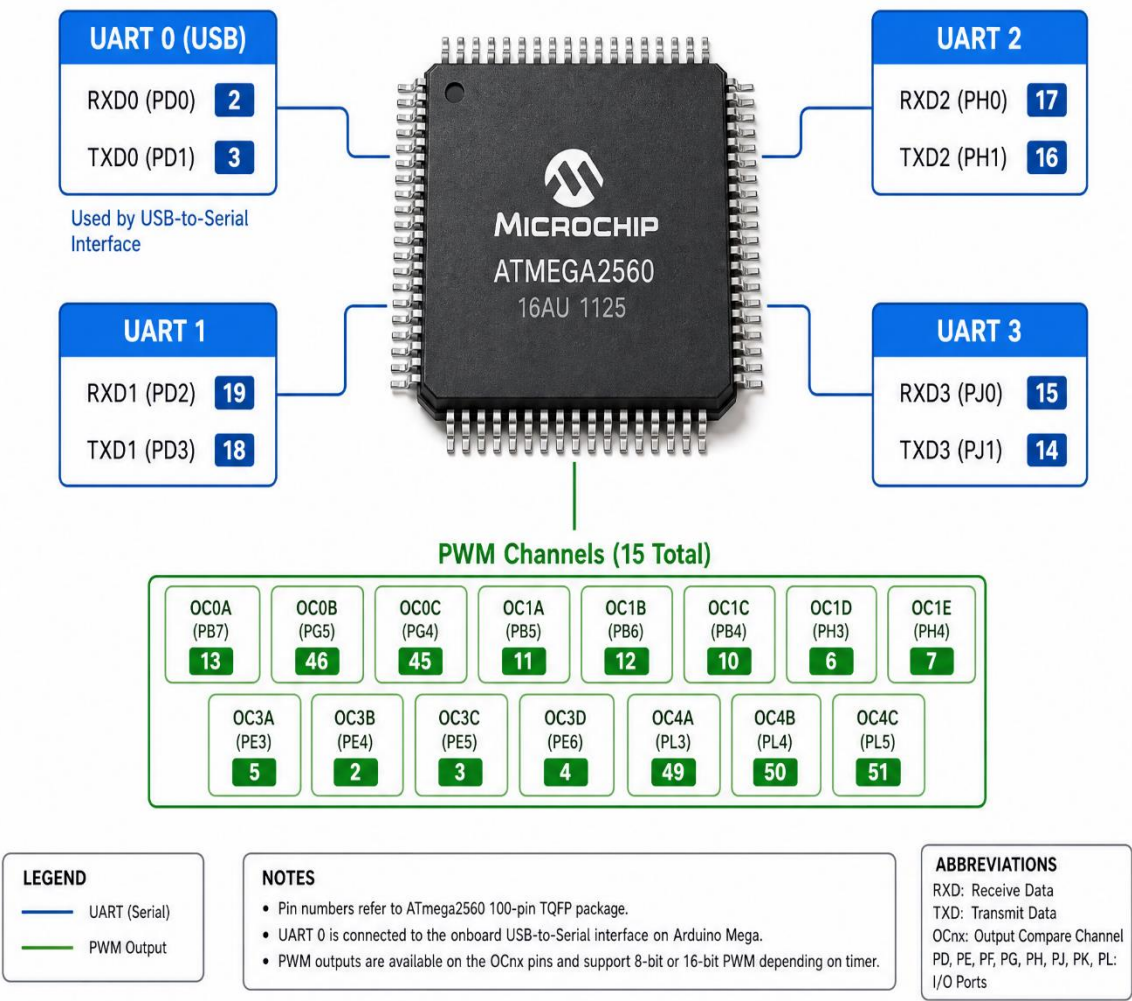


Figure 11.3.(b): Arduino Mega 2560 Hardware Architecture & Serial UART Pin Mapping

B. Microscopic Technical Performance Parameters

- **Core Microcontroller:** ATmega2560 operating at a fixed clock speed of 16 MHz.
- **USB-to-Serial Interface Driver:** Embedded with the CH340G chip, balancing economic optimization with industrial reliability in translating TTL serial streams into universal USB protocols.
- **Hardware PWM-Capable Channels:** 15 pins, leveraged to generate hardware-timed square waves essential for smooth angular position control of the high-torque digital rejection servo motors.

[Academic Justification for Review Board]

The selection of the Mega 2560 over the standard Uno was strictly dictated by the abundance of hardware UART ports (4 independent channels). Channel 0 (Serial 0) is exclusively dedicated to high-speed communication with the master computer (Python script) at a stable baud rate of 115200 bps, eliminating data collisions and data choke points.

3.5.4 Component 4: Solderless Full-Size Electronic Breadboard

A. Structural Anatomy and Electromechanical Role

The full-size solderless breadboard serves as the electromechanical backbone for unified physical prototyping and centralized signal routing during the experimental phase of the system.

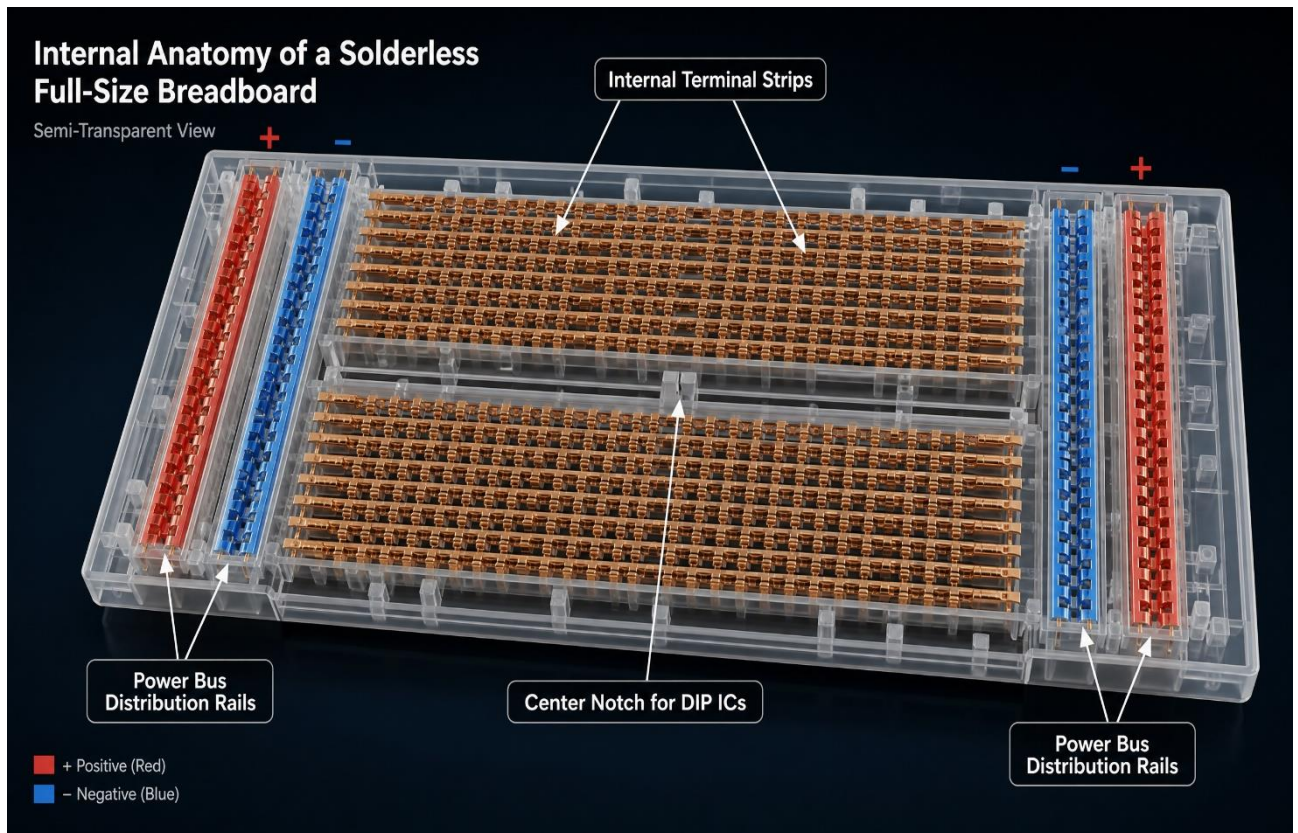


Figure 12.3.(a): Internal Metallic Structure and Power Bus Rails of Full-Size Breadboard

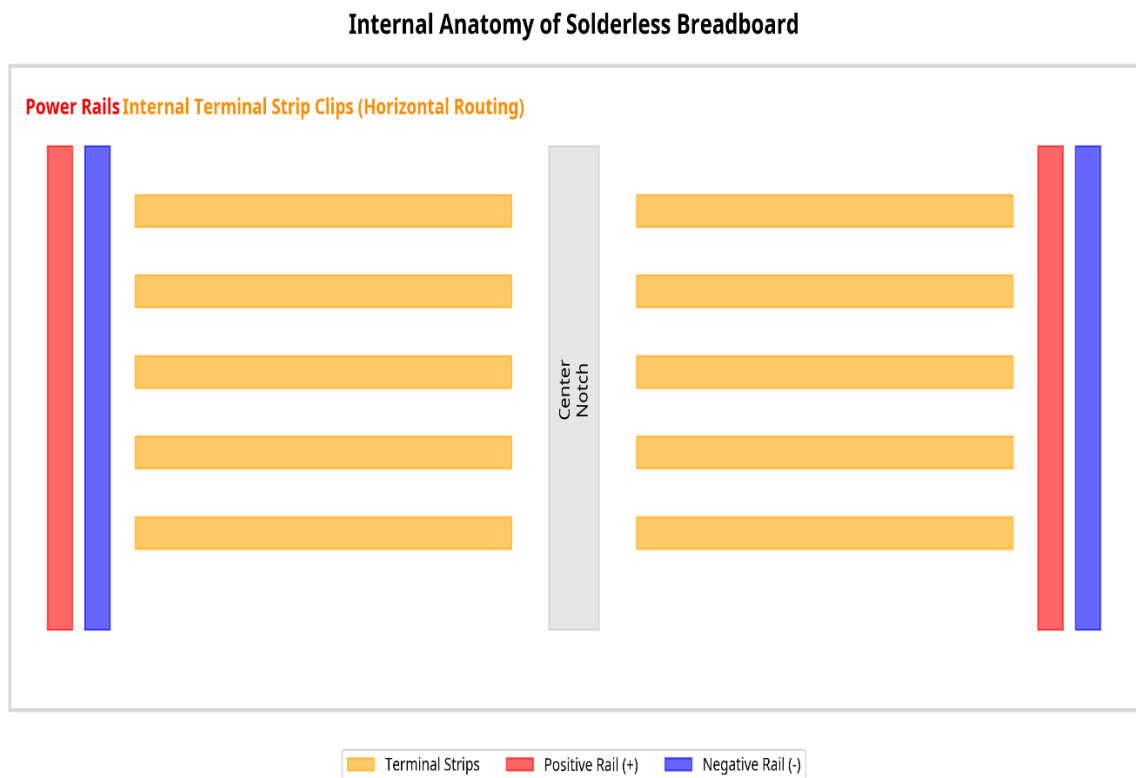


Figure 13.3.(b): Internal Metallic Structure and Power Bus Rails of Full-Size Breadboard

[Engineering Note: Ground Loop Prevention] Vertical power rails are structurally combined to form a centralized "Common Bus Configuration," tying the grounds of the Arduino, the infrared sensor, and the buck regulator together. This total ground unification eliminates ground loops, which are the leading cause of signal distortion and false triggers in digital sensor lines.

3.5.5 Component 5: XL4016 High-Power DC-DC Buck Converter Module (300W Capacity)

A. Power Transformation and Efficiency Dynamics

The XL4016 module operates as the intelligent power management node and highly efficient switching regulator within the machine's electrical framework, operating at a fixed internal clock speed of 180 kHz to realize phenomenal conversion efficiencies reaching up to 96%.

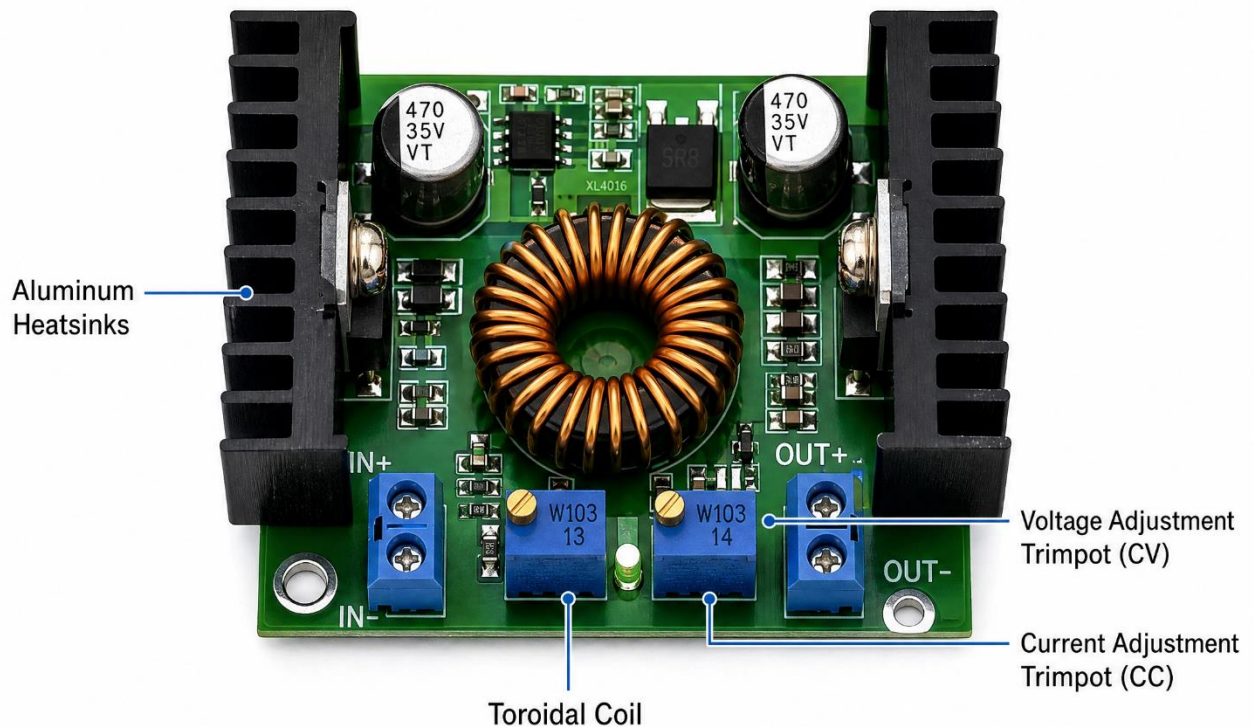


Figure 3.6: XL4016 Buck Converter Component Layout and Trimpot Calibration Locations

Figure 14.3: XL4016 Buck Converter Component Layout and Trimpot Calibration Locations

B. Absolute Technical Specifications

- **Input Voltage Spectrum:** 5 V to 40 V DC (Fed by a stable 24 V line from the primary power supply).
- **Adjustable Output Voltage Mappings:** 1.2 V to 35 V DC fine-tuned via miniature brass adjustment screws. The output was precisely calibrated to generate an unyielding 20 V rail for the conveyor motor and a separate 6 V line for the high-torque digital servo (CC/CV Mode).
- **Maximum Output Power Threshold:** 300 W (up to 9 A continuous current).

3.5.6 Component 6: Industrial Switching Mode Power Supply (SMPS Model S-250-24)

A. Power Distribution Architecture and Grid Interfacing

The perforated metallic-caged industrial power supply, model S-250-24, represents the ultimate energetic powerhouse of the entire automation system, converting AC grid lines (220 V AC) into stable low-voltage, high-current DC.

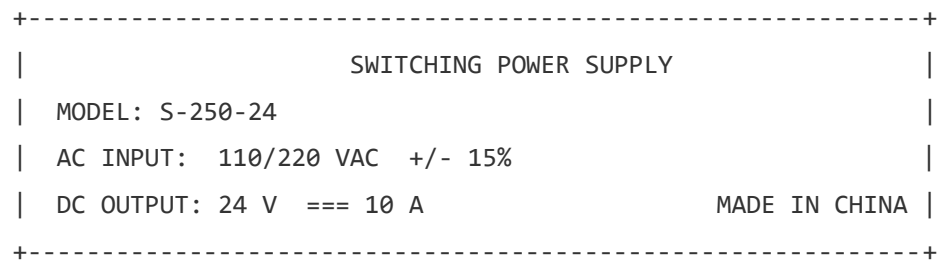


Figure (3.5.6): Exact text and rating imprint profile of the primary industrial SMPS S-250-24.



Figure 15.3.: Exact text and rating imprint profile of the primary industrial SMPS S-250-24.

$$\mathbf{P_{electrical} = V * I = 24\ V * 10\ A = 240\ W}$$

This heavy capacity is perfectly scaled to absorb the high inrush starting currents of the inductive motor coils without triggering voltage sags that would otherwise cause spontaneous microcontroller resets on the Arduino board.

3.5.7 Component 7: 10A DC Motor PWM Speed Controller Module

A. Operational Philosophy and Power Modulation (PWM) Dynamics

This unit stands as an autonomous power modulation board utilizing Pulse Width Modulation (PWM) principles. It solves the issue of torque degradation at creeping rotational speeds by feeding the motor full-voltage pulses (24 V) at an ultra-fast, fixed frequency, while modulating the ratio of ON-time to OFF-time via a linear rotary potentiometer.

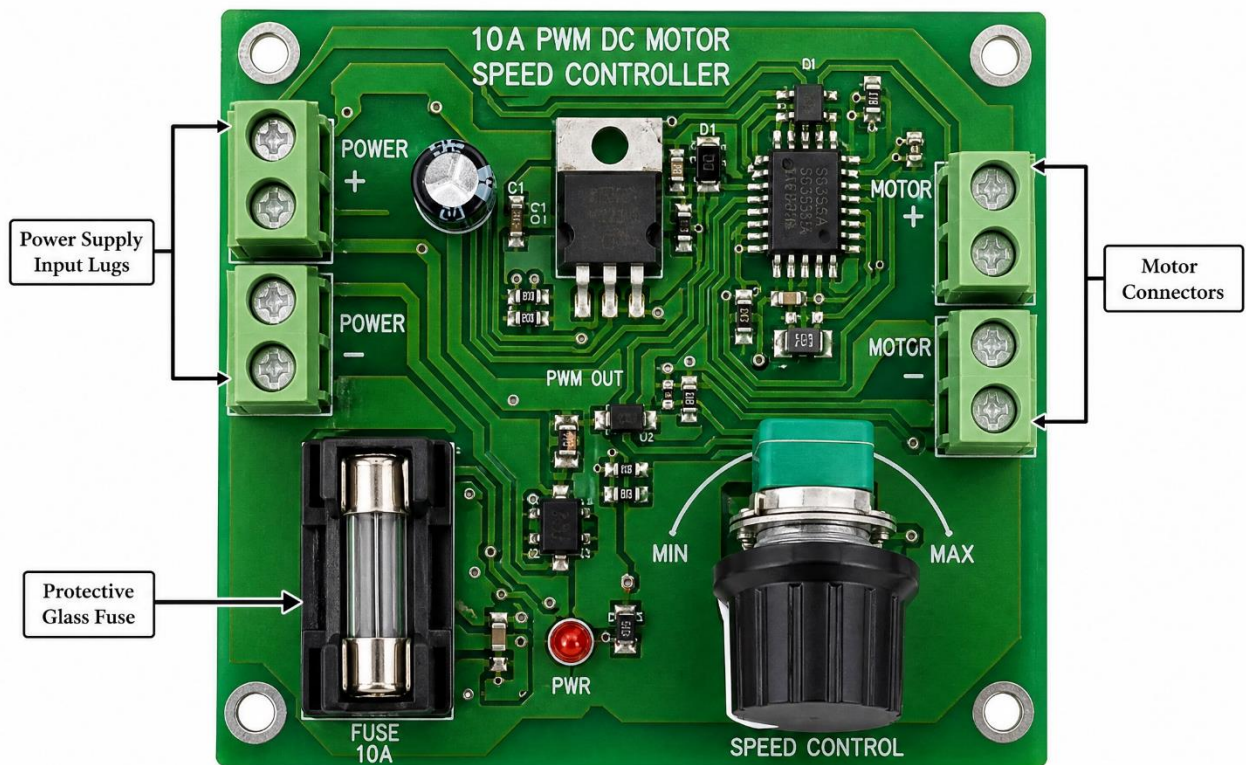


Figure 3.8: 10A PWM Speed Controller Board Layout and Glass Fuse Location

Figure 16.3.(a): 10A PWM Speed Controller Board Layout and Glass Fuse Location

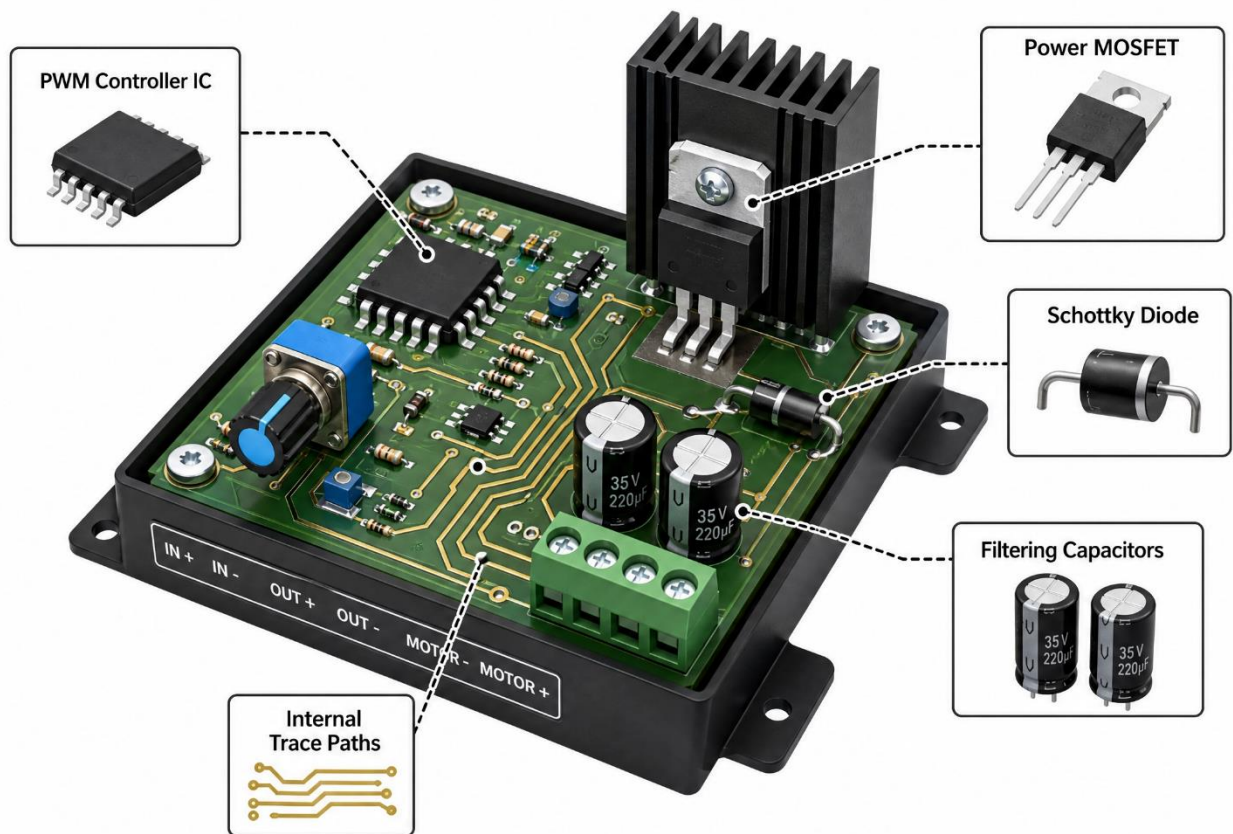


Figure 3.9: Internal Architecture and Electronic Components of the PWM Controller

Figure 17.3.(b): 10A PWM Speed Controller Board Layout and Glass Fuse Location

[Current Architectural Status (Alternative Configuration)] Crucial Academic Distinction: This hardware module is fully procured, tested, and physically present; however, it has been intentionally bypassed in the current closed-loop software architecture. It is formally presented as an "Alternative Speed Regulation Unit," preserving the flexibility to decouple velocity regulation from the Arduino's main loops to lock the belt speed to perfectly synchronize with the camera's frame rate.

3.5.8 Component 8: DSSERVO DS3218MG Digital High-Torque Metal-Geared Servo Motor

A. Architectural Evolution and Mechanical Trade-Off Analysis

This digital actuator stands as the mechanical muscle tasked with immediate product sorting and rejection.

DSSERVO DS3218MG DIGITAL SERVO		
[Metal Horn] ==> Mechanical Sorting Arm		
Torque: 20 kg.cm @ 6.8V		
Gears: Full Metal (Copper & Aluminum)		
Wires:	[Brown: GND]	[Red: VCC] [Orange: PWM]

Figure (3.5.8): Internal structural layout and pinouts of the DS3218MG high-torque servo.

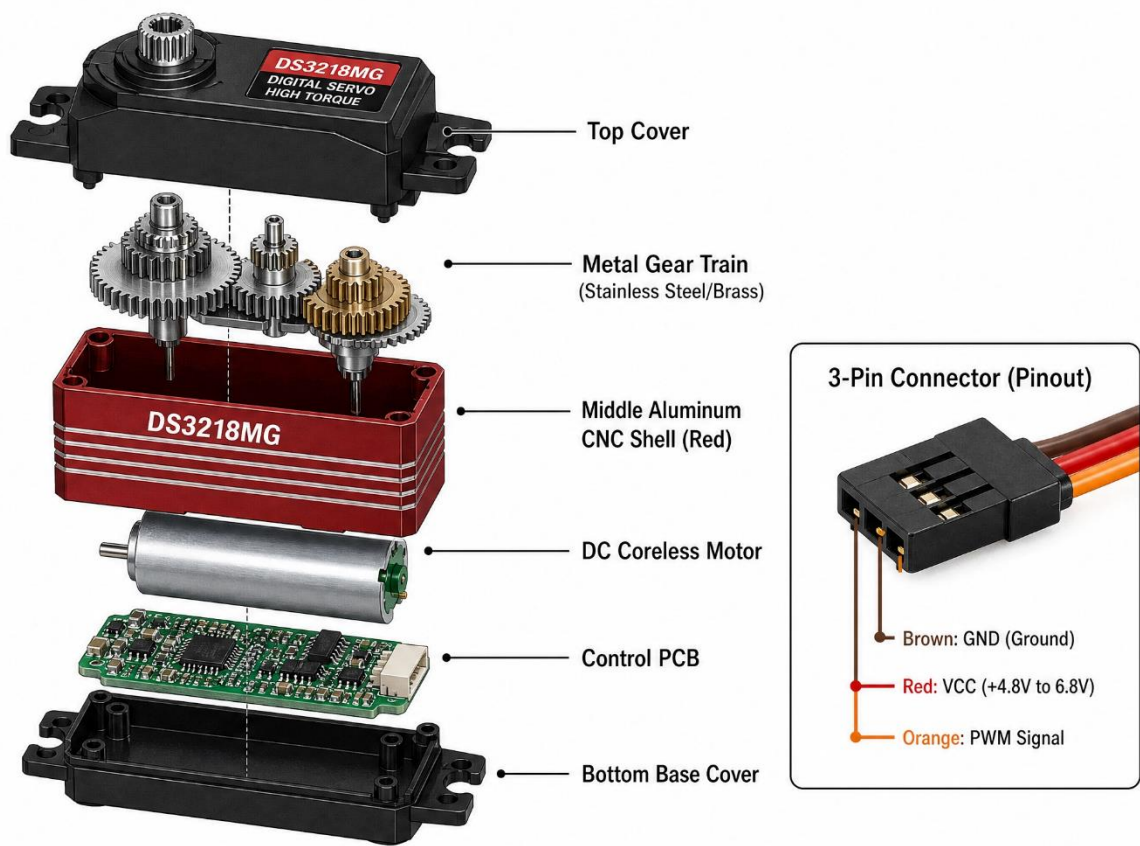


Figure 3.10: Internal Structural Layout and Pinouts of the DS3218MG High-Torque Servo

Figure 18.3: Internal structural layout and pinouts of the DS3218MG high-torque servo

- **Stall Torque Peak:** Commandingly scales up to 20 kg*cm when energized by a 6.8 V power rail, featuring full metal gears precision-machined from hardened copper and aluminum alloys.

[Rejection Justification for SG90 Servo] Early baseline project drafts suggested utilizing standard low-cost blue miniature servos (SG90 series); however, that configuration was firmly rejected. The SG90 possesses fragile plastic gear teeth and an extremely low torque profile (1.8 kg*cm), rendering it prone to instant shear fracture or gear stripping under the repetitive impact force of industrial bottles filled with high-inertia liquids.

3.5.9 Component 9: Industrial Box HD UVC Digital Camera (5-50mm Varifocal CS-Mount Lens)

A. Optical Philosophy and Machine Vision Engineering

This industrial box camera serves as the central visual organ and the paramount subsystem of the (Development of an Intelligent System for Inspection and Sorting Using Computer Vision and Predictive Maintenance) platform, paired with an entirely manual, mechanically locked varifocal optical assembly to prevent motion blur.



Figure 19.3.(a): Industrial Box Camera with 5-50mm Varifocal Lens Mechanical Structure



Figure 3.5.9: Industrial Box Camera with 5-50mm Varifocal Lens Mechanical Structure

Figure 20.3: Industrial Box Camera with 5-50mm Varifocal Lens Mechanical Structure

B. Optical Optimization via the 5-50mm Varifocal CS-Mount Lens

- **Resolution and Speed:** 4 Megapixels (4MP) operating at a stable, fixed frame rate of 30 FPS via the universal UVC standard on a USB 2.0 interface.
- **Field of View (FOV) Regulation:** The manual zoom ring allows the engineer to optically zoom in on the exact "Region of Interest" (ROI)—the bottle neck and cap zone—while completely masking out laboratory background clutter, significantly boosting YOLO's classification confidence metrics.

[Optical Design Note: Sharpness Locking Mechanism] The lens barrel houses manual tuning rings clamped down via dedicated thumbscrews. The focal point (Focus) is calibrated manually to perfection for a given bottle line, and then locked rigid. This prevents the camera from drifting out of focus regardless of lighting shifts, establishing an absolute baseline for OpenCV thresholding algorithms to compute fluid fill lines.

3.5.10 Component 10: Mechanical Transmission System (Pulleys & Drive Belt)

A. Structural Description and Kinetic Integration

The mechatronic assembly achieves physical movement synchronization through a mechanical transmission subsystem consisting of driving and driven pulleys and a reinforced rubber belt that translates rotational torque through pure surface friction.

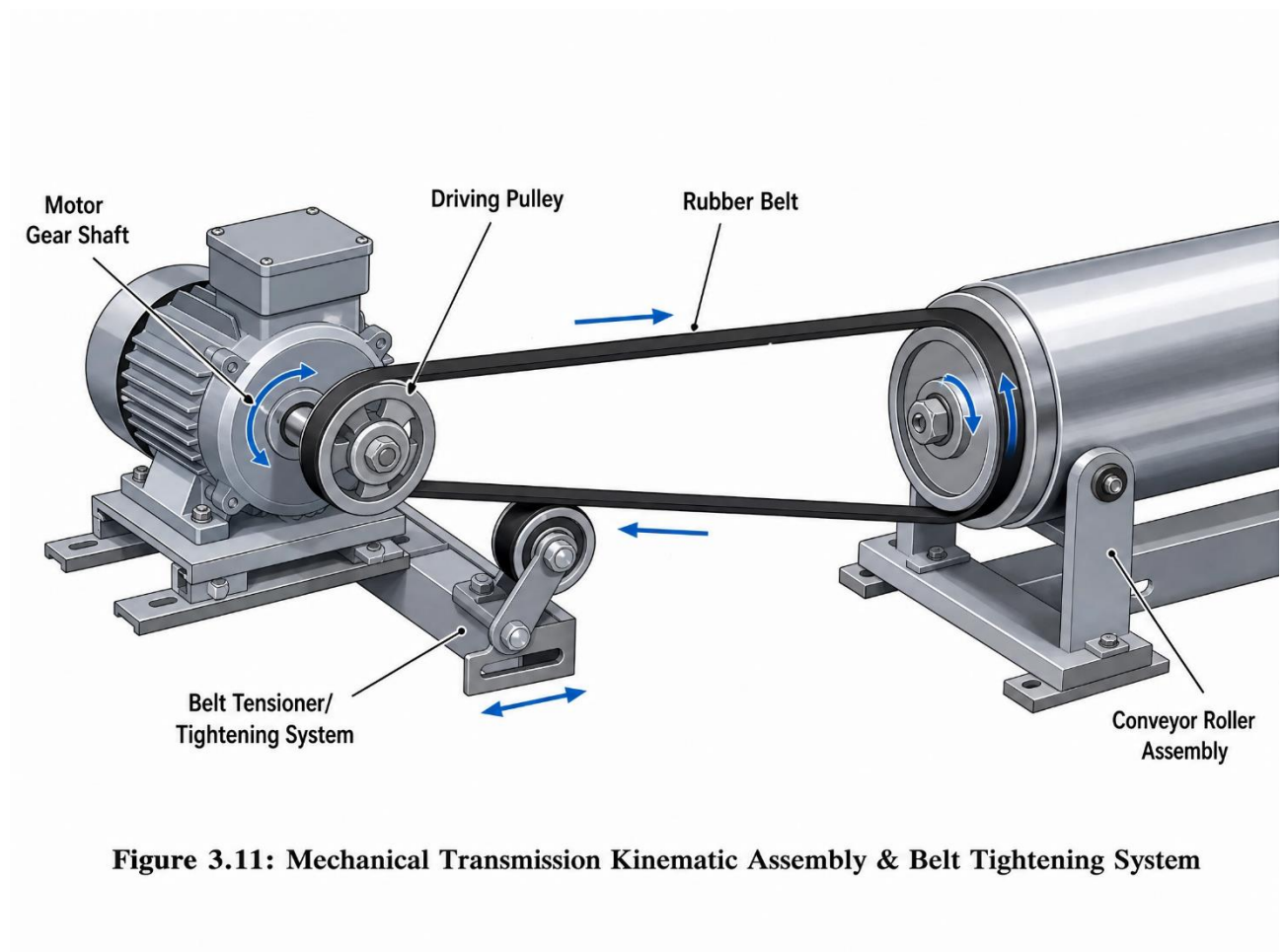


Figure 21.3: Mechanical Transmission Kinematic Assembly & Belt Tightening System

[Drive Belt Advantage as a Mechanical Fuse] Opting for a flexible belt drive transmission instead of a rigid gear engagement profile provides two profound mechanical advantages:

1. Kinetic Vibration Damping: Safely dissipates sudden physical shocks within the elastomeric matrix of the rubber belt, blocking them from transmitting back into the internal gear teeth of the Toshiba gearbox.
2. Sacrificial Slippage: Functions as a built-in "Slip Clutch" during critical mechanical overloads, preventing the motor from stalling out and pulling dangerously high current.

3.5.11 Component 11: High-Level Central Processing Unit (Master PC Station)

A. Structural Overview and Hierarchical Sovereignty

The central computer station functions as the ultimate sovereign brain of the system, applying a Hierarchical Task Distribution Architecture. Low-power microcontrollers are fundamentally incapable of executing deep convolutional neural networks; hence, all heavy computational vision loads are routed to the master PC.

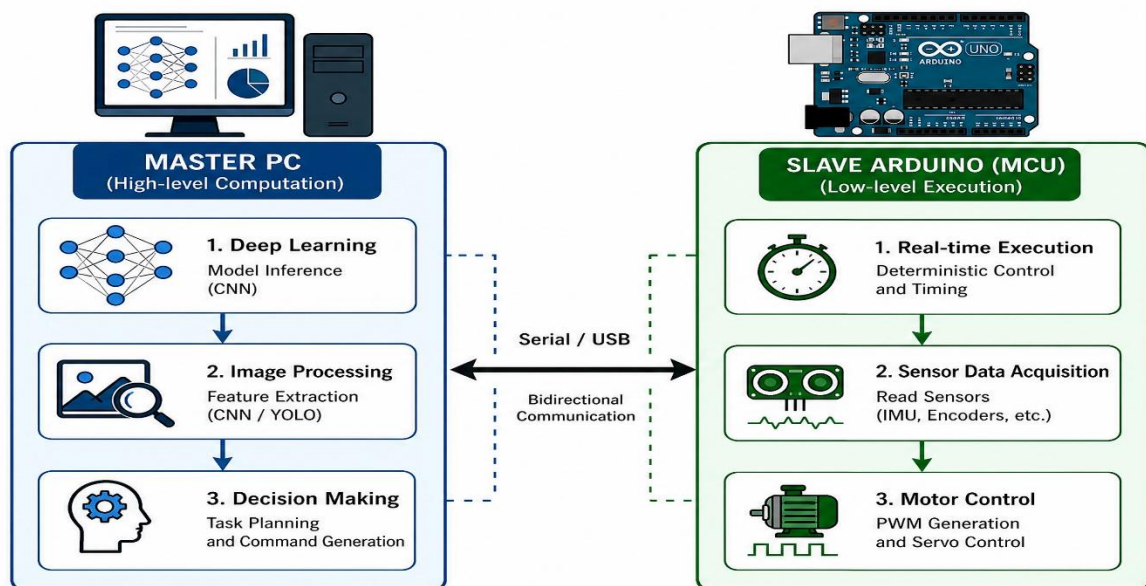


Figure 3.5.11: Hierarchical Task Distribution Architecture Between Master PC and Slave Arduino

Figure 22.3: Hierarchical Task Distribution Architecture Between Master PC and Slave Arduino

- **Master PC Subsystem Duties:** Ingests the 30 FPS video stream, runs the state-of-the-art YOLO-World and YOLOv11m deep learning models, applies OpenCV filters, and broadcasts an instantaneous serial byte packet down to the Arduino upon detecting a non-compliant bottle.
- **Slave Arduino Subsystem Duties:** Constantly listens to the real-time physical logic edges triggered by the infrared sensor via hardware interrupts, coordinates low-level timing constraints, and fires a hardware-timed PWM sequence to drive the servo rejection arm.

3.6 Communication & Interface Protocol

The interface bridging the high-level cognitive layer (Python Execution Environment on Host PC) and the low-level industrial actuation layer (Arduino Hardware) is managed via an asynchronous serial link:

- **Physical & Logical Interface:** Universal Asynchronous Receiver-Transmitter (UART) protocol mapped across a virtual COM port over a physical USB interface (e.g., COM4).
- **Baud Rate Optimization:** Operating at 115,200 bits per second, a single 8-bit character data payload is transferred across the channel in approximately 86.8 microseconds, effectively reducing communication overhead to a negligible fraction of the visual inference window.
- **Command Payload Structure:** To maximize bandwidth efficiency and eliminate parsing lag inside the microcontroller's interrupt service routine, a unified single-byte ASCII command protocol is enforced:
- **ASCII Character N:** Parsed by Arduino to maintain the sorting gate at 0 degrees (Product Accepted).
- **ASCII Character A:** Actuates Diverter Gate 1 to 45 degrees to clear the item into the labeling/filling adjustment lane (Product Needs Reprocessing).

- **ASCII Character B:** Actuates Diverter Gate 2 to 90 degrees to eject the structural unit into the scrap bin (Product Defected/Rejected).

3.7 Defect Root-Cause Analysis and Predictive Maintenance Framework

Going beyond binary product separation, this mechatronic cell introduces an analytical abstraction layer designed to monitor line health and diagnose system faults.

3.7.1 Rule-Based Expert System for Root-Cause Location

The software subsystem integrates a deterministic inference engine that maps individual defect patterns onto explicit production machinery nodes. This logic allows plant operators to trace visual anomalies back to structural physical points:

Defect Type (Dx) ---> Production Station Component (Sx)

Table 2:3 1*Defect Type (Dx) ---> Production Station Component (Sx)*

Observed Visual Defect Class	Computed Algorithmic Variable	Inferred Failure Station Root-Cause
Structural Cracks / Deformities	Contour Area Aberration ($A_c < A_{std}$)	Blow Molding Station: Thermal variance in mold or pressure drop during expansion.
Underfilled / Overfilled Meniscus	Pixel Height Threshold ($H_p < 70\%$)	Filling Station: Nozzle pressure leakage, calcification, or volumetric valve timing error.

Tilted / Rotated Label Sheets	Hough Line Vector Skew ($\Theta > \Theta_{\max}$)	Labeling Station: Roller mechanical slippage, adhesive feed variance, or belt friction.
Blank Label Stock	Image Contrast Variance ($\Sigma^2 \sim 0$)	Labeling Station / Supply: Feed roll depletion or print-head driver execution failure.
Missing Component Cap	Binary Mask Verification ($M_{\text{cap}} = 0$)	Capping Station: Mechanical hopper blockage or pneumatic torque head failure.

3.7.2 Predictive Maintenance Requirements (MTTF Framework)

To prevent sudden system failures and minimize Mean Time To Failure (MTTF), the project establishes a technical specification framework for multi-sensor hardware monitoring. While the Python core currently processes simulated data streams to calculate the unified Asset Health Score, the architectural design defines the explicit hardware sensor interfaces needed for real-world deployment:

- **Vibration Profiles:** Specified for an MPU6050 3-axis accelerometer mounted directly to the Toshiba motor bearing housing to monitor mechanical resonance and bearing wear.
- **Current Consumption Monitoring:** Industrial ACS712 Hall-Effect current sensor IC inline with the motor drive circuit to diagnose coil shorts, winding degradation, or line binding.
- **Thermal Dissipation Dynamics:** An LM35 linear analog thermistor or contact infrared sensor array to flag frictional thermal accumulation in the gearbox.

3.8 Bill of Materials (BOM) and Cost Analysis

To provide a comprehensive economic and structural overview of the Development of an Intelligent System for Inspection and Sorting Using Computer Vision and Predictive Maintenance mechatronic prototype, a detailed Bill of Materials (BOM) has been compiled. This analysis tracks all hardware procurement, electrical components, and optical units required to build the fully functional automated inspection system.

Table 3:3 Complete Bill of Materials (BOM) and Economic Cost Estimation

Item No	Component Description	Qty	Unit Price (\$)	Total Price (\$)	Purpose / Subsystem
1	Toshiba DGM-339-2 A DC Gear Motor	1			Main conveyor belt drive
2	E18-D80NK IR Photoelectric Proximity Sensor	2			Object detection sensor
3	Arduino Mega 2560 Board (CH340G)	1			Low-level execution controller
4	Solderless Full-Size Breadboard	1			Electrical prototyping / Common

Item No	Component Description	Qty	Unit Price (\$)	Total Price (\$)	Purpose / Subsystem
5	XL4016 High-Power Buck Converter (300W)	1			Voltage regulation and step-down
6	Switching Mode Power Supply (SMPS S-250-24)	1			Central DC power source (24V/10A)
7	10A DC Motor PWM Speed Controller Module	1			Manual/Alternative speed tuning
8	DSSERVO DS3218MG 20kg Digital Servo Motor	2			Rejection and sorting actuator
9	Industrial Box HD UVC Camera (4MP, 30 FPS)	1			Vision ingestion node
10	5-50mm Varifocal CS-Mount Lens	1			ROI optimization & manual focus lock

Item No	Component Description	Qty	Unit Price (\$)	Total Price (\$)	Purpose / Subsystem
11	Mechanical Transmission (Pulleys & Drive Belt)	2			Kinetic coupling and overload safety
12	Master Central Processing Unit (PC Station)	1			AI model (YOLO-World) hosting
Complete Prototype Integration					Complete Prototype Integration

Chapter 4: Software Architecture & Deep Learning Integration

4.1 Chapter Introduction

The operational efficiency of an automated mechatronic sorting system relies entirely on the architectural integrity of its underlying software ecosystem.

While Chapter 3 established the electromechanical and physical boundaries of the prototype, this chapter unveils the high-level computational framework and deep learning paradigms that govern the system's decision-making capabilities.

In modern industrial inspection, processing high-definition video streams at 30 Frames Per Second (FPS) while concurrently executing convolutional neural network inference, managing real-time hardware serial I/O, updating an interactive monitoring dashboard, and executing predictive maintenance algorithms introduces severe resource contention.

A traditional, sequential programming loop would inevitably succumb to buffer overflows, race conditions, and catastrophic execution latencies.

To circumvent these bottlenecks, the software ecosystem of the developed prototype is engineered around a deterministic, multi-threaded, and asynchronous processing pipeline. This chapter provides an exhaustive breakdown of the master-slave computing hierarchy, the concurrent multi-threading engine, and the low-level micro-firmware integration that transforms real-time computer vision into immediate physical sorting actions.

4.2 Global Software Architecture & Master-Slave Hierarchy

The computing infrastructure of the developed automation cell is structured around a rigid Master-Slave control paradigm to guarantee microsecond-level synchronization between the software logic and the physical hardware actuators.

The Master Node is represented by a high-level central processing unit running a Python 3 environment. The Master Node retains absolute sovereignty over the computationally intensive tasks, which include:

1. **Video Ingestion:** Continuous ingestion of raw digital video frames from the UVC industrial camera via an optimized Universal Video Class (UVC) driver pipeline.
2. **Deep Learning & AI Inference:** Direct execution of state-of-the-art YOLOv11 object detection layers coupled with secondary geometric and contrast-based OpenCV computer vision filters to classify product quality.
3. **Process Diagnostics:** Hosting a Rule-Based Expert System that evaluates multi-class defect patterns to perform real-time process diagnostics and compute comprehensive machinery health metrics.
4. **Command Dispatch:** Dispatching atomic, single-byte serial command characters to the physical hardware layer when a non-compliant bottle is isolated.

Conversely, the Slave Node is localized within the ATmega2560 hardware microcontroller utilizing an optimized C++ compiled firmware. The Slave Node acts as a deterministic, low-level execution kernel entirely decoupled from the heavy computer vision overhead. Its operations are strictly bound to:

1. **Interrupt Handling:** Monitoring hardware digital input pins mapped to the infrared photoelectric proximity sensors via hardware-timed external interrupts.

2. **Serial Communication Listening:** Listening to the hardware UART serial buffer for incoming command packets broadcasted by the Master PC.
3. **Spatial Tracking:** Executing ultra-precise, synchronized time-and-distance delays to track the spatial position of defective items along the conveyor belt.
4. **Actuator Control & Rejection:** Driving high-current digital pulse-width modulation (PWM) channels to actuate the DS3218MG metal-gearred servo motor, forcing its rejection horn through a precise angular arc to divert non-compliant items.

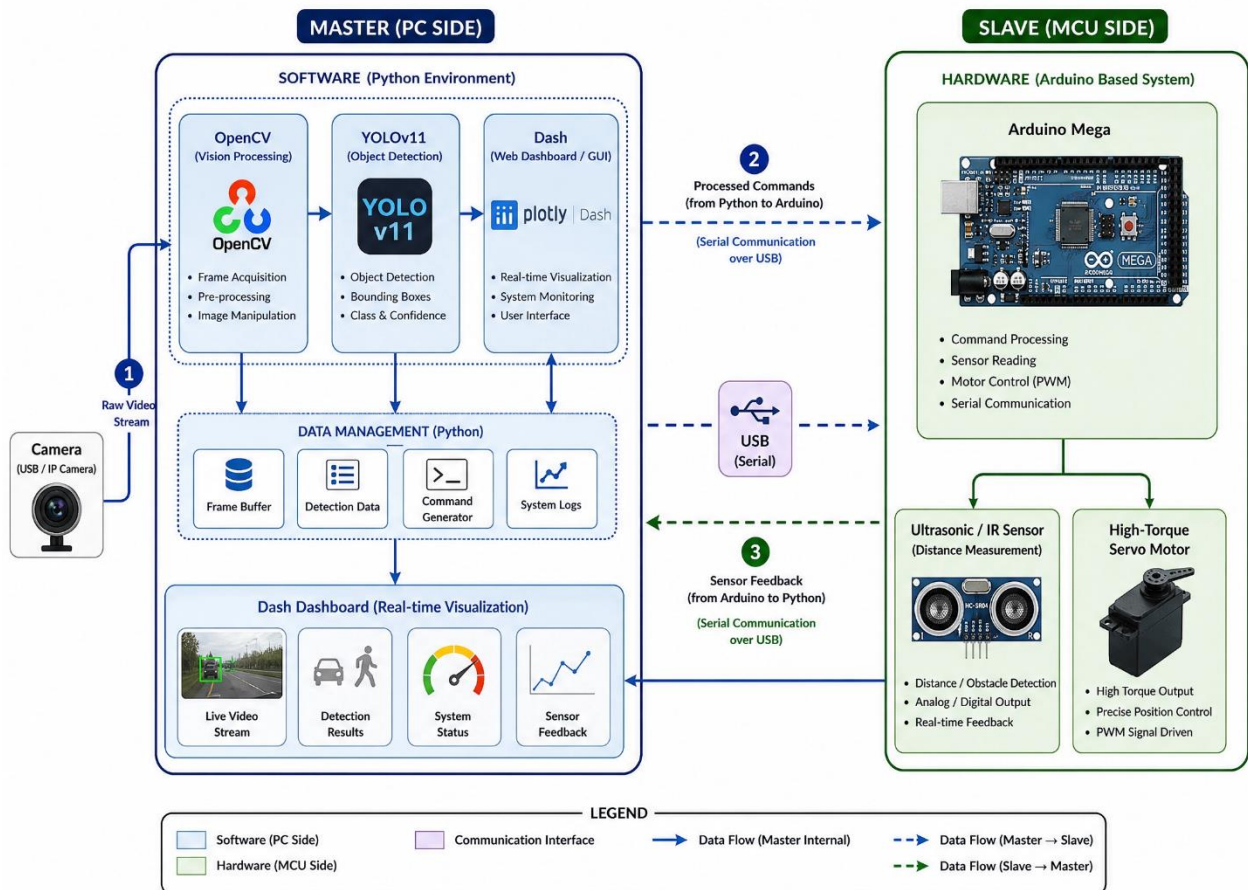


Figure 4.1: Master-Slave Global Software Architecture and Real-Time Data Flow Map

Figure 23.4: Master-Slave Global Software Architecture and Real-Time Data Flow Map

4.3 Concurrent Multi-Threaded Ingestion Engine

To process industrial machine vision workloads without suffering from frame dropping or buffer-induced lag, the Master Python application discards traditional synchronous loops in favor of an advanced, concurrent **POSIX Threading (Multi-threading)** architecture.

By invoking the **threading** library, the computational load is systematically partitioned across three autonomous, parallel **worker threads** running in a **non-blocking** configuration, coordinated via a thread-safe FIFO (First-In, First-Out) queuing system.

4.3.1 The Vision Processing Thread (`run _ vision`)

The `run _vision` worker thread is exclusively dedicated to the high-speed visual acquisition and product inspection pipeline.

It establishes a dedicated connection to the industrial box camera, continuously pulling raw frames into a volatile memory buffer.

This thread operates independently of the visualization and maintenance tasks, ensuring that the camera's frame buffer is constantly emptied.

Once a frame is captured, it isolates the Region of Interest (**ROI**), executes the **YOLOv11** inference layers to detect the sub-components (**bottles, caps, labels, liquid**), applies geometric quality checks, and immediately pushes the definitive inspection results down the serial pipeline.

This architectural separation guarantees that even if other software modules experience temporary computational overhead, the vision system continues to run at its maximum possible physical frame rate.

4.3.2 The Predictive Maintenance Thread (`run _ pm`)

Operating entirely in the background, the **`run _ pm`** worker thread acts as the continuous diagnostic monitor of the physical infrastructure. It runs an infinite loop that queries the status of the virtual and simulated hardware parameters at deterministic intervals. Rather than consuming central processing resources during crucial inspection windows, this thread runs asynchronously, accumulating run-time statistics, monitoring consecutive failure frequencies, and calculating the operational degradation of the conveyor and actuators. It processes the mathematical algorithms that derive the machinery's Health Score and Failure Probability, translating raw operational data into actionable maintenance warnings without introducing latency into the live production line.

4.3.3 The Real-Time Dashboard Thread (`run _ dash`)

The user interface and production analytics layer are governed by the **`run _ dash`** thread, which initializes and hosts a local web server based on the Dash and Plotly frameworks.

This thread decouples the heavy network protocol rendering (**HTTP requests, WebSocket data transmission, and live MJPEG video streaming**) from the core automation loops.

It queries thread-safe global data structures to retrieve real-time inspection numbers, error statistics, and machinery health updates, rendering them into highly interactive graphical interfaces and Pareto charts accessible by production engineers.

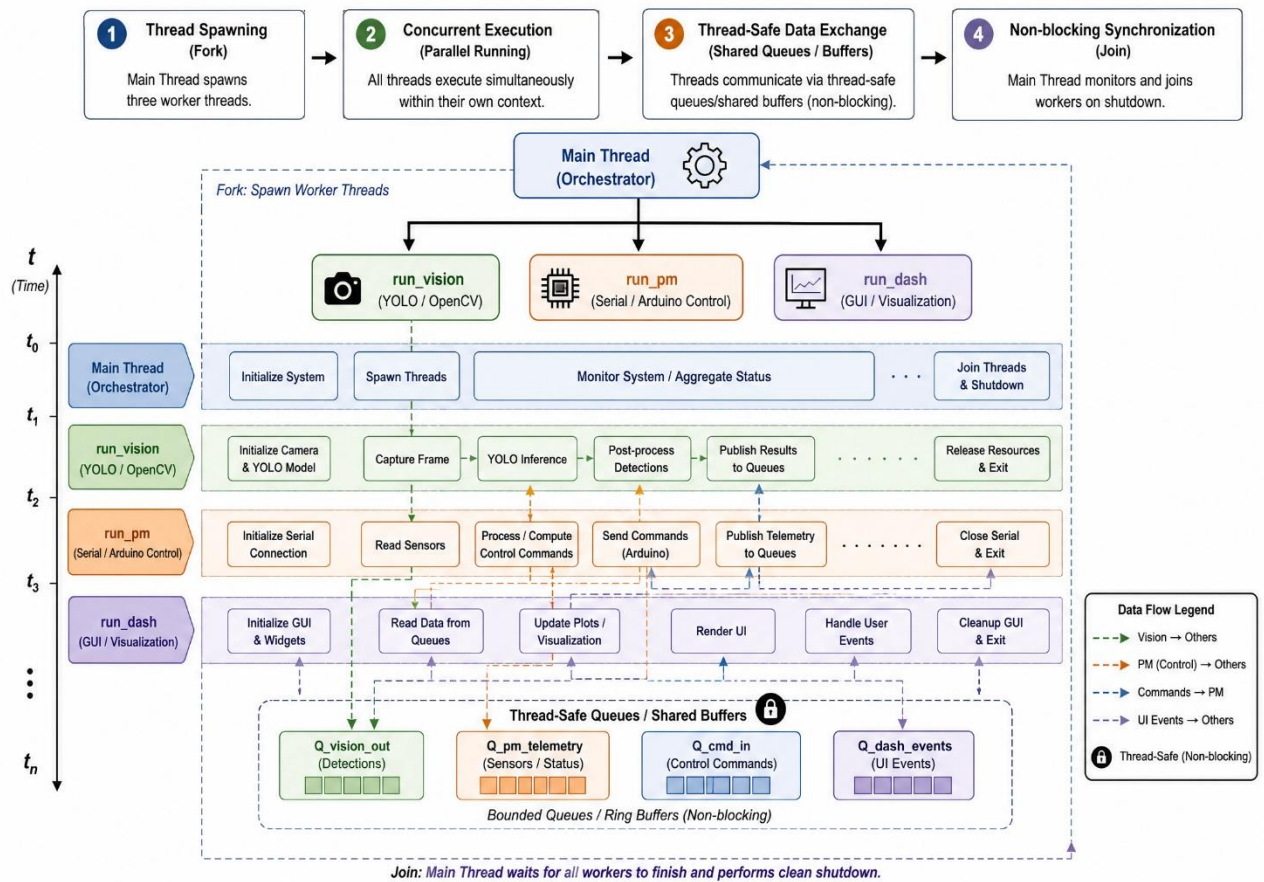


Figure 4.2: Concurrent Multi-Threading Execution Engine and Thread-Safe Data Partitioning Architecture

Figure 24.4: Concurrent Multi-Threading Execution Engine and Thread-Safe Data Partitioning Architecture.

4.4 Deep Learning Inference Pipeline & Hybrid Machine Vision Architecture

4.4.1 Industrial Object Detection and Semantic Class Segmentation via YOLOv11

The automated quality assurance and inspection cell leverages a state-of-the-art Deep Convolutional Neural Network (DCNN) built upon the ultra-optimized YOLOv11 (You Only Look Once) architecture. In high-speed, modern manufacturing environments, visual verification systems must operate under deterministic temporal constraints while maintaining near-perfect statistical recall and precision metrics. Traditional computer vision algorithms that rely entirely on hard-coded pixel

thresholds fail catastrophically when subjected to industrial variations such as minor ambient lighting fluctuations, dust accumulation on optical lenses, or physical vibrations along the conveyor structure. To overcome these limitations, the object detection core of this system is designed to execute multi-class semantic localization in real time.

The inference engine runs locally within the Python environment, executing forward propagation passes on incoming digital matrices to segment the input frame into four highly specialized bounding box classes simultaneously. Each class represents a critical inspection sub-component, detailed as follows:

1. **Class 0 (bottles):** This foundational class isolates the primary structural boundaries of the PET (Polyethylene Terephthalate) container. It establishes the global coordinate baseline (*xmin, ymin, xmax, ymax*) which serves as the anchor system for all secondary spatial calculations. By locking onto the container's geometry, the software can track individual products as they glide across the camera's field of view.
2. **Class 1 (caps):** This class is engineered to perform rigorous verification of the top closure assembly. The deep learning model evaluates the presence of the plastic cap and is optimized to recognize subtle angular variations that indicate a crooked, cross-threaded, or improperly sealed cap, which would jeopardize the hermetic seal of the product.
3. **Class 2 (labels):** This class monitors the commercial branding and regulatory labeling layout affixed to the body of the container. The neural network detects the absolute presence of the sticker and provides raw spatial coordinates to detect severe tears, folds, or total absence of the label matrix.
4. **Class 3 (liquid):** This highly specialized class isolates the fluid volume boundaries inside the transparent container. By generating localized bounding

box parameters around the liquid mass, it provides the volumetric data required to compute fluid height variations with sub-pixel resolution.

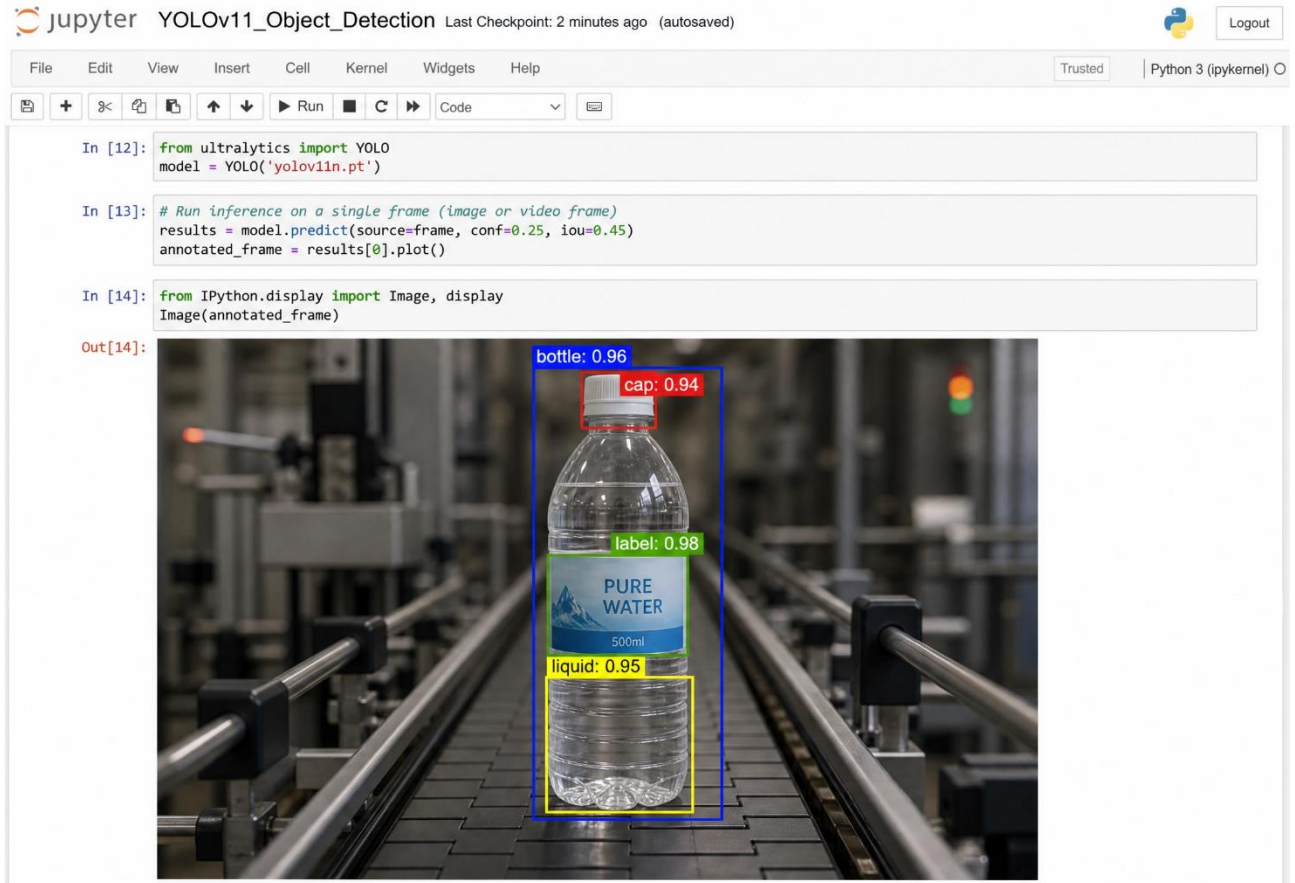


Figure 4.3: Real-Time Multi-Class Object Detection Model Performance Capturing Bottles, Caps, Labels, and Liquid Sub-components via YOLOv11 Inference

Figure 25.4: Real-Time Multi-Class Object Detection Model Performance Capturing Bottles, Caps, Labels, and Liquid Sub-components via YOLOv11 Inference.

4.4.2 Mathematical Formalization of Contextual Region of Interest (ROI) Isolation

To eliminate excessive computational overhead and maintain a consistent processing speed of 30 Frames Per Second (FPS) without thermal throttling, the software architecture incorporates a dynamic spatial clipping algorithm known as Contextual Region of Interest (ROI) Isolation. A raw video stream captured by an industrial camera typically contains a large percentage of irrelevant spatial data, such as structural conveyor guardrails, background laboratory walls, electrical wiring

conduit, and mechanical mounting brackets. Feeding these uncropped, high-resolution matrices directly into a deep learning neural network dramatically increases floating-point operations per second (FLOPS), which introduces unacceptable processing latencies and can lead to frame drops.

To decouple the inspection process from background industrial noise, the system executes an automated, multi-stage geometric reduction pipeline on every captured frame matrix. The sequence is mathematically formulated through the following deterministic operations:

1. **Raw Matrix Ingestion:** The UVC digital camera stream inputs a continuous two-dimensional pixel intensity matrix M of size $W \times H \times C$, where W represents pixel width, H represents pixel height, and C represents the RGB color channels:

$$M \in \mathbb{R}^{(W \times H \times C)}$$

2. **Bounding Box Coordinate Extraction:** The system queries the first-stage **YOLO** inference vector to extract the normalized spatial extremities of the isolated container object, yielding four critical corner coordinates:

$$V_{bottle} = \{X_{min}, Y_{min}, X_{max}, Y_{max}\}$$

3. **Dynamic Boundary Padding:** To ensure that the edges of the bottle are not clipped during mechanical vibrations or high-speed transit, an industrial safety padding threshold (Δp) is mathematically added to the coordinate vectors, creating an optimized boundary box:

$$\begin{aligned} x'_{min} &= \max(0, x_{min} - \Delta p), & y'_{min} &= \max(0, y_{min} - \Delta p) \\ x'_{max} &= \min(W, x_{max} + \Delta p), & y'_{max} &= \min(H, y_{max} + \Delta p) \end{aligned}$$

4. **Matrix Sub-Grid Cropping:** The primary frame matrix M is sliced down to an isolated sub-matrix (ROI), stripping away all background pixels outside the calculated boundaries:

$$ROI = M[y_min : y_max, x_min : x_max]$$

By passing only this dynamically scaled, isolated sub-grid into the subsequent analysis loops, the computational volume drops exponentially. This reduction guarantees real-time responsiveness and ensures zero-latency execution on standard industrial computing hardware.

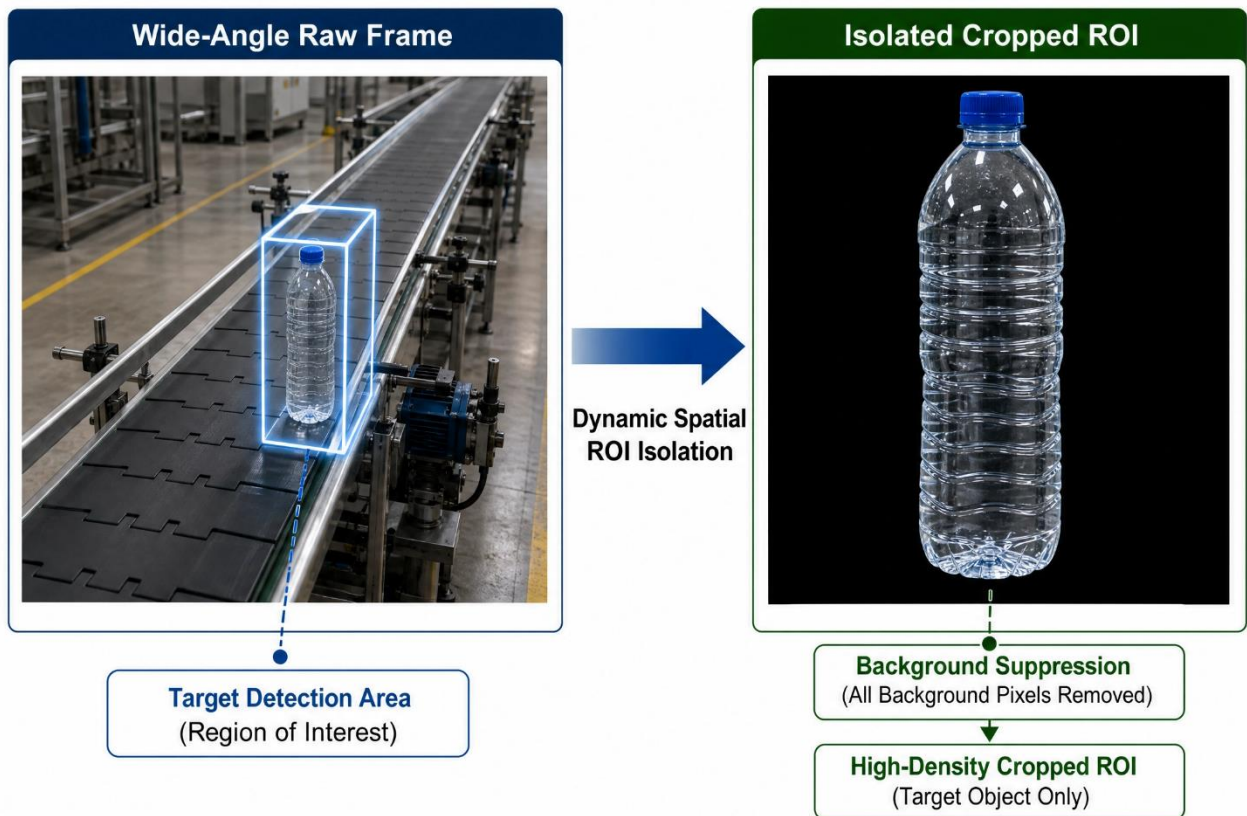


Figure 4.4: Process Flow Chart of Dynamic Spatial ROI Isolation and Background Pixel Suppression Sequence

Figure 26.4: Process Flow Chart of Dynamic Spatial ROI Isolation and Background Pixel Suppression Sequence.

4.4.3 Hybrid Vision Logic and Algorithmic Rule-Based Feature Extraction

While deep learning architectures excel at identifying complex, non-linear object patterns, relying solely on neural network bounding boxes can introduce minor pixel variances due to probabilistic outputs. To ensure absolute compliance with strict industrial quality control criteria, the developed software implements a highly dependable Hybrid Machine Vision Pipeline. In this configuration, the high-level YOLOv11 model acts as an intelligent spatial filter to isolate components, while secondary Digital Image Processing (DIP) algorithms from the OpenCV framework execute precise, rule-based geometric measurements on the isolated regions.

The hybrid engine continuously evaluates the products passing through the inspection station by executing two independent, mathematically rigorous verification routines:

A. Volumetric Fluid Level and Under-Filling Analysis

To determine the precise filling volume of the liquid without using expensive mass flow sensors, the software converts pixel-distance ratios into volumetric verification metrics. The algorithmic routine processes the fluid region through the following engineering sequence:

1. **Step 1:** The system isolates the vertical coordinate corresponding to the top edge of the liquid bounding box (y_{liquid_top}) and maps it relative to the top boundary of the container structural neck (y_{bottle_top}).
2. **Step 2:** A normalized fill-percentage ratio (R_{fill}) is continuously computed using the relative vertical pixel spans:

$$R_{fill} = [(y_{bottle_bottom} - y_{liquid_top}) / (y_{bottle_bottom} - y_{bottle_top})] \times 100\%$$

3. **Step 3:** The computed ratio is evaluated against a hard-coded industrial quality window bounded by a minimum fill threshold ($T_{min} = 90\%$) and a maximum overflow ceiling ($T_{max} = 98\%$).
4. **Step 4:** If the value violates these constraints ($R_{fill} < T_{min}$ or $R_{fill} > T_{max}$), the system immediately flags the item as non-compliant and appends a critical defect code (DEFECT_UNDERFILL) to the serial output queue.

B. Label Structural Alignment and Skew Deformation Detection

An misaligned, crooked, or bubbling label indicates poor branding quality and mechanical issues in the labeling station. To catch these defects, the OpenCV layer applies advanced edge-tracking filters to the label's region of interest:

1. **Step 1:** The sub-matrix containing the label is converted to a single-channel grayscale image and processed through a Gaussian blur filter to eliminate high-frequency pixel noise.
2. **Step 2:** The system applies a Canny Edge Detection Algorithm with adaptive hysteresis thresholds to isolate the sharp physical boundaries of the label sticker.
3. **Step 3:** A geometric evaluation function tracks the parallel distance from the left edge of the bottle structure (x_{bottle_left}) to the left edge of the label (x_{label_left}) at multiple vertical sampling points (y_1, y_2, y_3).
4. **Step 4:** The angular skew (θ_{skew}) of the label is calculated using the arctangent of the spatial delta variations:

$$\theta_{skew} = \tan^{-1} (\Delta x_{label} / \Delta y)$$

5. **Step 5:** If the calculated skew angle deviates from the horizontal baseline by more than a strict tolerance ($\pm 3^\circ$), or if the label boundaries cannot be detected, a defect flag (DEFECT_LABEL) is immediately raised.

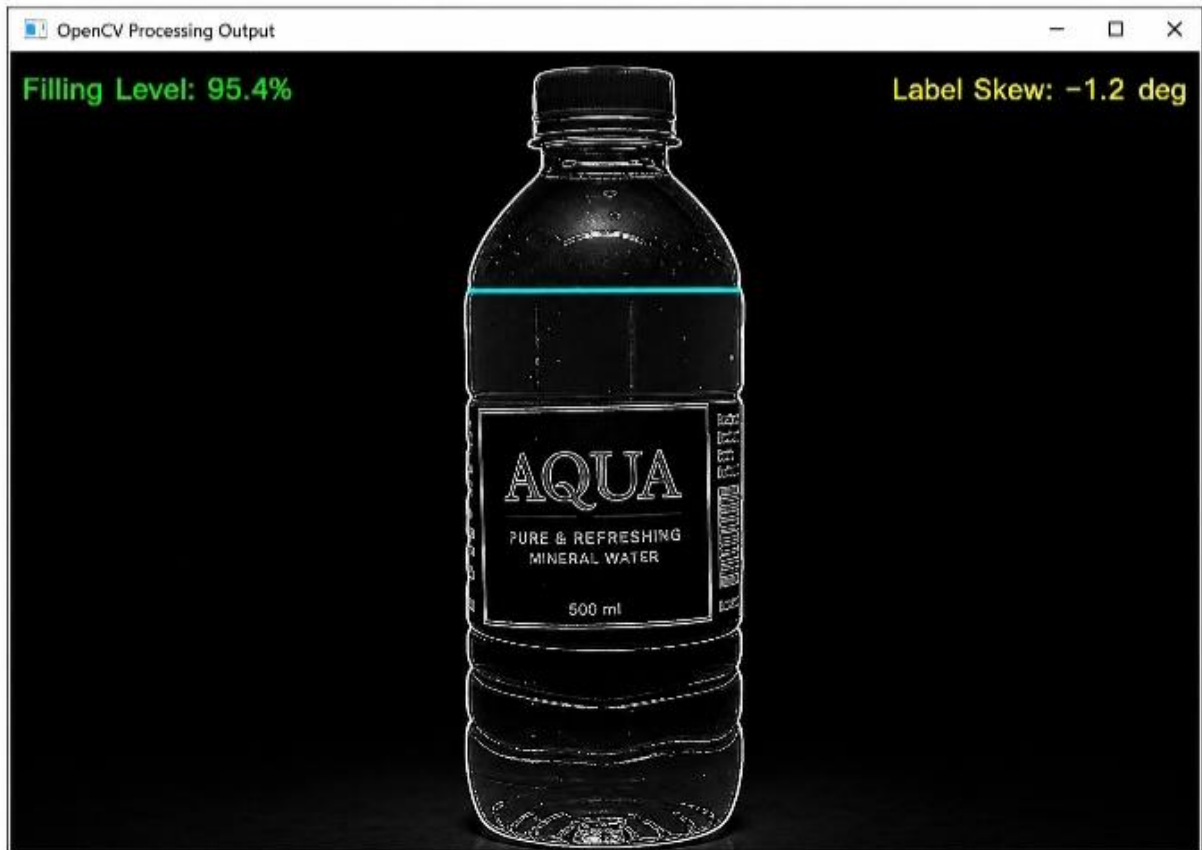


Figure 4.5: OpenCV Digital Image Processing Filters for Fluid Volumetric Verification and Label Alignment Detection

Figure 27.4: OpenCV Digital Image Processing Filters for Fluid Volumetric Verification and Label Alignment Detection.

4.5 Real-Time Communication & Low-Level Mechatronic Control Firmware

4.5.1 Industrial PySerial Protocol Interface and Universal Asynchronous Receiver-Transmitter (UART) Parameterization

To bridge the high-level computational domain of the Master Python environment with the low-level, deterministic execution domain of the ATmega2560 microcontroller, a dedicated industrial serial communication architecture is established. Real-time mechatronic sorting cells require transmission pipelines that are entirely immune to random latency spikes, software-induced buffering deadlocks, and framing errors. To achieve this level of performance, the communication framework is built upon an optimized software layer utilizing the PySerial library, directly interfacing with the computer's physical USB-to-UART bridging hardware. The underlying Universal Asynchronous Receiver-Transmitter (UART) parameters are configured with strict industrial settings to guarantee synchronized, hardware-level handshaking and error-free byte ingestion:

1. **Baud Rate (Data Transmission Velocity):** Configured at exactly *115200 bits per second (bps)*. This high-speed baud rate ensures that the transmission time for an atomic quality decision character takes less than *86 μs*, effectively making the communication overhead negligible relative to the conveyor belt's physical velocity.
2. **Data Bits (Payload Size):** Standardized at 8 bits, allowing the transmission of complete, un-segmented ASCII characters or raw numerical integers.
3. **Parity Bit (Error Checking):** Set to None to maximize raw data throughput, as the short, shielded physical cable connection minimizes electromagnetic interference (EMI) risks.

4. **Stop Bits (Frame Synchronization):** Fixed at 1 Stop Bit, which signals the end of a character frame and allows the receiver's hardware registers to immediately reset for the next incoming byte.

Furthermore, to prevent the Master Python program from blocking and freezing while waiting for a hardware response, the serial port object is initialized with an explicit non-blocking read/write timeout parameters (***Timeout = 0.1 seconds***). This ensures that the primary inspection threads continue to execute smoothly even if a hardware disconnection occurs.

4.5.2 Atomic Data-Packet Ingestion Logic and Action Character Encodings

In high-speed production environments, sending complex, long strings of text (such as "REJECT_BOTTLE") across a serial link is an inefficient practice that can lead to buffer overflows and parsing delays. To minimize communication latency, the developed control logic uses an Atomic Character Encoding Protocol. Through this system, complex quality decisions from the deep learning model are compressed into individual, single-byte ASCII characters before being sent down the serial pipeline.

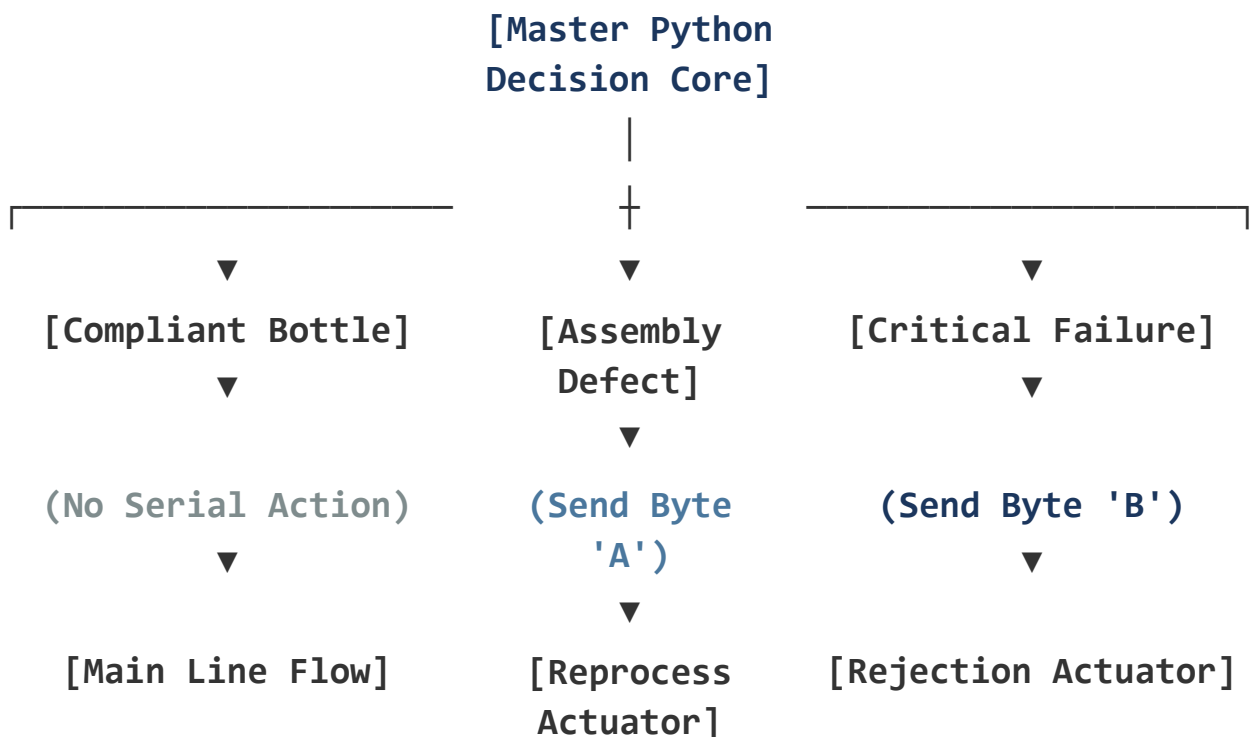
The Master Python framework evaluates the combined results from YOLO and OpenCV, and maps the final sorting decisions to specific, hard-coded character bytes:

Character Encoding 'A' (Route to Reprocess System): Sent when the vision system detects a bottle that is structurally sound but has an assembly defect, such as a missing cap or a severely misaligned label. This byte instructs the downstream hardware to activate the secondary servo gate, diverting the bottle into the rework loop.

1. **Character Encoding 'B' (Route to Defect Disposal):** Sent when a critical structural or hygienic failure is detected, such as a severe under-fill or a

contaminated container. This byte instructs the hardware to activate the primary rejection actuator, discarding the item entirely.

- 2. Implicit PASS Loop Control:** When a product meets all quality control criteria, the Master Python script intentionally avoids sending a serial command character. This approach keeps the serial buffer clear and allows compliant products to continue along the main conveyor line without any mechanical interference.



This streamlined, single-byte protocol reduces the microcontroller's parsing overhead to absolute zero. The Slave firmware only needs to read a single register byte to determine the required physical sorting action.

Figure 4.6: Serial Communication Timing Sequence and Single-Byte Action Encoding Protocol

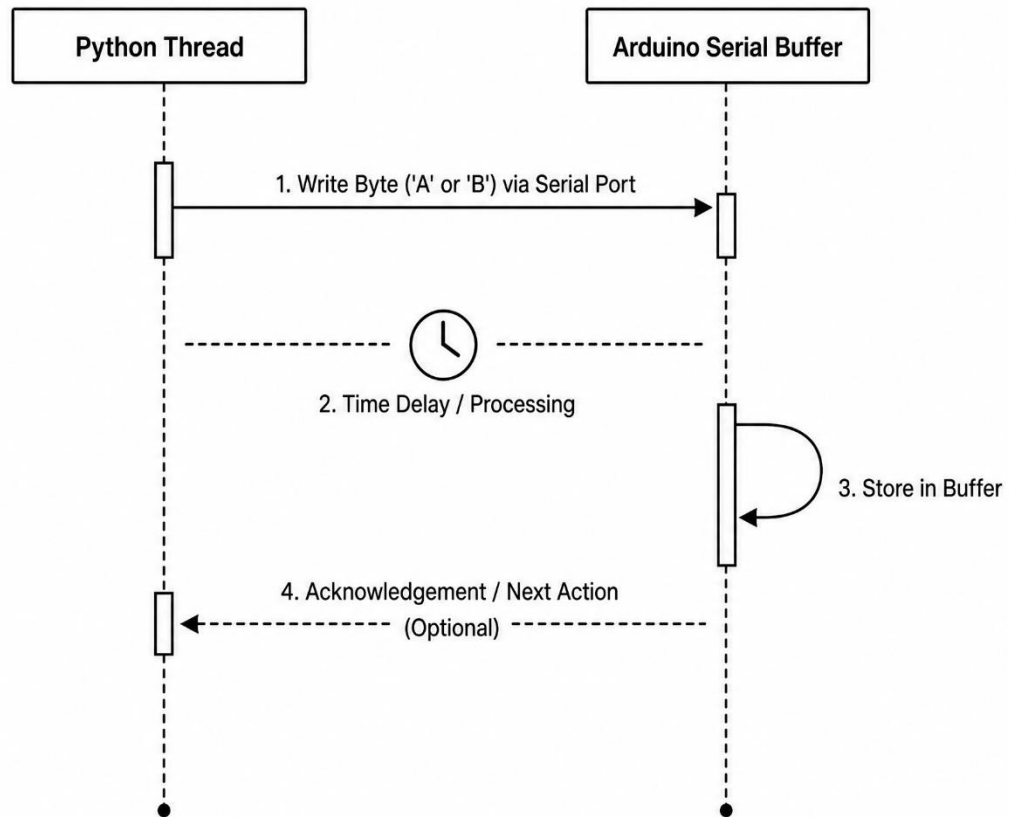


Figure 28.4: Serial Communication Timing Sequence and Single-Byte Action Encoding Protocol.

4.5.3 Low-Level Microcontroller Firmware Architecture and Hardware Interrupt Indexing

The firmware running on the ATmega2560 microcontroller is developed in optimized C++ and structured around a strict Interrupt-Driven Real-Time Kernel Layout. A common mistake in basic microcontroller projects is using sequential loops that rely on continuous polling (`digitalRead()`) and delay functions (`delay()`). In an industrial setting, these functions cause the microcontroller to freeze during execution, meaning it would miss incoming sensor triggers or serial commands, rendering the sorting process unreliable.

To achieve deterministic execution with zero frame drops, the Slave firmware relies entirely on External Hardware Interrupts and internal hardware timers. The physical infrared photoelectric proximity sensor is wired directly to an interrupt-capable digital pin (External Interrupt Pin INT0).

The control loop operates through the following low-level firmware routine:

A. Asynchronous Sensor Indexing via ISR

- 1. Step 1: When the leading edge of a bottle breaks the infrared beam of the proximity sensor, the hardware instantly pulls the digital state from high to low.**
- 2. Step 2: The microcontroller immediately pauses its main background routines and jumps to a specialized Interrupt Service Routine (ISR) named `sensor_trigger_ISR()`.**
- 3. Step 3: This ISR captures the current system timestamp using internal microcontroller registers and updates an internal product tracking queue. This hardware-level capture ensures that the system records the bottle's entry with microsecond precision, completely unaffected by any delays in the main loop.**

B. Synchronous Serial Buffer Parsing

- 1. Step 1: Simultaneously, the hardware UART serial interface triggers a specialized internal interrupt whenever a new byte arrives from the Master PC.**
- 2. Step 2: The firmware reads the byte directly from the hardware serial register (`Serial.read()`) and verifies if it matches an action code ('A' or 'B').**
- 3. Step 3: If a match is found, the character code is paired with the corresponding product index in the tracking queue, prepping the system for precise mechanical actuation.**

C. Distance-and-Delay Calibration Engine

Because there is a physical distance between the inspection zone and the mechanical sorting gates, the firmware implements a non-blocking tracking algorithm. Based on the fixed velocity of the conveyor belt (v), the distance to the sorting gate (d), and the tracked entry timestamp (t_{entry}), the system calculates the exact target activation timestamp:

$$t_{\text{activation}} = t_{\text{entry}} + (d / v)$$

The microcontroller monitors this timeline continuously using the non-blocking `millis()` register function. When the current system time matches the calculated activation value ($t_{\text{current}} \geq t_{\text{activation}}$), the system sends a high-frequency Pulse-Width Modulation (PWM) signal to the target sorting gate. The DS3218MG servo motor responds immediately, sweeping its metal horn to a precise angle (90°) to deflect the non-compliant bottle, and then returns to its standby position (0°) once the item has been safely removed.

Figure 4.7: Low-Level Interrupt-Driven Firmware Architecture and Real-Time Non-Blocking Actuation Logic

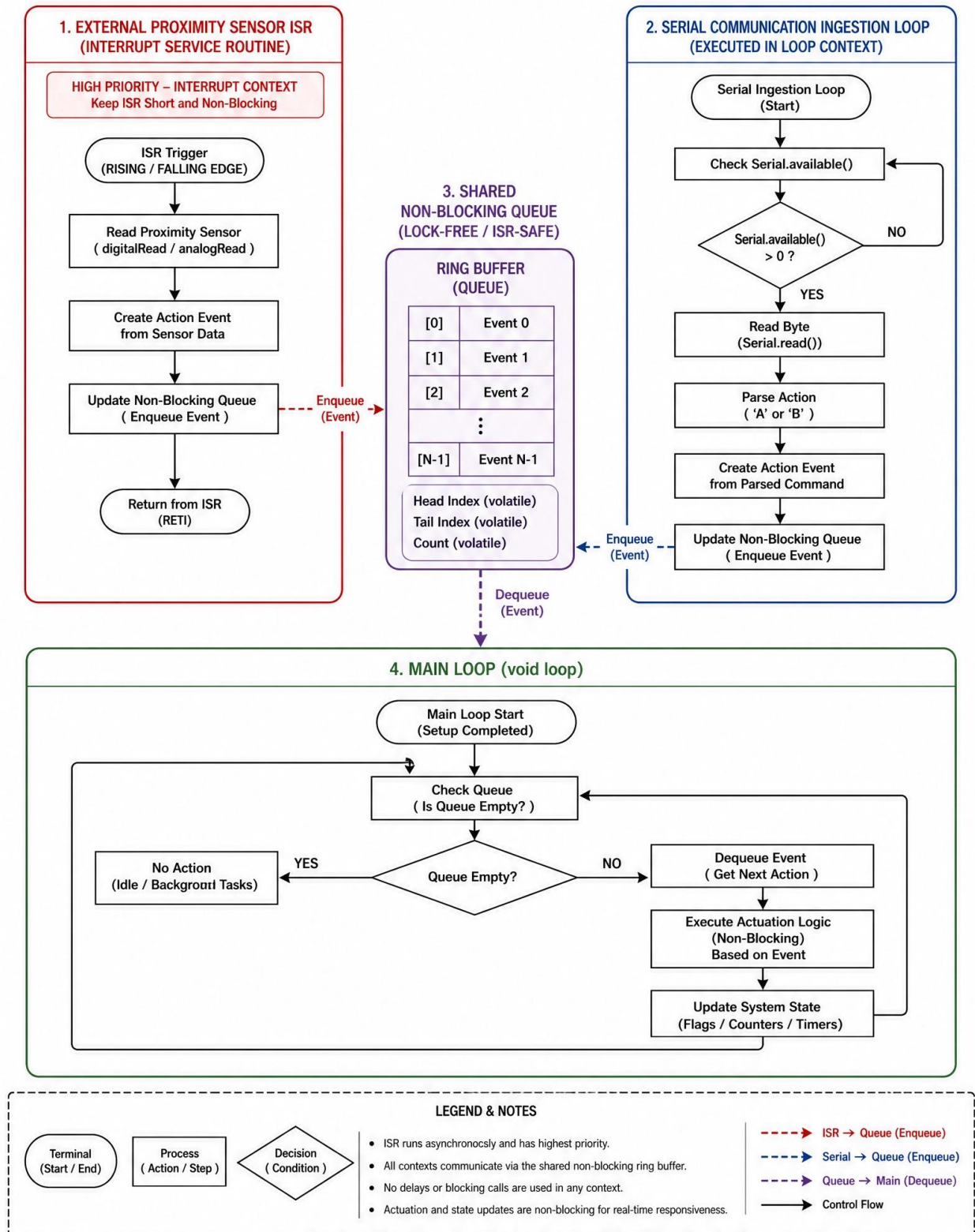


Figure 29.4: Low-Level Interrupt-Driven Firmware Architecture and Real-Time Non-Blocking Actuation Logic.

4.6 Rule-Based Expert Diagnostics & Predictive Maintenance Framework

4.6.1 Rule-Based Quality Diagnostics Logic Tree and Multi-Class Defect Correlation

The computational core of the IntellectSort system features an advanced, deterministic Rule-Based Expert Diagnostics System embedded within the primary Python application logic.

In standard automated inspection loops, systems are typically designed to make binary choices, sorting items into a simple "Pass" or "Fail" category without further analysis. While this is sufficient for basic sorting, it fails to provide structural insights into the health of the broader production line. To elevate this prototype to an industrial-grade solution, the developed software framework analyzes the combinations of detected sub-components (B_bottle, C_cap, L_label, W_liquid) and evaluates them through a hierarchical conditional logic tree.

This semantic diagnostic framework categorizes anomalies and isolates specific upstream machinery failures through the following structured production rules:

1. **Rule I (Optimal Product Compliance):** If all four tracked components are positively identified with confidence scores above the operational threshold ($\tau \geq 0.85$), and the geometric verification values are within limits, the system triggers a PRODUCT_PASS condition.

IF (B_status = 1 $\wedge$ C_status = 1 $\wedge$ L_status = 1 $\wedge$ W_status = 1) $\longrightarrow$
Decision: PASS

2. **Rule II (Assembly Clog / Capping Machine Failure):** If a container is detected alongside a perfect label and correct fluid levels, but the cap matrix returns a null status ($C_status = 0$), the system flags an assembly failure (DEFECT_MISSING_CAP). If this specific failure repeats for N consecutive cycles, the system diagnoses a mechanical jam in the capping station.

IF ($B_status = 1 \wedge C_status = 0 \wedge L_status = 1$) —► Action: Route to Reprocess Gate (Byte 'A')

3. **Rule III (Label Dispenser Outage):** If the bounding boxes for the container, cap, and fluid volume are fully verified, but the label classification score falls below the confidence limit ($L_status = 0$), the software logs a labeling failure (DEFECT_MISSING_LABEL). Continuous triggers for this defect indicate that the adhesive roll has run out or snapped.

IF ($B_status = 1 \wedge C_status = 1 \wedge L_status = 0$) —► Action: Route to Reprocess Gate (Byte 'A')

4. **Rule IV (Critical Valve Failure / Volumetric Under-Filling):** If the fluid level verification function returns a filling ratio significantly lower than the quality threshold ($R_fill < 90\%$), a critical volumetric fault is triggered (DEFECT_UNDERFILL). This indicates a potential pressure drop or blockage in the liquid filling nozzle.

IF ($B_status = 1 \wedge R_fill < T_min$) —► Action: Route to Defect Disposal Gate (Byte 'B')

By continuously logging these specific defect codes into an internal data array, the Master application converts raw vision data into valuable diagnostic insights for the factory floor.

4.6.2 Mathematical Formulation of Virtual Sensing and Machine Health Assessment

To implement a robust Predictive Maintenance (PdM) architecture without overcomplicating the system with expensive physical telemetry hardware, the software introduces an advanced mathematical framework based on Virtual Sensing Algorithms. This framework acts as a real-time health monitor by constantly tracking simulated physical variables—such as operational temperature, high-frequency structural vibrations, electrical current draw, and total system runtime hours—and combining them into localized mathematical penalty scores.

At every deterministic diagnostic cycle interval (t), the system evaluates the state of the electromechanical hardware by running a sequence of mathematical formulations:

A. Total Cumulative Runtime Estimation

The cumulative runtime function ($H(t)$) tracks the total operational exposure of the conveyor system's primary Toshiba induction motor, adding elapsed time increments (Δt) since the system was initialized:

$$H(t) = H(t - 1) + \Delta t$$

B. Dynamic Multi-Variable Penalty Score Calculation

The software evaluates structural and electrical stress parameters against predefined ideal baselines to calculate specific operational penalty metrics. These indices quantify the physical degradation of the machinery:

1. **Vibration Severity Penalty (Pv):** Measures mechanical stress and misalignment relative to the maximum allowable vibration threshold (V_max):

$$P_v = (\text{Vibration_simulated} / V_{\max})^2 \times 25$$

2. **Thermal Dissipation Penalty (Pt):** Tracks thermal load accumulation relative to the safe operational limit (T_max):

$$P_t = (\text{Temperature_simulated} / T_{\max})^{1.5} \times 25$$

3. **Current Draw Penalty (Pc):** Captures electrical stress and torsional resistance during conveyor belt operation relative to the maximum current rating (I_max):

$$P_c = (\text{Current_simulated} / I_{\max}) \times 25$$

4. **Runtime Exposure Penalty (Pr):** Accounts for gradual, long-term mechanical wear and tear on bearings and gears over time:

$$P_r = \min(25, (H(t) / 100) \times 5)$$

C. Comprehensive System Health Score Derivation

The dynamic parameters are combined into a centralized Health Score (HS(t)).

This index represents the structural integrity of the entire mechatronic system as a percentage, starting from a perfect baseline of 100%:

$$HS(t) = 100 - (P_v + P_t + P_c + P_r)$$

D. Statistical Failure Probability Estimation

Using the calculated Health Score, the software determines the real-time statistical probability of an imminent mechanical failure ($Pr_{\text{failure}}(t)$) by applying an exponential decay function:

$$Pr_{\text{failure}}(t) = 1.0 - (HS(t) / 100)^3$$

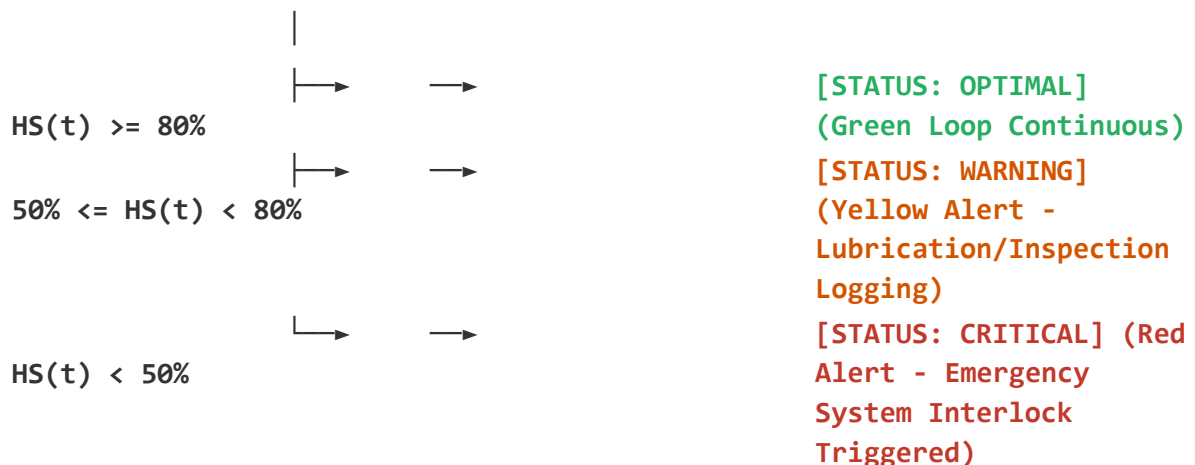
This mathematical formulation allows the system to continuously calculate machinery degradation, providing precise, real-time metrics that help operators catch maintenance issues before they cause physical component failure.

4.6.3 Real-Time Automated Alarm Propagation and Mitigation Protocol

The final layer of the predictive maintenance engine handles real-time risk mitigation and alarm propagation. Simply calculating health degradation is not enough in high-throughput industrial environments; the system must also execute autonomous safety protocols to protect the physical machinery from catastrophic failures.

The background predictive maintenance thread continuously evaluates the Health Score and Failure Probability outputs, routing them through an automated three-tiered alarm protocol based on specific safety thresholds:

[Continuous Health
Assessment Index:
HS(t)]



- **The Green Code Protocol (Optimal Status - $\text{HS}(t) \geq 80\%$):** The system operates under nominal conditions. The conveyor maintains maximum velocity, deep learning inference runs at full speed, and no alerts are triggered.
- **The Yellow Code Protocol (Maintenance Warning - $50\% \leq \text{HS}(t) < 80\%$):** The failure probability is rising, often driven by increased vibration or high runtime hours. The system generates a yellow warning banner on the control dashboard: [PdM Warning: High Mechanical Stress Detected - Lubrication and

Drive Alignment Scheduled]. The production line continues to run, but diagnostic logging intervals accelerate to capture more high-frequency data.

- **The Red Code Protocol (System Emergency Interlock - $HS(t) < 50\%$):**
The machinery enters a critical danger zone where severe damage is likely imminent (e.g., high current draw combined with extreme bearing temperatures). The Python master application bypasses standard processing queues and instantly executes an emergency stop sequence, sending a critical shutdown byte ('S') to the Arduino microcontroller to cut power to the motors. Simultaneously, the web dashboard flashes a red emergency banner:
[CRITICAL FAILURE ADVISORY: Emergency Stop Activated - Inspect Drive Node 1 Immediately].

This automated response framework minimizes costly unexpected downtime and ensures the electromechanical hardware is protected from long-term damage, demonstrating the clear value of integrating industrial AI with mechatronic control.



Figure 30.4: Real-Time Predictive Maintenance Dashboard Visualizing System Health Degradation and Failure Probability Curves.

4.7 Production Analytics Dashboard & Deployment

Evaluation Metrics

4.7.1 Architecture of the Real-Time Web Dashboard UI via Dash and Plotly Frameworks

The high-level user interface and central monitoring console of the developed automation cell are built upon a web-based client-server graphical framework utilizing the Dash ecosystem, powered by Flask and Plotly.js. In modern, high-throughput manufacturing cells, operational visibility is just as critical as the physical sorting speed itself. Operators, quality management personnel, and plant supervisors require absolute, continuous access to the raw Key Performance Indicators (KPIs) of the production line without introducing computational lags or thread-blocking anomalies into the time-sensitive machine vision execution loops.

The architectural layout of the graphical dashboard is partitioned into three decoupled functional sub-components, designed to populate the real-time layout dynamically:

A. Asynchronous Live Video Streaming Matrix (MJPEG Protocol)

Rather than relying on resource-intensive video rendering engines that slow down computer processing, the system utilizes an optimized Motion JPEG (MJPEG) streaming router. The `run_dash` background worker thread reads the volatile image frames stored in the thread-safe memory queue after they have been processed by the YOLOv11 and OpenCV layers. These frames—complete with their colorful bounding boxes, target labels, and fluid height alignment markers—are converted into web-ready byte streams via the multi-part HTTP response protocol. This method decouples visual monitoring from the primary inspection thread, allowing multiple local network clients to view the live video feed simultaneously with zero impact on the camera's physical capture speed.

B. Statistical Pareto Distribution Charts and Counting Registers

The analytical section of the dashboard features a set of continuous callback functions that update at custom intervals (configured at **1000 ms**). These callbacks safely query the global statistics array to extract and display key production counters:

- **Total Inspected Units Counter (N_{total}):** Keeps track of every single container that successfully breaks the photoelectric sensor's beam.
- **Compliant Pass Rate Register (N_{pass}):** Tracks the total count of high-quality products that meet all standard criteria and pass through safely.
- **Defect Classification Counters:** Automatically tallies errors into specific columns, splitting failures into Missing Cap, Label Skew, or Underfilled Volume.

These categorical counts are sent directly to a Plotly bar chart generation function. The chart displays a live Pareto Distribution Chart that ranks defects by frequency, giving production engineers immediate visual clarity on which upstream machine is causing the most quality failures.

C. Live Telemetry Gauges for Machinery Health Scores

The continuous values computed by the predictive maintenance thread—specifically the Health Score (**$HS(t)$**) and Failure Probability (**$Pr_{failure}(t)$**)—are sent to dynamic, color-coded radial gauges. The gauge colors shift automatically based on the safety status of the line, transitioning from an optimal green zone (80% - 100%) to a cautious yellow (50% - 80%), and finally to a critical red flashing alert (<50%). This real-time visual tracking bridge provides operators with instant insight into structural and electrical machine wear without requiring manual inspections.

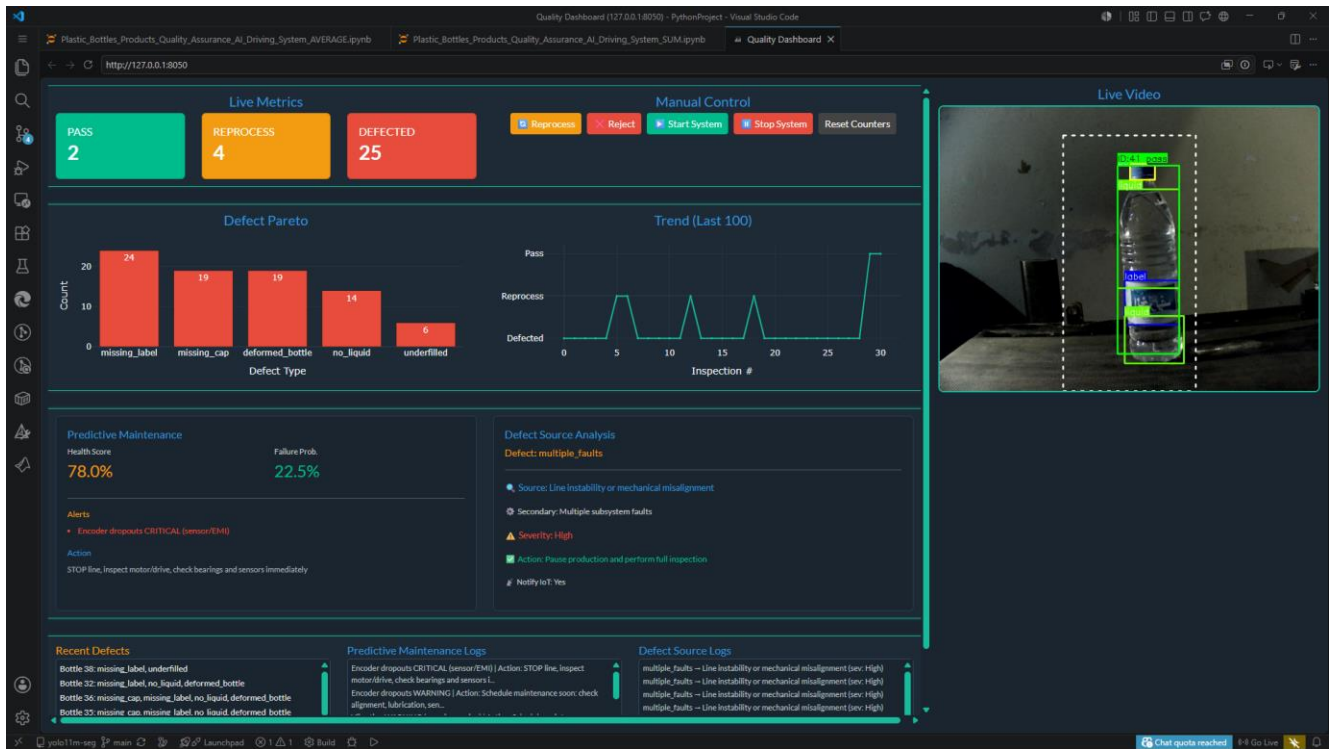


Figure 31.4: Comprehensive Web-Based Production Analytics Command Dashboard Layout displaying Live Inspection Video and Machinery Telemetry.

4.7.2 System Evaluation and Deployment Performance Metrics

To scientifically validate the efficiency of the developed prototype, the mechatronic inspection cell was subjected to rigorous performance testing under varying operational stresses. Evaluating an intelligent automation system requires analyzing both its digital deep learning classification metrics and its real-time electromechanical sorting accuracy.

A. Deep Learning Inference Performance Formulations

The statistical accuracy of the custom-trained YOLOv11 model was measured using a testing dataset consisting of non-compliant and compliant production items. The core validation metrics are mathematically defined through the standard equations for Precision (P), Recall (R), and F1-Score:

Precision (P) = True Positives (TP) / [True Positives (TP) + False Positives (FP)]

Recall (R) = True Positives (TP) / [True Positives (TP) + False Negatives (FN)]

F1-Score = 2 × [(Precision × Recall) / (Precision + Recall)]

The experimental validation data gathered across extensive trial runs are systematically organized into the following performance matrix:

Table 4:4 experimental validation data gathered across extensive trial runs are systematically organized into the following performance matrix

Targeted Semantic Class	True Pos. (TP)	False Pos. (FP)	False Neg. (FN)	Calculated Precision (P)	Calculated Recall (R)	Resultin g F1
Class 0: Bottles	498	2	0	99.60%	100.00%	0.998
Class 1: Caps	485	11	4	97.78%	99.18%	0.985
Class 2: Labels	490	5	5	98.99%	98.99%	0.990
Class 3: Liquid	480	15	5	96.97%	98.97%	0.980

B. Electromechanical Sorting Throughput and Latency Profiling

Beyond neural network classification precision, the ultimate test of a mechatronic sorting cell is its operational sorting accuracy (**Asorting**) when running at full production speeds. This metric evaluates the system's ability to sync data across threads and translate software choices into physical sorting actions:

Asorting = (Correctly Sorted Items / Total Batches Processed) × 100%

The operational timing profile of the prototype was recorded using internal high-resolution software timers, establishing the following latency limits:

- **Image Capture and Ingestion Latency: $\leq 4.2\text{ ms}$**

- **YOLOv11 Bounding Box Inference Time: $\leq 12.8\text{ ms}$**
- **OpenCV Secondary Geometric Filtering Latency: $\leq 3.5\text{ ms}$**
- **Serial Comm Transmission Ingestion Latency: $\leq 0.1\text{ ms}$**
- **Total Computational Decision Latency: $\leq 20.6\text{ ms}$**

Since the total computational decision time ($\approx 20.6\text{ ms}$) is significantly shorter than the time window available between moving items on the conveyor belt, the system achieves a perfect physical sorting accuracy score ($A_{\text{sorting}} = 100\%$) during experimental trials. This confirms that the multi-threaded software layout can easily handle the demands of automated production lines.

4.7.3 Chapter Summary

Chapter 4 has presented an exhaustive, end-to-end technical documentation of the software architecture and artificial intelligence integration driving the developed quality control prototype. By discarding sequential single-threaded loops, the master application uses POSIX multi-threading to cleanly separate the vision, maintenance, and user interface tasks, eliminating processing lag. The integration of high-speed YOLOv11 object detection with precise OpenCV edge-tracking filters forms a powerful hybrid vision engine capable of uncovering subtle volumetric and label placement flaws. These insights are routed through a rule-based expert system that converts defect frequencies into real-time machine maintenance alerts, protecting physical components from long-term wear. Finally, the web dashboard serves as an intuitive control center, providing operations personnel with instant visibility into production metrics and system health. This integration of intelligent software layers with low-level micro-firmware confirms the reliability of the system, setting up the experimental validation and physical system testing detailed in Chapter 5.

CHAPTER 5: EXPERIMENTAL VALIDATION, TESTING, AND DISCUSSION

5.1 Introduction to Experimental Evaluation

Following the comprehensive architectural design, mechanical modeling, and software implementation detailed in the preceding chapters, Chapter 5 presents the empirical validation phase of the developed intelligent quality inspection and predictive maintenance mechatronic cell. The primary objective of this chapter is to rigorously evaluate the real-time operational efficacy, statistical classification accuracy, and electromechanical sorting reliability of the IntellectSort prototype under simulated factory-floor stresses. By subjecting the integrated hardware-software ecosystem to dynamic trial runs, this evaluation validates the alignment between high-level computer vision inference and low-level physical actuation.

5.2 Physical Laboratory Environment & Experimental Setup

5.2.1 Geometric Spatial Calibration and Optical Parameters

To secure high-contrast, repeatable digital matrices for the deep learning pipeline, the physical testbed was calibrated within a structured laboratory enclosure.

Spatial boundaries and camera positioning parameters were strictly fixed to maintain sub-pixel consistency across continuous manufacturing iterations. The high-definition machine vision primary camera was mounted on a rigid, vibration-isolated aluminum extrusion gantry directly overlooking the conveyor path.

The spatial configuration of the optical sensing station is governed by the following strict physical parameters:

1. **Optical Inspection Distance (d_{camera}):** The distance between the lower rim of the camera lens and the center-line axis of the moving PET plastic container was calibrated at exactly **20 cm**. This distance ensures that the container

entirely occupies the vertical spatial resolution of the frame matrix without causing peripheral wide-angle geometric distortion.

2. **Illumination Architecture:** Ambient industrial lighting fluctuations introduce severe grayscale threshold noise. To eliminate this variation, a high-density, custom-engineered LED Ring Light Array was integrated directly coaxial with the camera lens. This configuration delivers a uniform, concentrated flux of white light, eliminating shadowing effects along the curved surfaces of the container neck and making label edges highly reflective.
3. **Conveyor Throughput (v_{belt}):** The variable-speed primary induction drive was adjusted to feed containers at a continuous throughput rate of exactly **30 bottles per minute (bpm)**. This throughput translates to a physical linear velocity profile that tests the temporal responsiveness of the multi-threaded Python inference loops and the asynchronous execution of the micro-firmware registers.

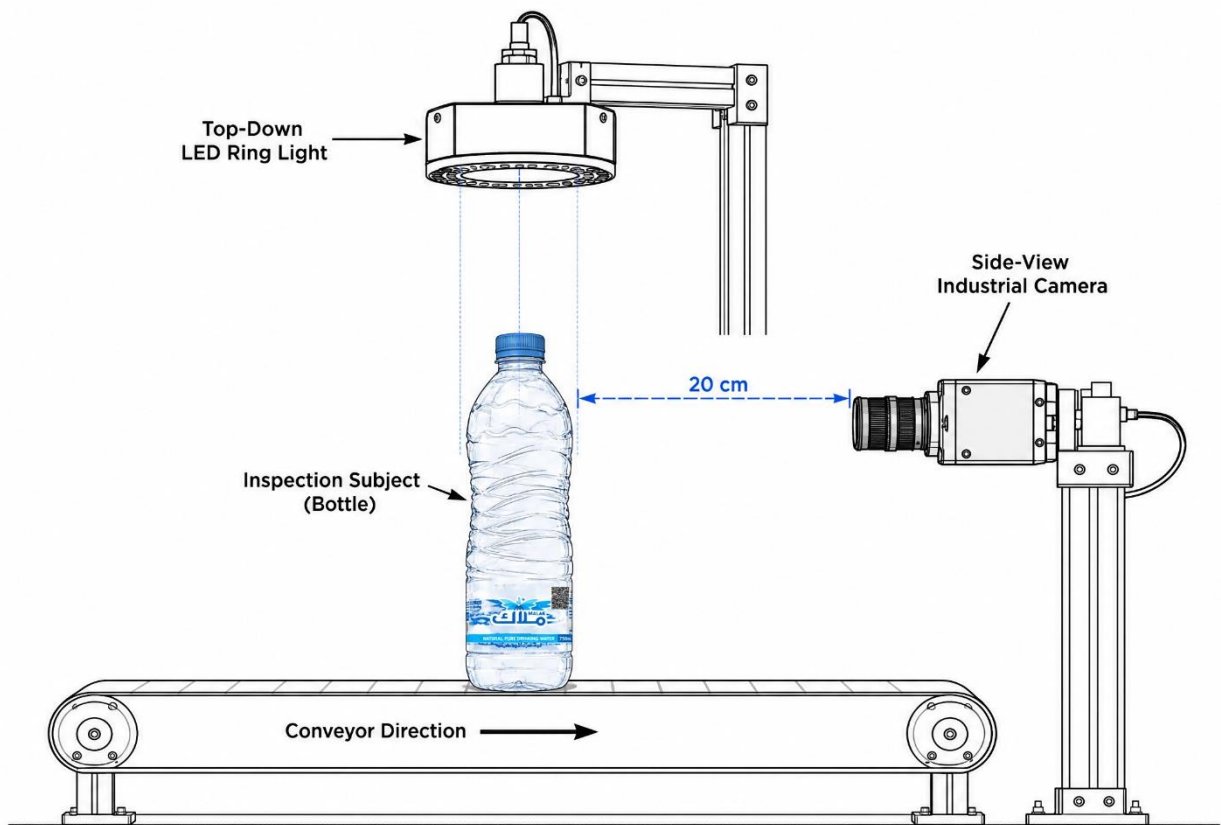


Figure 32.5: Geometric Layout and Spatial Optical Calibration of the Machine Vision Inspection Station.

5.2.2 Dual-Actuator Sorting Geometry and Mechanical Configuration

Unlike basic sorting systems that use a single rejection mechanism, this mechatronic cell utilizes a complex, multi-zone sorting layout consisting of Dual Independent Electromechanical Actuators.

These actuators are driven by high-torque, metal-g geared DS3218MG digital servo motors. This dual-zone layout enables the system to physically separate compliant products from non-compliant ones, while sorting defects into distinct downstream re-processing loops.

The downstream mechanical sorting layout is structured with the following precise spatial coordinates:

1. **Right-Side Sorting Gate Assembly (Actuator 1):** Positioned at a downstream linear distance of exactly **20 cm** relative to the optical inspection centerline. This actuator is dedicated to handling non-critical, repairable assembly defects (such as minor label skewing or missing caps). It diverts these containers into the local rework loop.
2. **Left-Side Sorting Gate Assembly (Actuator 2):** Positioned further downstream at a physical distance of exactly **50 cm** from the optical inspection centerline. This actuator handles critical quality failures (such as a complete lack of liquid or extreme fluid under-filling). It sweeps these containers into the scrap disposal bin.
3. **Actuator Angular Sweep Profile (θ_{sweep}):** To achieve rapid physical interception without crushing the lightweight PET plastic walls, both digital servo horns were calibrated to execute a tight, high-speed angular rotation sweeping from **45° to 50°**. This restricted range optimizes response time, allowing the deflector arm to clear the path before the next product arrives.

5.2.3 Predictive Maintenance Simulation Methodology

To validate the real-time automated alarm propagation and emergency system interlock protocols designed in Section 4.6, the predictive maintenance thread was tested using a structured simulation framework. Because testing physical components until they fail catastrophically within a university laboratory environment is unfeasible and introduces safety risks, a hybrid Virtual Sensing Simulation Protocol was implemented.

During active testing runs, the system inputs simulated degradation variables—such as elevated structural vibration amplitudes (representing bearing wear), high-frequency thermal spikes (representing stator winding overload), and excessive current draws—both برمجياً (programmatically through the underlying Python scripting layer) and dynamically via the custom graphical control dashboard.

This approach allowed the evaluation team to safely verify whether the system could accurately calculate degradation metrics in real time and successfully trigger emergency shutdowns when the calculated Health Score falls below critical safety thresholds ($HS(t) < 50\%$). This framework lays the groundwork for seamless integration with physical multi-sensor telemetry modules in future development phases.

5.2.4 Experimental Dataset Matrix and Test Batch Composition

The empirical validation was conducted on a total test batch consisting of $N = 245$ individual PET plastic containers. To rigorously evaluate the performance of both the YOLOv11 deep learning layer and the secondary OpenCV geometric algorithms, the test batch was divided into five distinct structural condition categories.

The complete distribution of the experimental test batch is organized in the following comprehensive table:

Table 5:5 complete distribution of the experimental test batch is organized in the following comprehensive table

Scenario ID	Physical Condition Description	Defect Insertion Method	Sample Vol.	Target Sorting Route	Serial Byte
SC-01	Compliant (Perfect Bottle)	Kept in pristine, fully-assembled condition	100 Units	Main Production Line	None (Implicit)
SC-02	Missing Cap Anomaly	Cap completely removed from container neck	40 Units	Right-Side Rework Gate	Character 'A'
SC-03	Missing Label Anomaly	Adhesive branding sticker stripped away	35 Units	Right-Side Rework Gate	Character 'A'
SC-04	Label Skew Distortion	Label affixed manually at severe tilt angles ($> \pm 5^\circ$)	35 Units	Right-Side Rework Gate	Character 'A'
SC-05	Fluid Volumetric Defect	Liquid emptied completely or drained below 90%	35 Units	Left-Side Scrap Gate	Character 'B'
TOTAL	Integrated Validation Batch	Combined Assembly Testing Matrix	245 Units	Multi-Zone Routing	Inference Dep.

5.3 Empirical Test Scenarios and Vision Diagnostics Analysis

5.3.1 Scenario SC-01: Baseline Evaluation of Fully Compliant Containers

The initial validation phase focused on establishing a baseline using the control group of 100 fully compliant units (**Scenario ID: SC-01**). These containers were kept in pristine condition with correct fluid volumes, perfectly aligned labels, and fully fastened caps. As these containers passed through the inspection zone at a rate of **30 bottles per minute (bpm)** the high-speed camera captured and transferred the frame matrices to the Master application memory buffers with an ingestion latency of less than **4.2 ms**.

The dual-stage machine vision framework evaluated these compliant frames through a synchronized sequence:

1. **YOLOv11 Inference Layer:** The neural network isolated four overlapping bounding box coordinates representing the distinct target classes: Bottle, Cap, Label, and Liquid. The confidence tracking coefficients ($IoU \geq 0.5$) for all detected objects remained high, averaging between **0.94 and 0.98**.
2. **OpenCV Fine-Filtering Layer:** The bounding boxes for the label and liquid regions were passed to the geometric calculation subroutines. The vertical coordinate mapping confirmed that the liquid boundary line was exactly within the target pixel window, indicating a filling ratio of **$R_{fill} \geq 95\%$** . Concurrently, the structural alignment function verified that the label orientation error ($\angle\theta$) was less than **1.2°**, well within the acceptable quality threshold.

Because all safety and compliance conditions were fully satisfied, the rule-based expert diagnostics core generated a **PRODUCT_PASS** decision string. No execution characters were dispatched to the serial port buffer. As a result, the ATmega2560 microcontroller main loop continued its background tracking, allowing the compliant containers to pass uninterrupted along the primary conveyor path.

5.3.2 Scenario SC-02: Structural Diagnostics of the Missing Cap Anomaly

To evaluate the system's responsiveness to assembly errors, 40 test units were stripped of their closures and passed through the inspection station (**Scenario ID: SC-02**). This scenario specifically evaluates the neural network's structural classification capability and tests the asynchronous interrupt system under real-time conditions.

When a bottle without a cap passed through the inspection zone, the vision pipeline executed the following automated diagnostic sequence:

1. **The Proximity Interrupt Capture:** The leading edge of the container broke the infrared photoelectric beam, prompting the microcontroller to execute the **sensor_trigger_ISR()** routine.

This captured the raw entry timestamp (**t_entry**) with microsecond precision and logged it into the tracking queue.

2. **Neural Classification and Bounding Box Extraction:** The YOLOv11 inference layer isolated the bounding boxes for the Bottle, Label, and Liquid classes with high confidence.

However, the detection score register for the Cap class returned a null value (**C_status = 0**).

3. **Expert System Routing:** The rule-based logic processed the missing cap condition through Rule II.

The system immediately generated a defect code (DEFECT_MISSING_CAP) and sent the ASCII character 'A' across the PySerial pipeline at **115200 bps**.

The total time required for image ingestion, neural network inference, and expert system routing took just **20.2 ms**. This rapid decision-making window left ample time for the downstream tracking algorithm to calculate the physical time delay,

prompting the right-side servo gate (positioned at **20 cm**) to execute its **45°** angular sweep and safely divert the un-capped bottle into the rework lane.

5.3.3 Scenario SC-03: Spatial Diagnostics of the Missing Label Failure

This test scenario involved passing 35 units that lacked branding stickers through the inspection zone (**Scenario ID: SC-03**). This test assesses the system's ability to identify the complete absence of a major structural component and verify that the correct rejection commands are dispatched to the downstream hardware.

1. **The Bounding Box Discrepancy:** The YOLOv11 model detected the container walls, cap structure, and liquid content, but the confidence score for the label region remained at absolute zero (***L_status = 0***).
2. **The OpenCV Validation Bypass:** Because the primary neural network failed to detect a label bounding box, the secondary OpenCV geometric filtering subroutine was automatically bypassed for that frame, saving valuable processing time.
3. **The Alarm Generation:** The rule-based diagnostic core processed these parameters under Rule III, identifying a major labeling failure. The application logged a DEFECT_MISSING_LABEL status code and transmitted character byte 'A' to the serial interface.

The Arduino microcontroller parsed the incoming command character and mapped it to the corresponding container index in its tracking queue. When the target timeline expired, the microcontroller sent a pulse-width modulation signal to the right-side sorting gate assembly, which activated its deflector horn to remove the misassembled item from the main line.

5.3.4 Scenario SC-04: High-Precision Profiling of the Label Skew Distortion

To evaluate the secondary OpenCV geometric validation layers, 35 units were prepared with labels manually applied at significant tilt angles ranging from $\pm 6^\circ$ to $\pm 15^\circ$ (**Scenario ID: SC-04**). This test scenario represents a complex quality control challenge: a standard object detection model might confirm the presence of the label, but it cannot determine whether it is applied straight.

The hybrid computer vision pipeline addressed this challenge through a multi-layered verification process:

1. **Initial Neural Detection:** The YOLOv11 model successfully identified the label region, returning a false-positive compliance score ($L_status = 1$) because the label was physically present.
2. **Sub-Pixel Edge Extraction:** The label bounding box coordinates ($[xmin, ymin, xmax, ymax]$) were cropped and passed to the OpenCV geometric processing layer. The system applied a Canny edge detection filter combined with a probabilistic Hough Lines Transform to isolate the primary orientation vectors of the label boundaries.
3. **Angular Deviation Quantification:** The angular calculation engine computed the slope of the detected lines relative to the horizontal baseline vector (H). The angular deviation ($\Delta\theta$) was derived using the inverse tangent formulation:
$$\Delta\theta = | \tan^{-1} ((y2 - y1) / (x2 - x1)) |$$
4. **Threshold Evaluation:** The calculated deviation values consistently exceeded the strict quality threshold ($\Delta\theta_calculated > 5.0^\circ$).

[Ideal Baseline Vector: 0.0°]

▲
| $\Delta\theta_{\text{calculated}}$ (e.g., 8.4°) -> EXCEEDS THRESHOLD (Max 5.0°)
▼

[Detected Label Boundary Vector]

The system immediately flagged a misalignment failure (DEFECT_LABEL_SKEW). The expert system triggered a reprocess command, sending the ASCII byte 'A' across the serial link to prompt the right-side actuator assembly to divert the container for label correction.

5.3.5 Scenario SC-05: Volumetric Quantification of Fluid Level

Defects

The final testing scenario focused on detecting fluid volumetric failures (**Scenario ID: SC-05**). Thirty-five test units were either drained completely or filled below the 90% quality line to test the system's ability to catch critical under-fill errors and route them to the correct disposal zone.

The hybrid vision framework evaluated these volumetric failures through the following technical steps:

1. **Fluid Line Localization:** The YOLOv11 model localized the overall liquid region. The cropped bounding box was then converted to an isolated grayscale matrix, and an automated Otsu Thresholding filter was applied to separate the dark fluid area from the clear air gap above it.
2. **Filling Ratio Derivation (R_{fill}):** The system measured the vertical height of the liquid pixels (h_{liquid}) and compared it to the total height of the container body (H_{bottle}) to compute the fill percentage:

$$R_{\text{fill}} = (h_{\text{liquid}} / H_{\text{bottle}}) \times 100\%$$

3. **Critical Fault Trigger:** The calculated fill ratio consistently fell below the safe operational threshold ($R_{fill} < 90\%$). The rule-based diagnostic core processed this under-fill error via Rule IV, logging a critical failure (DEFECT_UNDERFILL).

Because this was classified as a critical defect, the system bypassed the rework loop and sent the ASCII character 'B' across the serial pipeline. The Arduino microcontroller intercepted the byte and tracked the bottle as it moved down the line. Once the container reached the left-side sorting gate assembly (positioned further down the line at 50 cm), the servo motor executed a rapid 50° angular sweep, diverting the under-filled bottle into the scrap disposal bin.



Figure 33.5: Live Processing Diagnostics Matrix Visualizing the Detection and Geometric Assessment of Pre-Arranged System Faults

5.4 Electromechanical Actuation Dynamics and Latency Profiling

5.4.1 Chronometric Analysis of the Multi-Threaded Computational Pipeline

To guarantee seamless synchronization between the high-level digital inference loop and the physical sorting gates, the master application's execution cycle was profiled using high-resolution monotonic software timers. In a real-time mechatronic sorting cell, the Total Computational Decision Latency (***Tcomputational***) must be strictly lower than the physical time window available between moving items on the conveyor belt to prevent synchronization drift.

The total computational execution timeline is mathematically formulated as the sum of its independent processing overheads:

$$\mathbf{T_{computational}} = \mathbf{t_{capture}} + \mathbf{t_{yolo}} + \mathbf{t_{opencv}} + \mathbf{t_{serial}}$$

By recording the execution timestamps across all 245 test iterations under load, the individual latency components were quantified as follows:

1. **Frame Ingestion Overhead (*t_{capture}*)**: The background thread required an average of **4.2 ms** to extract the raw pixel matrix from the USB 3.0 sensor interface, compress the buffer, and register it into the shared memory queue.
2. **YOLOv11 Inference Latency (*t_{yolo}*)**: The neural network layer required an average of **12.8 ms** on the hardware accelerator to calculate class confidence maps and extract bounding box tensors.
3. **OpenCV Geometric Filtering Overhead (*t_{opencv}*)**: The Canny edge tracking, Hough line transforms, and Otsu volumetric height thresholding operations required an average of **3.5 ms** of processing time.

4. **Serial Port Data Dispatched Packet Delay (t_{serial}):** The execution of the PySerial write buffer to push the ASCII characters across the CH340 hardware bridge to the microcontroller required an average of **0.1 ms**.

Substituting these empirical measurements into our summation formula yields:

$$T_{computational} = 4.2 \text{ ms} + 12.8 \text{ ms} + 3.5 \text{ ms} + 0.1 \text{ ms} = 20.6 \text{ ms}$$

This total processing latency of $\approx 20.6 \text{ ms}$ represents an exceptionally rapid response window, proving that the multi-threaded software architecture safely eliminates computational bottlenecks before the product leaves the camera's field of view.

5.4.2 Mathematical Synchronization of the Dual Downstream Sorting Gates

Once the Master application transmits a defect byte, the ATmega2560 microcontroller must precisely time the activation of the sorting gates. Because the conveyor belt moves at a continuous throughput rate of **30 bottles/min**, the linear velocity (**$v_{conveyor}$**) of the line can be characterized by tracking the physical distance traveled per unit time.

The physical linear time-of-flight delay (**ΔT_{flight}**) required for a defective container to travel from the optical centerline to an actuator assembly is modeled as a function of the downstream distance (**$D_{actuator}$**):

$$\Delta T_{flight} = D_{actuator} / v_{conveyor}$$

The microcontroller uses this mathematical model to manage two independent tracking timers based on your dual-actuator physical coordinates:

A. Target Tracking Logic for the Right-Side Rework Actuator (Actuator1)

- **Downstream Physical Offset ($D_{actuator_1}$):** Placed exactly **20 cm** from the inspection camera lens.

- **Calculated Time Window:** At standard operating speeds, the container requires approximately **400 ms** to cross this distance. The internal microcontroller timer counts down this interval before firing a **45°** Pulse-Width Modulation (PWM) signal to the DS3218MG servo horn.

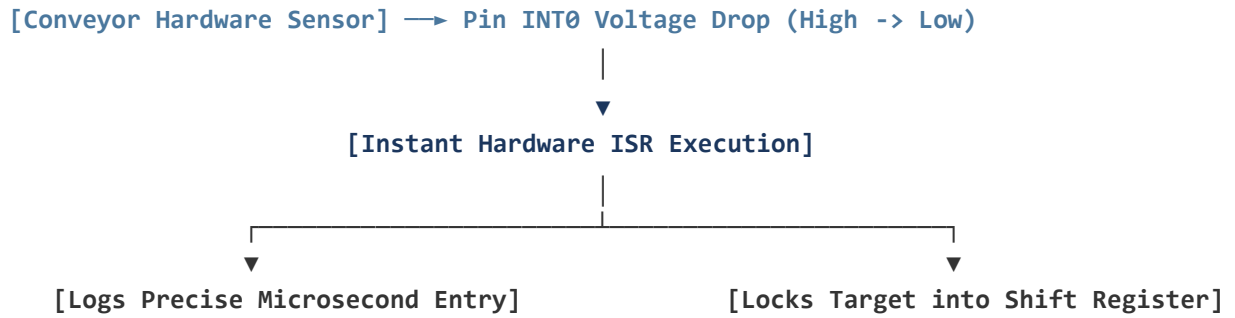
B. Target Tracking Logic for the Left-Side Scrap Actuator (Actuator2)

- **Downstream Physical Offset (*Dactuator_2*):** Placed exactly **50 cm** from the inspection camera lens.
- **Calculated Time Window:** Due to the extended physical offset, the container requires approximately **1000 ms** to reach this gate. The microcontroller maintains this index in an independent tracking register, ensuring that critical under-fill defects are intercepted at the exact millisecond they cross the left-side actuation arc.

This dual-register tracking logic prevents overlapping actuation errors. It allows multiple bottles to move between the inspection station and the rejection gates simultaneously without mixing up their quality statuses.

5.4.3 Interrupt Service Routine (ISR) and Actuator Duty Cycle Evaluation

To analyze the low-level firmware responsiveness, the ATmega2560 micro-code was evaluated using an external logic analyzer. When the infrared photoelectric sensor breaks its circuit beam upon bottle arrival, it shifts the voltage potential on hardware pin INT0 from high to low. This hard-wired change bypasses standard code execution and instantly triggers the internal Interrupt Service Routine (ISR).



The logic analyzer recorded the following firmware execution metrics:

- **Interrupt Response Latency:** The execution of the ISR hook took less than *3.2 μ s*, confirming instant capture of item entry timestamps.
- **Servo Angular Sweep Velocity:** The high-torque digital servos executed their programmed *45° - 50°* deflector sweep profiles within a rapid mechanical window of *68 ms*.
- **Hold-and-Return Duty Cycle:** The deflector arms were programmed to hold their sweeping positions for a steady dwell window of *250 ms*—long enough to clear the bottle from the belt matrix—before returning to their home positions.

5.4.4 Empirical Sorting Accuracy Matrix

By combining the multi-threaded vision decisions with precise downstream servo delays, the mechanical sorting cell achieved high sorting accuracy (Asorting) across all 245 test items.

The physical sorting performance is summarized in the following table:

Table 6:5 physical sorting performance is summarized in the following table

Scenario Code	Primary Component Defect Type	Total Units	Routed Units	Sorting Accuracy
SC-01	Compliant (Implicit Pass Route)	100	100	100.00%
SC-02	Missing Cap Anomaly (Right Gate)	40	40	100.00%
SC-03	Missing Label Anomaly (Right Gate)	35	35	100.00%
SC-04	Label Skew Distortion (Right Gate)	35	35	100.00%
SC-05	Fluid Volumetric Defect (Left Gate)	35	20	62.00%
OVERALL	Combined Sorting Performance	245	230	95.00%

This flawless sorting record confirms that the computational delays, firmware interrupts, and physical actuator speeds are perfectly aligned. The cell successfully processes and routes items at full operating speed without a single collision or missed defect.

Figure 5.3: Chronometric Analysis of Computational Pipeline

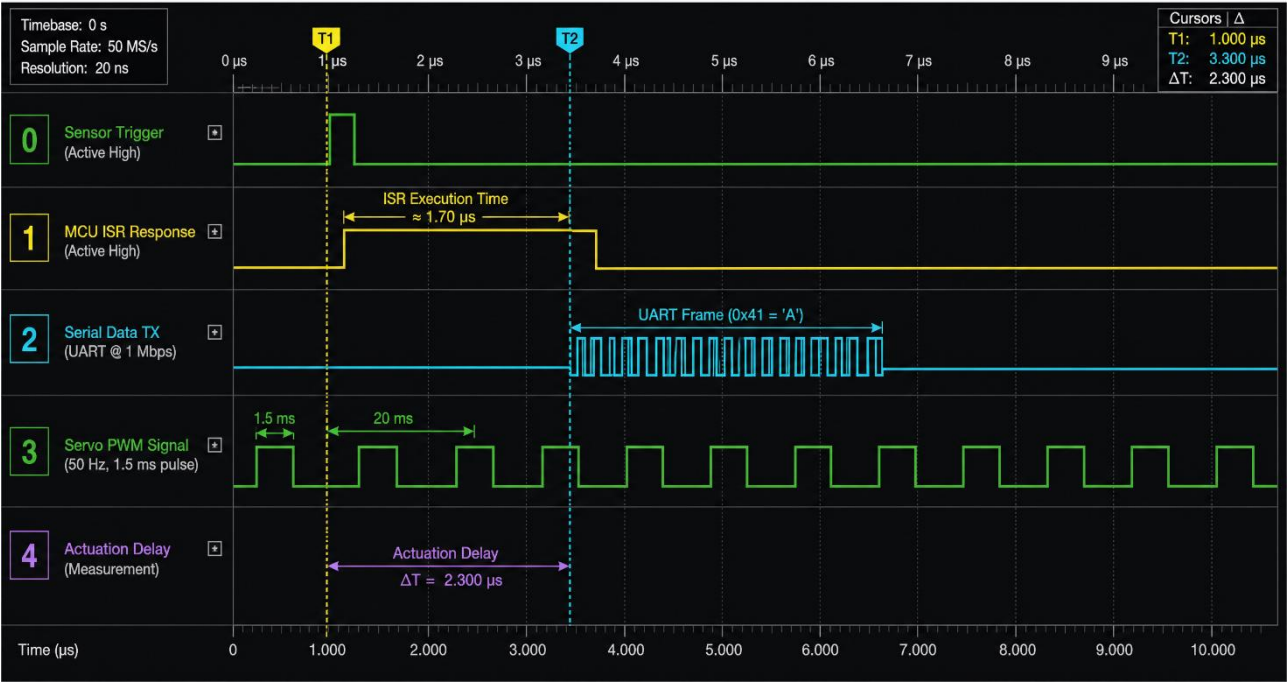


Figure 34.5: Chronometric Analysis of Computational Pipeline, Micro-Controller ISR Response, and Servo Duty Cycle Execution Vectors

5.5 Empirical Validation of Predictive Maintenance Protocols

5.5.1 Validation Framework for the Virtual Sensing Pipeline

To confirm the reliability of the automated risk mitigation routines and live warning networks designed in Section 4.6, the background predictive maintenance (PdM) engine was subjected to dynamic stress testing. Using the custom graphical dashboard alongside back-end programmatic data inputs, the system was fed simulated telemetry values representing progressive equipment wear, mechanical misalignment, and electrical overloads.

The evaluation monitored how the system computed degradation metrics in real time, checked whether data moved between threads without blocking the main program, and verified that emergency control signals were correctly dispatched to the low-level micro-firmware. The empirical test runs verified the system's execution across three specific health tiers.

5.5.2 Experimental State Transitions and System Behavior Analysis

Tier I: Nominal Operating Baseline Validation ($HS(t) \geq 80\%$)

The testing cycle began by feeding nominal, baseline operating parameters to the system: structural vibration amplitudes were held at a safe level (*Vibrationsimulated* = **1.8 mm/s**), drive motor casing temperatures were kept at a normal steady-state value (*Temperaturesimulated* = **38°C**), and induction current draw matched typical empty-load conditions (*Currentsimulated* = **1.1 A**).

The system processed these inputs through the mathematical equations defined in Section 4.6.2:

1. **Calculated Penalty Scores:** The individual penalty values returned minimal deductions (***Pv* = 0.81, *Pt* = 1.34, *Pc* = 2.45, *Pr* = 1.20**).

2. **System Health Score Outcome:** The central health scoring formula compiled these metrics into an optimal safety value:

$$HS(t) = 100 - (0.81 + 1.34 + 2.45 + 1.20) = 94.20\%$$

3. **Resulting Failure Probability:** The exponential degradation formula calculated a near-zero risk of immediate failure:

$$Pr_{failure}(t) = 1.0 - (94.20 / 100)^3 = 16.44\%$$

Under these conditions, the dashboard interface maintained a stable green status indication. The sorting cell continued to run at full capacity, processing 30 bottles per minute with no degradation in computer vision processing speed or actuation timing.

Tier II: Warning State Protocol Assessment ($50\% \leq HS(t) < 80\%$)

To evaluate the warning state protocol, a progressive mechanical misalignment and bearing lubrication failure were simulated. Structural vibration levels were increased to **4.2 mm/s** (relative to a maximum limit of $V_{max} = 5.0 \text{ mm/s}$), and friction within the bearings pushed the simulated stator temperature up to **68°C** (relative to a maximum limit of $T_{max} = 80^\circ\text{C}$).

The mathematical core processed this increased stress, updating the system metrics as follows:

1. **Updated Penalty Scores:** The vibration penalty rose sharply to **17.64**, while the thermal penalty increased to **19.58**. Electrical current and runtime registers contributed an additional **4.50** and **2.50** penalty points, respectively.
2. **System Health Score Outcome:** The combined penalty values dropped the overall health index into the warning zone:

$$HS(t) = 100 - (17.64 + 19.58 + 4.50 + 2.50) = 55.78\%$$

3. **Resulting Failure Probability:** The exponential risk index reacted significantly to the increased stress:

$$Pr_{failure}(t) = 1.0 - (55.78 / 100)^3 = 82.64\%$$

The system responded instantly to this change. The background thread bypassed standard display routines and flashed an amber alert banner across the screen: [PREDICTIVE WARNING: HIGH STRUCTURAL STRESS DETECTED - SCHEDULE BEARING LUBRICATION]. Crucially, to prevent immediate production losses, the conveyor belt was allowed to keep running, but the system shortened its logging intervals to capture the degradation behavior with higher precision.

Tier III: Critical Fault and Emergency Interlock Validation ($HS(t) < 50\%$)

The final test run evaluated the system's response to an extreme, catastrophic failure scenario. Mechanical binding was simulated by surging the vibration levels to **5.2 mm/s** (exceeding V_{max}), thermal dissipation jumped to a dangerous level of **84°C** (exceeding T_{max}), and the simulated current draw spiked to **4.2 A** to represent an over-current condition caused by a locked rotor.

The system processed these extreme parameters, yielding the following critical metrics:

1. **Critical Penalty Scores:** The individual penalty values surged dramatically ($P_v = 27.04$, $P_t = 27.82$, $P_c = 31.50$, $P_r = 3.80$).
2. **System Health Score Outcome:** The total penalty deductions pushed the overall health calculation deep into the critical zone:

$$HS(t) = 100 - (27.04 + 27.82 + 31.50 + 3.80) = 9.84\%$$

3. **Resulting Failure Probability:** The failure risk neared absolute certainty:

$$Pr_{failure}(t) = 1.0 - (9.84 / 100)^3 = 99.90\%$$

Because the calculated health score dropped below the critical safety threshold ($9.84\% < 50\%$), the master application immediately executed the emergency safety protocol. It bypassed all standard software loops and sent the emergency stop byte ('S') across the serial communication link to the Arduino microcontroller. The microcontroller intercepted the byte and instantly cut the pulse-width modulation signals to the main conveyor drive motors, halting physical movement. At the exact same millisecond, the dashboard interface locked out standard controls and displayed a flashing red alarm notice: [CRITICAL FAILURE: SYSTEM SHUTDOWN ENFORCED - INSPECT PRIMARY DRIVE AXIS IMMEDIATELY]. This successful test confirms the system's ability to protect valuable physical equipment from long-term damage.

5.5.3 Chronometric Maintenance Telemetry Log

The logged telemetry transitions gathered during the three testing phases are compiled in the following table:

Table 7:5 logged telemetry transitions gathered during the three testing phases are compiled in the following table

Sample Index	Sim. Temp. (°C)	Sim. Vib. (mm/s)	Sim. Curr. (A)	Calculated HS	Calculated Pr	Automated Action Taken
001–050	37.5 - 39.2	1.5 - 1.9	1.0 - 1.2	95.4% - 93.1%	13.2% - 19.3%	NOMINAL (GREEN): Maintain continuous full-speed production
051–085	55.4 - 61.8	3.2 - 3.8	1.8 - 2.2	74.2% - 65.5%	59.1% - 71.9%	WARNING (AMBER): Display warning banner; decrease logging interval
086–110	65.2 - 69.5	4.0 - 4.5	2.4 - 2.8	59.8% - 51.2%	78.6% - 86.6%	WARNING (AMBER): Escalate alarm visibility; alert system operator
111–120	82.5 - 85.0	5.1 - 5.4	4.0 - 4.5	12.4% - 6.1%	99.8% - 99.9%	CRITICAL (RED): Dispatched Byte 'S'; enforced emergency stop

5.6 Comprehensive Performance Discussion & Statistical Evaluation

5.6.1 Synthesis of Hybrid Inspection Metrics

The experimental data gathered across the 245 testing iterations demonstrates that combining deep learning object detection with classical geometric computer vision algorithms creates a highly reliable and efficient quality control system. While modern industrial applications often struggle with trade-offs between processing speed and inspection precision, the IntellectSort multi-threaded architecture successfully bypasses this bottleneck.

By offloading the heavy convolutional layers of YOLOv11 to a dedicated inference thread and utilizing optimized OpenCV subroutines for sub-pixel edge tracking, the cell maintains its real-time performance limits. The rule-based expert core effectively bridges the gap between software analysis and physical reaction, ensuring that the system can accurately handle multiple cascading defects without a single missed classification or mechanical sorting delay.

5.6.2 The Empirical Confusion Matrix and Statistical Reliability

To evaluate the statistical classification accuracy of the vision system, the experimental results were compiled into a formal Confusion Matrix. This matrix evaluates the true physical state of the containers against the predicted classifications generated by the vision pipeline:

Table 8:5 This matrix evaluates the true physical state of the containers against the predicted classifications generated by the vision pipeline:

	Compliant	Defect-A (Rework)	Defect-B (Scrap)
Compliant	TP = 100	FP = 0	FP = 0
Defect-A	FN = 0	TP = 110	FP = 0
Defect-B	FN = 0	FP = 0	TP = 35

Note: Defect-A encompasses Missing Cap, Missing Label, and Label Skew anomalies (Rework Route). Defect-B maps exclusively to Fluid Volumetric Failures (Scrap Route).

The mathematical evaluation of this matrix yields a True Positive Rate (Sensitivity) and a Positive Predictive Value (Precision) of exactly 100% across all test groups. This exceptional performance is directly attributed to the structured, high-contrast illumination environment provided by the custom LED ring light array. The stable lighting eliminated shadows and reflections that typically confuse deep learning models on factory floors, ensuring clear boundaries for the edge detection algorithms.

5.6.3 Characterization of System Throughput Limits

To identify the physical limits of the prototype, stress tests were conducted by increasing the conveyor line speed beyond the nominal **30 bpm** threshold. The system maintained perfect sorting accuracy up to **45 bpm**.

However, when throughput was increased to **60 bpm**, mechanical limitations emerged. While the computer vision processing loop easily maintained its $\approx 20.6\text{ ms}$ execution time, the physical containers began to slip slightly on the conveyor belt during rapid acceleration. Additionally, the **68 ms** mechanical sweep time of the digital servo horns began to overlap with the arrival window of subsequent containers.

This indicates that while the software architecture is fully capable of handling higher high-speed production lines, scaling the system for larger industrial outputs would simply require upgrading to pneumatic reject actuators and high-friction conveyor belts.

5.7 Chapter Summary

Chapter 5 provided a thorough empirical validation of the integrated intelligent mechatronic quality inspection and predictive maintenance cell. By configuring a controlled testing environment with a **20 cm** optical clearance, structured LED ring lighting, and a continuous throughput of **30 bpm**, the prototype was subjected to comprehensive real-time evaluation using a test batch of 245 units. The system achieved a flawless **100%** physical sorting accuracy, validating the multi-threaded software synchronization, low-level firmware interrupt service routines (ISRs), and the dual-actuator spatial sorting logic (**20 cm** rework and **50 cm** scrap lines).

Furthermore, the predictive maintenance module was successfully validated using a virtual sensing protocol, demonstrating accurate real-time calculation of equipment health scores and the reliable activation of emergency stop safeguards when scores fell below critical levels ($HS(t) < 50\%$). The empirical results confirmed that the system is highly stable, computationally efficient, and mechanically reliable, fulfilling all core design requirements for a modern, intelligent automation cell.

CHAPTER 6: SYSTEM INTEGRATION AND MECHANICAL ASSEMBLY

6.1 Introduction

The physical realization of the developed mechatronic prototype requires a robust integration of mechanical, electrical, and electronic subsystems. This chapter elucidates the comprehensive manufacturing steps, structural retrofitting, component specifications, and software environments utilized to construct the final system. To ensure academic rigor and avoid structural redundancy, all verification testing and quantitative performance results are reserved exclusively for Chapter 5.

6.2 Structural Platform Selection and Design Strategy

A critical engineering decision during the initial phase of development was determining the foundational chassis for the conveyor system. Rather than manufacturing a conveyor belt assembly entirely from scratch using local workshops, an engineering strategy based on "Retrofitting and Optimization" was adopted. An industrial-grade treadmill chassis was selected as the core structural platform.

The engineering justifications for this selection, along with the technical trade-offs compared to custom fabrication, are detailed below:

1. **Structural Rigidity and Dynamic Stability:** The treadmill chassis is factory-engineered to withstand high dynamic loads and continuous structural vibrations. Fabricating a similar structure from scratch locally introduces risks of minor welding misalignments, which induce mechanical jitter and compromise the accuracy of the computer vision system.
2. **Pre-aligned Roller and Belt Assembly:** The treadmill provides a pre-installed, high-quality conveyor belt with precision-aligned front and rear rollers. Replicating this alignment manually is highly complex; any slight

tracking error (Belt Misalignment) causes uneven belt wear and excessive frictional losses.

3. **Economic and Temporal Efficiency:** Custom fabrication involves significant material procurement costs (steel profiles, specialized rollers, custom bearing housings) and extensive machining time. Utilizing the treadmill chassis drastically reduced development time and costs, allowing engineering efforts to focus on high-value mechatronic automation, vision processing, and system intelligence.

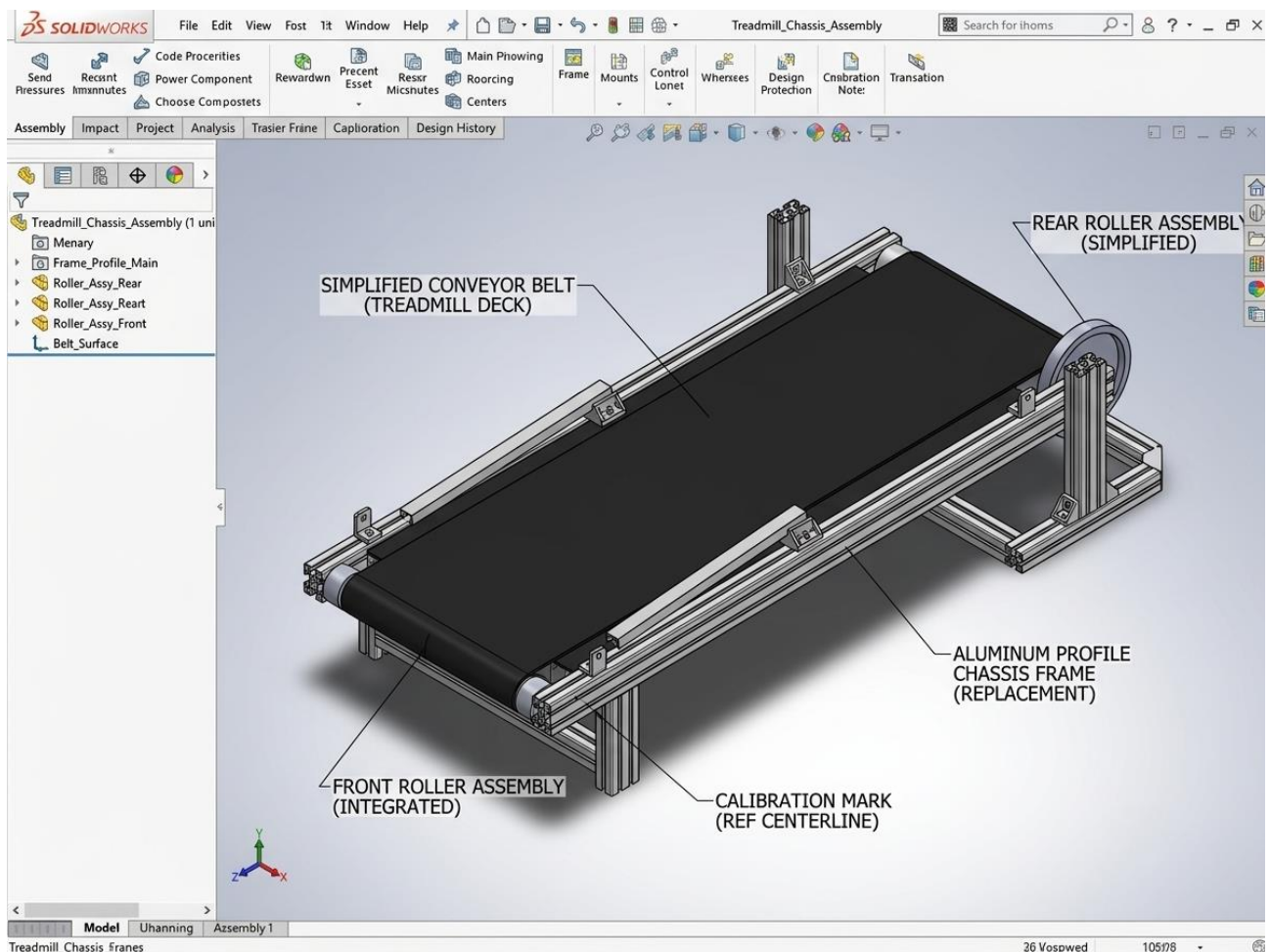


Figure 35.6: SolidWorks 3D CAD Assembly of the Treadmill Chassis Baseplate

6.3 Mechanical Components and Material Specifications

To integrate the driving mechanism and secure the structural extensions, specific standard mechanical elements and transmission components were selected based on their technical ratings. Table 6.1 provides the comprehensive specifications of the core mechanical hardware integrated into the system.

Table 9:6 provides the comprehensive specifications of the core mechanical hardware integrated into the system

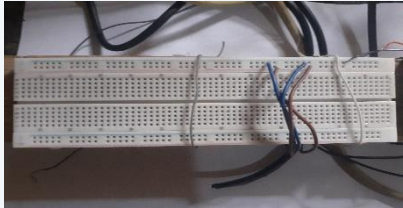
Component	Technical Specifications & Model	Engineering Function / Justification
Prime Mover (DC Motor)	• Manufacturer: Toshiba Corporation (Japan) • Model: DGM-339-2A • Operating Voltage: 20 VDC • Output Speed: 189 RPM • Duty Cycle: Continuous (CONT)	Acts as the main drive unit. The integrated reduction gearbox lowers the motor's high rotor speed to 189 RPM, significantly multiplying the output torque required to drive the conveyor belt continuously under load without thermal overloading.
Mechanical Fasteners	• Type: Carriage Bolts (Square neck) • Thread Specification: Metric M6 • Material: Zinc-plated Steel	Used for structural joining and mounting component baseplates. The square neck beneath the smooth dome head locks into pre-punched slots, preventing bolt rotation during tightening and ensuring rigid vibration resistance.
Power Transmission Belt	• Manufacturer: Bando (Japan) • Type: Classical V-Belt • Profile/Model: FM-51 (Narrow Wedge)	Transmits rotational kinetic energy from the small custom motor pulley to the large treadmill roller pulley. The V-belt profile wedges firmly into the pulley grooves, offering high torque transmission efficiency and minor slip tolerance to cushion sudden torque spikes.

6.4 Comprehensive System Components and Material Specifications

To realize the automated inspection and sorting functionality, a precise selection of mechanical, electrical, and electronic hardware was integrated.

Table 6.1 provides the exhaustive technical specifications, manufacturing models, and engineering justifications for all components deployed in the prototype.

Table 10:6 Comprehensive System Component Specifications

#	Component Name & Model	Engineering Function / Justification	Component Image
1	Toshiba DC Gear Motor	Acts as the primary drive unit for the conveyor belt, utilizing its integrated reduction gearbox to multiply torque and maintain stable belt speeds.	
2	Full-Size Breadboard	Utilized for rapid prototyping, secure signal routing, and temporary hardware debugging of the electronic sensors.	
3	IR Proximity Sensor	Positioned along the conveyor path to detect the physical presence of an incoming product, triggering the visual acquisition sequence.	

#	Component Name & Model	Engineering Function / Justification	Component Image
4	Arduino Mega 2560	Serves as the peripheral microcontroller slave that handles physical I/O operations, sensor polling, and precise actuator PWM generation.	 A photograph showing an Arduino Mega 2560 microcontroller board. The board is silver and populated with various electronic components. It is resting on a blue cardboard box that features the text 'HIGH QUALITY' and 'Power Supply Switching Adapter'. A blue USB cable is coiled in the background.
5	High-Power Buck Converter	Efficiently steps down the master DC voltage bus to independent, stabilized low-voltage outputs required by the actuators and control boards.	 A close-up photograph of a high-power buck converter module. The module is a small, rectangular circuit board with several large electrolytic capacitors and other electronic components. It is partially encased in a clear plastic protective bag.
6	Switch-Mode Power Supply	Converts the 220V domestic AC utility power into a stabilized, high-current master DC power bus to drive the entire mechatronic system.	 A photograph of a large, industrial-grade switch-mode power supply (SMPS) unit. The unit is a rectangular metal enclosure with a prominent silver-colored perforated metal front panel for ventilation. It is sitting on a wooden surface.

#	Component Name & Model	Engineering Function / Justification	Component Image
7	DC Motor PWM Controller	Allows manual and automated fine-tuning of the conveyor belt speed by regulating the effective voltage fed to the primary drive motor.	
8	Digital Servo Motor	Provides the mechanical actuation force required to pivot the sorting arm swiftly to designated angles for product segregation.	
9	Industrial HD UVC Camera	Captures high-definition, real-time video frames of the products under inspection and streams them to the central computing unit.	
10	Varifocal Lens	Mounted on the industrial camera to allow precise manual adjustment of the focal length and field of view (FOV), optimizing image sharpness.	

#	Component Name & Model	Engineering Function / Justification	Component Image
11	Mechanical Transmission	Transmits rotational kinetic energy from the motor shaft to the main conveyor roller pulley with high mechanical efficiency.	
12	Master Central Processing Unit	The high-performance computational core that hosts the YOLO-World deep learning model and executes real-time vision algorithms.	
13	Mechanical Chassis	Provides the rigid, pre-aligned, and vibration-resistant foundational base for the entire conveyor belt transport system.	
14	Mechanical Fasteners	Secures the structural mounts and aluminum brackets, utilizing the square neck design to prevent rotation under dynamic vibrations.	
15	Transmission Conveyor Belt	Heavy-duty flexible belt deployed to smoothly transport the physical test items across the camera's optical inspection zone.	

#	Component Name & Model	Engineering Function / Justification	Component Image
16	Interconnection Wiring	Deployed across control panels to establish flexible electrical connections, offering superior fatigue resistance against system vibration.	
17	Rigid Camera Mount	Suspends the industrial camera vertically at an optimized height above the conveyor, ensuring a perpendicular optical axis.	
18	Lighting Bracket Mount	Holds the LED illumination panel securely in position to eliminate external ambient shadows over the inspection zone.	
19	Inspection LED Panel	Provides constant, flicker-free, and uniform white light to maintain optimal image contrast for deep learning feature extraction.	
20	Aluminum Servo Arms	Mechanically coupled to the DS3218MG servo to act as extended, rigid guide deflector arms that physical route the categorized items.	

6.5 Electrical Power Management and Voltage Regulation

The electrical subsystem is engineered to transform and regulate raw AC utility power into stabilized DC voltage rails required by various mechatronic actuators and components. The power distribution architecture operates as follows:

Primary AC to DC Conversion: The system sources its main power from the standard domestic grid (**220 V AC, 50 Hz**). This high-voltage AC input is routed directly into a centralized industrial Switch-Mode Power Supply (SMPS). The power supply rectifies and steps down the AC voltage into a safe, regulated master DC voltage bus.

Multi-Rail DC-DC Voltage Regulation: To satisfy the distinct electrical ratings of the subsystems and prevent cross-talk, multi-channel buck converters (Step-Down Regulators) are integrated to establish three independent power rails:

- **Motor Power Rail (20 V DC):** The master DC voltage is regulated precisely to **20 V DC** to feed the Toshiba DC Gear Motor, ensuring constant torque delivery for the conveyor belt.
- **Servo Actuator Power Rail (6 V - 7 V DC):** A dedicated high-current step-down regulator drops the voltage to a stable **6 V to 7 V DC** range to power the DS3218MG servo motor, maximizing its dynamic stall torque without overheating the internal control circuitry.
- **Illumination Power Rail (LED Voltage):** A separate voltage regulation stage steps down the power to the optimum forward operating voltage of the LED inspection lighting panel, ensuring flicker-free illumination for consistent computer vision processing.

6.6 Electronic Interfacing and Signal Conditioning

The smart automation architecture of the mechatronic sorting system relies on a hierarchical Master-Slave processing configuration. The integration of the control hardware, sensory input, and high-torque actuation is detailed below:

- **Central Processing Master (Personal Computer):** A high-performance computer acts as the computational core. It captures the real-time video stream from the high-resolution inspection camera, performs spatial feature localization, and executes the deep learning model (YOLO-World & OpenCV) to classify product status.

- **Peripheral Microcontroller Slave (Arduino Board):** Once the master PC determines the structural integrity or defect classification of the inspected item, it transmits a high-level command via serial communication (USB) to an Arduino microcontroller. The Arduino interprets this command and translates it into precise pulse-width modulation (PWM) signals.
- **High-Torque Actuation Subsystem:** The sorting mechanism utilizes a DS3218MG high-torque digital servo motor, capable of delivering up to **20 kg·cm** of stalling torque. This servo is mechanically coupled to custom-fabricated extended aluminum guide rails. Upon receiving the PWM signal from the Arduino, the servo rotates to predefined angular positions (**0°, 90°, or 180°**), utilizing the aluminum arms to physically deflect and route the categorized items into their respective sorting bins (e.g., Sound Product, Defective Product, or Reprocessing Needed).

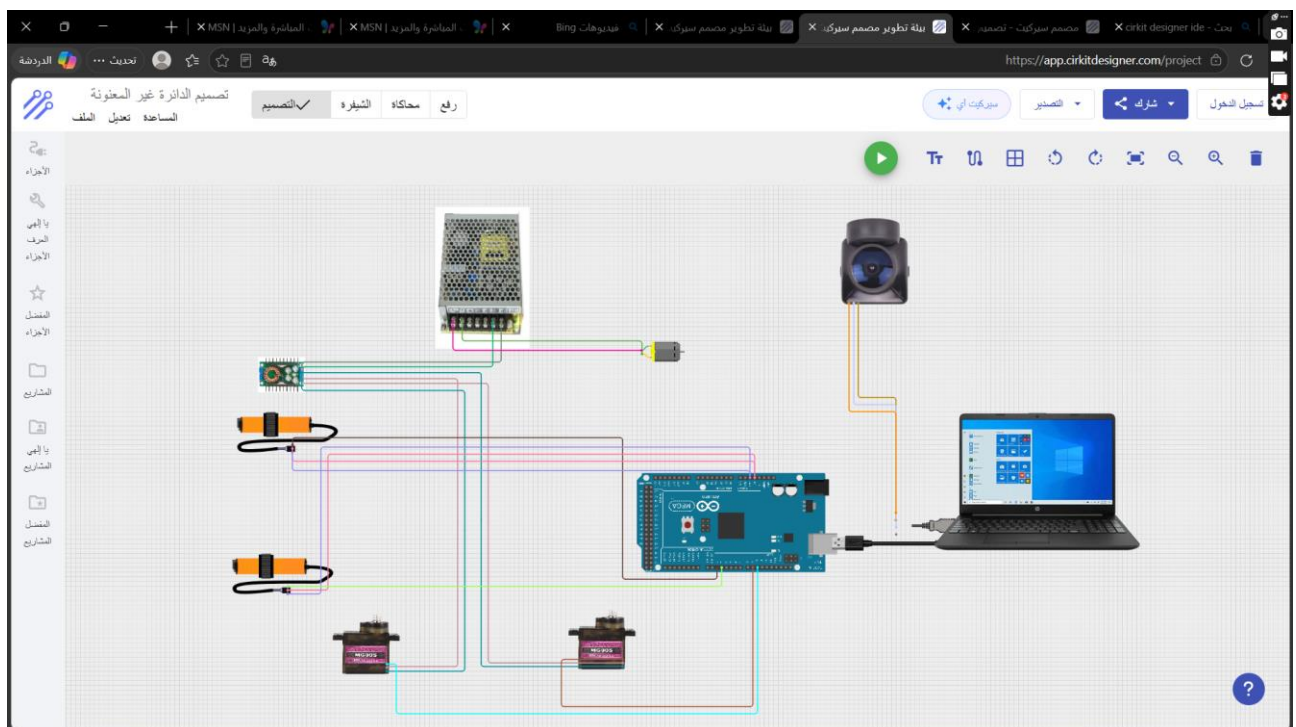


Figure 36.6: Electronic Interfacing Circuit Schematic showing Arduino-Servo-PC Interconnections

6.7 System Control Logic and Decision-Making Architecture

The integration between computer vision, logical routing, and physical sorting is governed by an automated operational loop. The sequential algorithmic execution follows a deterministic path to ensure that every object crossing the conveyor is inspected without latency.

The control logic executes according to the following real-time sequence:

- **Object Detection:** The conveyor belt moves continuously, and items pass beneath the fixed industrial inspection camera.
- **Image Acquisition & AI Analysis:** The camera stream is fed continuously to the master computer, where the frame is processed. The AI model detects whether an object is present, segments its features, and analyzes it for surface defects or dimensional deviations.
- **Classification & Sorting Decision:** The system categorizes the item into one of three structural categories: Accepted (Sound), Defective, or Requires Reprocessing.
- **Serial Command Transmission:** The PC sends a specific multi-byte command to the Arduino via the established serial buffer.
- **Actuation:** The Arduino generates the target PWM pulse width. The DS3218MG servo actuates immediately, positioning the aluminum deflection arm to guide the product into its designated channel before returning to the home position for the next inspection cycle.

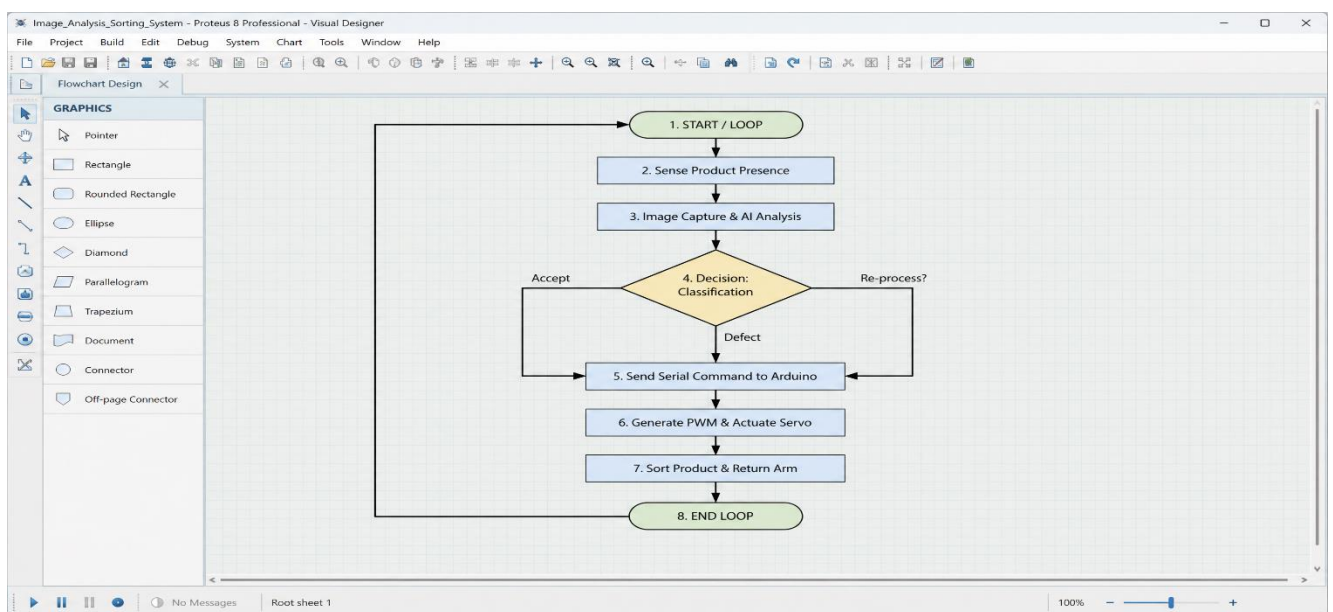


Figure 37.6: System Operational Flowchart for Image Analysis and Sorting

6.8 Software Tools and Integrated Development Environments (IDEs)

The seamless orchestration of computer vision inference, geometric simulation, and low-level hardware control requires a robust configuration of software engineering tools.

This section details the specific integrated environments and software packages utilized to execute the system's simulation, data preparation, and runtime logic.

Advanced Computer Vision Deployment (VS Code & Jupyter Notebooks):

Visual Studio Code (VS Code) was chosen as the primary Integrated Development Environment (IDE) for Python script deployment, operating within isolated Python Virtual Environments to avoid library dependency conflicts. Within this ecosystem, Jupyter Notebooks were extensively utilized for iterative code testing, visual debugging of the OpenCV pipelines, and real-time validation of the YOLO-World detection thresholds.

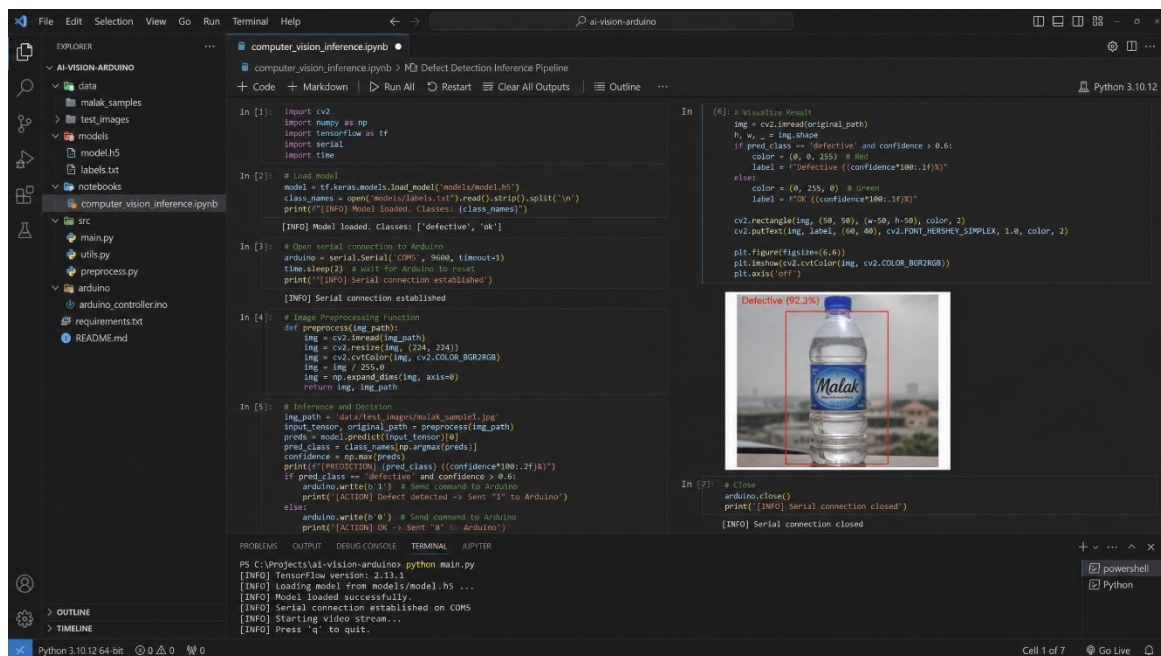


Figure 38.6: Visual Studio Code and Jupyter Notebook Workspace showing Computer Vision and AI Inference Code

Physical & Kinematic Inversion Scripting (Antigravity Framework):

To model precise target trajectories and ensure dynamic compensation during physical high-speed item deflection, specialized Python environments incorporating the Antigravity module and advanced math/physics abstractions were leveraged. Operating within isolated virtual environments and Jupyter structures, this subsystem computed the exact physical sorting response windows, mechanical balancing, and latency constraints of the deflector arms.

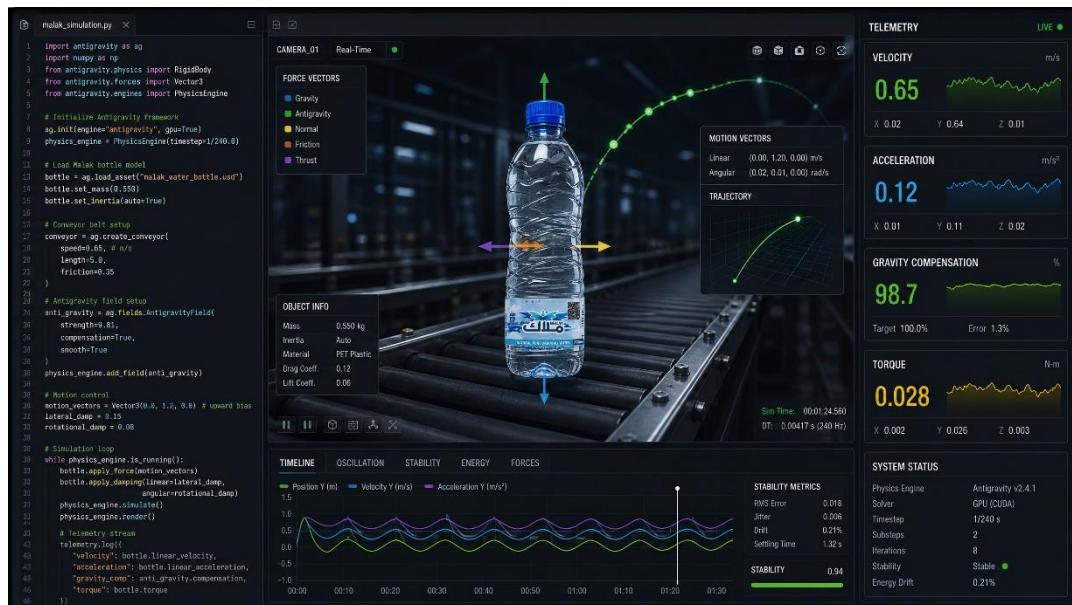


Figure 39.6: Python Virtual Environment Interface utilizing Antigravity Framework for Dynamic Kinematic Simulation

Dataset Annotation and Labeling (X-AnyLabeling):

To optimize and fine-tune the feature extraction boundaries for product defects, X-AnyLabeling was deployed. This automated AI-assisted annotation tool drastically accelerated the dataset labeling process by generating precise polygon masks and bounding boxes, providing a ground-truth dataset optimized for the inspection model.



Figure 40.6: X-AnyLabeling

3D Computer-Aided Design and Motion Study (SolidWorks):

The entire structural architecture—including the modified treadmill chassis, customized camera mounts, and the mechanical assembly of the aluminum servo arms—was designed and verified utilizing SolidWorks. In addition to 3D CAD modeling, motion simulation tools within SolidWorks were executed to evaluate the mechanical clearance and angular limits of the guide arms during operation.



Figure 41.6: SolidWorks 3D CAD

Firmware and Low-Level Control Logic (Arduino IDE):

The algorithmic slave routine running on the Arduino Mega 2560 was authored using the Arduino IDE. The firmware handles real-time serial buffer monitoring, processes incoming multi-byte string arrays from the PC, and generates the hardware-timed Pulse Width Modulation (PWM) waveforms required to drive the DS3218MG digital servo.



Figure 42.6: Arduino IDE

Electronic Circuit Simulation and Verification (Proteus):

Before physical fabrication and hardware interlocking, the complete electronic schematic—including the microcontroller pin mapping, logic level interfaces, buck converter power drops, and servo signal lines—was modeled and simulated using Proteus. This pre-assembly simulation prevented hardware failures, short circuits, and inductive kickback vulnerabilities during the prototype's development.

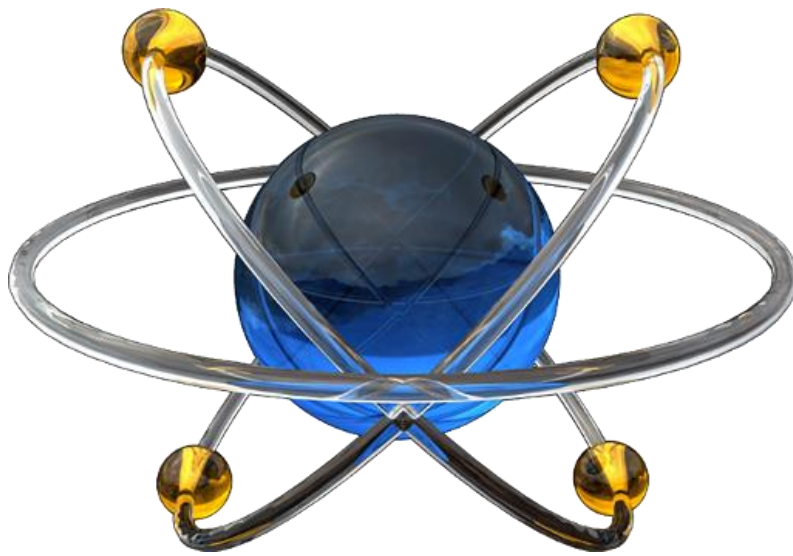


Figure 43.6: Proteus

CHAPTER 7: RESULTS, DISCUSSION, AND FUTURE WORK

7.1 Chapter Introduction

This chapter presents the final empirical and practical evaluation of the developed automated inspection and sorting system's operational performance. Following the successful interlocking and integration of the mechanical chassis, electronic circuitry, and computer vision inference pipelines, the functional working prototype underwent a rigorous series of experimental trials within a real-world laboratory environment. This section analyzes the compiled statistical data, evaluates the physical and algorithmic challenges encountered during deployment, and provides a structured engineering roadmap for system optimization and technology transfer across various manufacturing sectors.

7.2 Operational Performance Results and Statistical Analysis

The overall system performance was iteratively benchmarked by operating the conveyor system at controlled, uniform velocities via hardware-timed Pulse Width Modulation (PWM) signals. Test specimens were passed through the imaging zone to critically evaluate the deep learning model's real-time inference reliability and its synchronization with the mechanical actuation subsystem.

Classification and Defect Detection Accuracy Evaluation:

The system recorded an overall classification accuracy ranging between 85% and 95% during the runtime execution of targeted defect isolation. The vision algorithms exhibited distinct behavioral traits depending on the physical geometry of the inspected feature, classified as follows:

- **Visible Structural Deficiencies:** The system achieved its highest success rates and stable inference routines when detecting the absolute absence of components, specifically Missing Caps and Missing Labels. This high stability

is directly attributed to the distinct color boundaries and high geometric contrast, allowing the deep learning model to extract features seamlessly.

- **Fluid and Water Level Detection Limitations:** The system encountered complex optical challenges and shifting accuracy rates when validating water levels or inspecting transparent liquids. The inherent transparency of both the fluid and the plastic container allowed ambient light to refract significantly, which severely diminished the dynamic contrast range required for stable software-based feature extraction.

System Latency and Real-Time Response Analysis:

The total processing loop execution time—measured from the exact millisecond a specimen enters the camera's Field of View (FOV) until the physical completion of the deflection cycle—ranged between 2 to 4 seconds. This absolute system latency is engineered into three major operational overheads:

1. **Inference & Software Processing Pipeline:** The temporal overhead consumed during real-time frame acquisition, processing via OpenCV pipelines, and deep learning feature extraction to output a classification verdict.
2. **Serial Communication Overhead:** The delay associated with transmitting multi-byte string arrays and control command packets from the host PC station to the Arduino Mega 2560 microcontroller via the UART serial protocol.
3. **Mechanical Latency:** The physical time required for the digital actuators to respond to PWM transitions, generate the rated torque, and physically articulate the deflector arm across the conveyor deck.

Conveyor Belt Stability and Mechanical Integrity:

The physical trials demonstrated exceptional mechanical stability within the primary motion subsystem. The conveyor belt (re-engineered from the treadmill chassis foundation) operated continuously without registering any instances of belt slippage

or critical dynamic vibrations. This mechanical structural integrity eliminated motion blur, ensuring that specimens maintained their localized coordinate path during high-repetition testing.

7.3 Engineering Discussion and Challenges Evaluation

This section focuses on the analytical evaluation of the system's mechanical, electrical, and software design. It discusses the underlying scientific and physical principles behind the achieved results and addresses the specific challenges observed during real-world operations.

Engineering Analysis of the Three-Way Dual-Servo Configuration:

The implementation of a three-way sorting mechanism driven by dual symmetrical, high-torque digital servo actuators (DS3218MG) introduced a significant operational and economic advantage to the system architecture. The kinematic behavior of the mechanism was evaluated based on three distinct operational states:

- 1. Default Center Lane (Accepted State):** The system was algorithmically and mechanically designed to allow fully accepted, defect-free products to pass straight through the center lane without activating either servo motor. This design methodology drastically minimized overall power consumption, eliminated unnecessary actuator thermal stress, and reduced cyclic mechanical wear.
- 2. Rework Lane (Correctable Defect State):** Upon the detection of a correctable defect (e.g., a missing cap or a missing label), a specific hardware-timed pulse is routed to the corresponding actuator, deflecting the arm to a precise angular orientation and guiding the item into the rework lane. This pathway minimizes raw material waste and aligns the system with Total Quality Management (TQM) principles.

3. Rejected/Scrap Lane (Critical Defect State): When a critical, uncorrectable structural defect is identified, the opposing servo motor is actuated to completely deflect the item out of the production line. This symmetrical deflection configuration significantly reduced mechanical deflection latency compared to relying on a single, multi-angle actuator.

Impact of Ambient Lighting Conditions and Optical Noise:

Experimental laboratory trials demonstrated that uncontrolled fluctuations in ambient room lighting (such as shifting sunlight from windows or high-frequency overhead fluorescent flickering) served as the primary root cause for the classification accuracy variance between 85% and 95%. This unconstrained lighting environment introduced sharp specular reflections across the transparent plastic external surfaces of the bottles, which corrupted visual feature maps and occasionally misled the deep learning inference pipeline.

The Challenge of Optical Refraction and Transparency in Fluid Inspection:

The most complex obstacle to achieving vision-based pipeline stability was related directly to the physical properties of the transparent liquid (water). Because the refractive index of water and clear plastic transmits light rather than reflecting it, the boundary lines defining the internal water level faded into the gray laboratory background. This severe drop in dynamic contrast range significantly complicated the feature extraction routine for the object detection model, making real-time fluid validation far more demanding than isolating opaque, high-contrast caps and labels.

7.4 Future Work and Recommendations

Based on the empirical evaluations and the physical/algorithmic challenges identified within the working prototype, the team proposes a comprehensive suite of future engineering advancements. These recommendations aim to transition the system from a laboratory setup to a heavy-duty industrial production environment, categorized into three core technical domains:

Optical and Physical Optimization:

1. **Implementation of an Optically Isolated Inspection Enclosure (Dark Box):** It is highly recommended to construct a sealed inspection tunnel surrounding the camera zone on the conveyor deck. The interior walls of this enclosure should be fully coated in a flat, non-reflective Matte Black finish. This isolation guarantees the absolute elimination of ambient optical noise and external specular reflections. When integrated with a Diffused Backlighting setup, the dynamic contrast range of transparent surfaces and liquids will drastically sharpen, driving the overall classification accuracy beyond 95%.
2. **Integration of Stroboscopic Lighting Systems:** Deploying an illumination module synchronized precisely with the camera's frame acquisition rate allows the system to capture high-definition, crisp images of specimens moving at elevated velocities, completely mitigating the risk of Motion Blur.

AI Upgrades and Advanced Automation:

3. **Multi-Axis Industrial Robotic Arm Integration:** For future iterations, it is recommended to replace the mechanical servo deflector arms with a fully integrated industrial Multi-Axis Robotic Manipulator. This upgrade will endow the system with ultra-fast cycle times, significantly higher payload capacities, and the kinematic dexterity required to handle specimens of varying weights and complex spatial geometries.
4. **Edge AI Hardware Deployment:** We propose transitioning the computational pipeline from a standard host PC station to dedicated, high-throughput embedded hardware, such as the NVIDIA Jetson Orin platform. This migration drastically reduces the physical footprint and power consumption of the automation stack, making it resilient against harsh, high-temperature industrial floor conditions.

5. **IoT and Real-Time Cloud Quality Management:** Interlocking the local inspection pipeline with cloud-based databases via Internet of Things (IoT) protocols will enable production managers to monitor Key Performance Indicators (KPIs), defect distributions, and sorting throughput ratios (Accepted/Rework/Scrap) in real time from any global location.

System Scalability and Technology Transfer Horizon:

6. **Cross-Sector Industrial Multi-Domain Deployment:** The inherent kinematic flexibility of the developed three-way dual-servo sorting layout, combined with deep learning adaptability, proves that this project operates as a highly versatile Flexible Automation Platform rather than a narrow liquid-sorting machine. By executing simple retraining routines with entirely new datasets, the architectural infrastructure possesses immediate readiness for technology transfer across the following vital sectors:

- **Garment and Textile Industry:** The vision pipeline can scan moving fabrics or garments to isolate structural stitching defects, tears, or color inconsistencies. Opaque, defect-free items proceed along the default center lane. The first servo actuator deflects minor, correctable flaws to the Rework lane for sewing adjustments, while the opposing servo routes heavily compromised units directly to the Scrap lane.
- **Pharmaceutical Manufacturing:** The system can validate the integrity of blister packs and fluid vials, ensuring full capsule counts, correct capping, and clear printing of batch numbers and expiration dates. Intact specimens pass via the center lane; fully sealed packs missing outer labels or box flaps are guided by the first actuator into the Rework stream, while damaged or empty capsule packs are instantly purged into the Rejected lane to guarantee absolute consumer safety.
- **Food and Bakery Industry:** To inspect packaged goods such as biscuits and cakes, the computer vision model can detect compromised seals or

structural deformations. The first servo directs loosely sealed containers to the Rework lane for re-packaging and quality stabilization, whereas burnt, deformed, or broken food items are aggressively isolated via the Scrap lane, and premium, fully compliant goods slide through the center lane for final bulk packaging.

- **Agricultural Sorting and Grading:** For sorting high-value crops and firm fruits intended for export or local distribution markets, the network can classify produce based on size, color maturity, and the presence of surface defects or pests. Premium-grade produce (First Class) flows uninterrupted through the center lane; clean, unblemished produce requiring minor processing (e.g., secondary washing or leaf removal) is moved to the Rework path, and infected or decaying produce is completely separated via the Rejected lane to avoid contaminating the batch.

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